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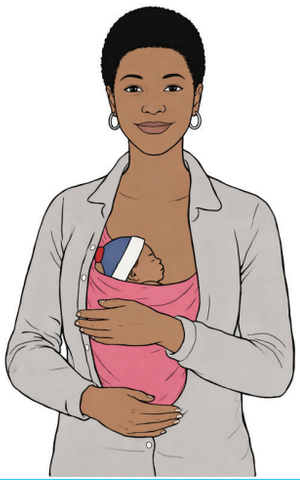
Ministry of Health

INTEGRATED MANAGEMENT OF NEWBORN & CHILDHOOD ILLNESS (IMNCI)

A guide for Healthcare Workers

2026 Edition











INTEGRATED MANAGEMENT OF NEWBORN & CHILDHOOD ILLNESS (IMNCI)

FOREWORD

Child deaths in Kenya remain unacceptably high. According to the Kenya Demographic Health Study (KDHS) 2022, the under-5 mortality is currently at 41 deaths per 1,000 live births, the Infant mortality at 32 deaths per 1,000 live births and the neonatal mortality at 21 deaths per 1,000 live births. Unfortunately, most of these deaths are attributed to preventable and treatable childhood illnesses such as diarrhoea, pneumonia (acute respiratory infections), malnutrition & anaemia, malaria, measles, HIV and tuberculosis. Additionally, a proportion of deaths in children occur in the neonatal period due to infections such as sepsis, complications of pregnancy and childbirth such as birth asphyxia, prematurity, low birth weight and other congenital anomalies and these account for 66% of infant deaths and 51% of under-5 deaths.

The Integrated Management of Newborn & Childhood Illness (IMNCI) strategy for delivering interventions that prevent and treat common causes of mortality in children was introduced in the mid-1990s by WHO/UNICEF as part of Child Survival Strategy. IMNCI has demonstrated improved and quality child health services. It recognizes that sick children often present with more than one symptom at the same time. These symptoms contribute to many childhood deaths if not comprehensively assessed, and treated. As a result, all sick children should be managed using an integrated approach in order to tackle the major drivers of child mortality, and thereby save lives.

The IMNCI strategy incorporates standard case management guidelines meant to improve skills of health care workers, as well as approaches for improving family and community health practices that ensure child survival, growth and development. IMNCI calls for better integration of child health programming across different ministries and sectors at the national and county level. As such, its implementation requires continuous strengthening of a number of elements of child health programs in planning and policy development, health systems strengthening, skilled human resources at all levels of care, health promotion and community-based care.

Importantly, IMNCI guidelines remain responsive to current research and recommendations; hence it is an important tool for health care professionals. The information is presented in a manner using the IMNCI 6 major steps namely; assessment, of treatment, treating the sick child or young infant, counseling the mother and providing follow up care.

Through the dissemination of IMNCI 2026 updated guidelines, it is envisaged that the technical information herein will provide impetus towards county level implementation and compliance at all levels of care in Kenya.



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TABLE OF CONTENTS

FOREWORD 1
ACKNOWLEDGEMENTS 2
ABBREVIATIONS 5
INTEGRATED MANAGEMENT OF NEWBORN AND CHILDHOOD ILLNESS (IMNCI) CHART BOOKLET 6
Background 6
What is IMNCI? 6
What is the IMNCI Chart booklet? 6
ASSESS, CLASSIFY AND IDENTIFY TREATMENT FOR THE SICK CHILD (AGE 2 MONTHS UP TO 5 YEARS)
CHECK FOR GENERAL DANGER SIGNS 8
DOES THE CHILD HAVE COUGH OR DIFFICULTY IN BREATHING? 9
DOES THE CHILD HAVE DIARRHOEA? 10
DOES THE CHILD HAVE FEVER? 12
DOES THE CHILD HAVE FEVER? (CHECK FOR MEASLES) 13
DOES THE CHILD HAVE AN EAR PROBLEM? 14
ASSESSMENT OF INFANTS UNDER 6 MONTHS AT RISK OF POOR GROWTH AND DEVELOPMENT 15
CHECK FOR ACUTE MALNUTRITION (6 - 59 MONTHS) 16
THEN CHECK FOR ANAEMIA 18
CHECK FOR HIV EXPOSURE AND INFECTION 19
SCREEN FOR TUBERCULOSIS 21
CHECK FOR CHILD'S DEVELOPMENTAL MILESTONES 22
ASSESS FOR INTERACTION, COMMUNICATION AND RESPONSIVENESS 23
IMMUNIZATION 25
VITAMIN A & DEWORMING 25
ASSESS FOR OTHER PROBLEMS THAT THE CHILD MAY HAVE and HAVE NOT BEEN DISCUSSED IN IMNCI 25
ASSESS THE MOTHER'S / CAREGIVER'S HEALTH NEEDS 25
TREAT THE CHILD
TEACH THE MOTHER/CAREGIVER TO GIVE ORAL DRUGS AT HOME
GIVE AN APPROPRIATE ORAL ANTIBIOTIC 26
DEWORMING USING ALBENDAZOLE 26
TREATMENT FOR UNCOMPLICATED MALARIA 27
GIVE VITAMIN A 27
GIVE IRON 27
GIVE ZINC SULPHATE 27
GIVE PARACETAMOL FOR FEVER OR EAR PAIN 27
TEACH THE MOTHER/CAREGIVER TO TREAT LOCAL INFECTIONS AT HOME
TREAT EYE INFECTION WITH TETRACYCLINE EYE OINTMENT 28
CLEAR THE EAR BY DRY WICKING 28
TREAT MOUTH ULCERS WITH GENTIAN VIOLET 28
TREAT FOR THRUSH WITH NYSTATIN 28
SOOTHE THE THROAT, RELIEVE THE COUGH WITH A SAFE REMEDY 28
GIVE THESE TREATMENTS IN A HEALTH FACILITY ONLY
TREATMENT OF VERY SEVERE DISEASE 29
GIVE AN INTRAVENOUS/ INTRAMUSCULAR ANTIBIOTIC 29
TREATMENT OF SEVERE MALARIA 29
PRESCRIBING OXYGEN 30
TREAT FOR POSSIBLE ASTHMA 30
MANAGE THE CHILD FOR LOW BLOOD SUGAR 31
RAPID ACTING BRONCHODILATOR 31
TREATMENT OF CONVULSIONS 31
REHYDRATION THERAPY AND FEEDING FOR DIARRHOEA
PLAN A: TREAT DIARRHOEA WITH NO DEHYDRATION 32
PLAN B: TREAT DIARRHOEA AT FACILITY WITH ORS (SOME DEHYDRATION) 32
TREAT HYPOVOLAEMIC SHOCK 33
PLAN C: TREAT SEVERE DEHYDRATION QUICKLY 33
REHYDRATION THERAPY FOR DIARRHOEA IN CHILDREN WITH SEVERE ACUTE MALNUTRITION 33
HIV TREATMENT
INITIAL ART REGIMEN 34
ART FOR INFANTS 0 – 4 WEEKS AND >37 WEEKS 34
ARVS DOSING CHARTS 34
ART FOR INFANTS <37 WEEKS 34
COTRIMOXAZOLE PROPHYLAXIS 35
ART FOR CHILDREN WITH TB CO-INFECTION 35
RISK CATEGORIZATION FOR MTCT CLIENTS TO DETERMINE INFANT PROPHYLAXIS 35
INFANT HIV PROPHYLAXIS
INFANT PROPHYLAXIS DOSING: ABC+3TC+ DTG REGIMEN FOR INFANTS AT HIGH RISK OF HIV ACQUISITION 36
NEVIRAPINE AND ZIDOVUDINE (BIRTH TO 6 WEEKS) AND EXTENDED NEVIRAPINE PROPHYLAXIS FOR INFANTS AT LOW RISK OF VERTICAL TRANSMISSION OF HIV, AND INFANTS CONTINUING INFANT PROPHYLAXIS 36
TB TREATMENT
ELIGIBILITY CRITERIA FOR PAEDIATRIC TB REGIMENS 39
DOSAGES FOR PAEDIATRIC TB TREATMENT (IMPROVED FORMULATIONS) DOSAGES FOR A CHILD UP TO 3.9KGS 40
REGIMEN FOR TREATMENT OF TB IN CHILDREN 40
DOSAGES FOR A CHILD BETWEEN 4 - 25KGS 40
PYRIDOXINE (VITAMIN B6) DOSING FOR CHILDREN ON TB TREATMENT 40
COUNSEL THE MOTHER
ASSESS CHILD'S FEEDING 41
CONDUCT THE APPETITE TEST 41
COUNSEL ON FEEDING PROBLEMS AND GUIDANCE ON INFANT FEEDING 42
GUIDANCE ON INFANT FEEDING IN HIV CONTEXT 42
FEEDING RECOMMENDATIONS FOR ALL CHILDREN DURING SICKNESS AND HEALTH, INCLUDING HIV EXPOSED CHILDREN ON ARV PROPHYLAXIS 43
RESPONSIVE CARE FOR EARLY CHILDHOOD DEVELOPMENT 44

ABOUT CARE FOR DEVELOPMENT PROBLEMS 45
GIVE GOOD HOME CARE FOR YOUR CHILD 46
ABOUT MOTHER'S OWN HEALTH 46
WHEN TO RETURN FOR SCHEDULED VISITS 47
WHEN TO RETURN IMMEDIATELY 47

GIVE FOLLOW - UP CARE

PNEUMONIA 48
WHEEZING 48
PERSISTENT DIARRHOEA 48
DYSENTERY 48
UNCOMPLICATED MALARIA 49
FEVER - NO MALARIA 49
EYE OR MOUTH COMPLICATIONS OF MEASLES 49
EAR INFECTION 49
FEEDING PROBLEM 50
MALNUTRITION 50
MODERATE ACUTE MALNUTRITION 50
SEVERE ACUTE MALNUTRITION WITHOUT COMPLICATIONS 50
ANAEMIA 50
HIV EXPOSED & INFECTED CHILDREN 50

IMNCI PROCESS FOR ALL YOUNG INFANTS (BIRTH UP TO TWO MONTHS)

ASSESS, AND CLASSIFY AND IDENTIFY TREATMENT FOR THE SICK YOUNG INFANT FROM BIRTH AGE UP TO 2 MONTHS

CHECK FOR POSSIBLE SERIOUS BACTERIAL INFECTION, VERY SEVERE DISEASE, PNEUMONIA OR LOCAL BACTERIAL INFECTION 52
THEN, CHECK FOR JAUNDICE 53
THEN CHECK FOR EYE CONDITIONS 53
DOES THE YOUNG INFANT HAVE DIARRHOEA OR SIGNS OF DEHYDRATION 54
THEN CHECK FOR SPECIAL TREATMENT NEEDS 55
CHECK FOR HIV EXPOSURE AND INFECTION 56
THEN CHECK FOR FEEDING PROBLEM, LOW WEIGHT FOR AGE OR LOW BIRTH WEIGHT 57
CHECK FOR LOW BIRTH WEIGHT IN YOUNG INFANT LESS THAN 1 WEEK 58
CHECK FOR DEVELOPMENTAL MILESTONES 58
ASSESS FOR OTHER ILLNESSES 58
CHECK FOR INTERACTION AND /OR COMMUNICATION AND RESPONSIVENESS 58
ASSESS MOTHER'S / CAREGIVER'S NEEDS 58
THEN CHECK IMMUNIZATION STATUS 58

TREAT THE YOUNG INFANT

ROUTINE CARE FOR ALL NEWBORNS AFTER DELIVERY 59
KEEP THE YOUNG INFANT WARM 59
TEACH THE MOTHER HOW TO KEEP THE YOUNG INFANT WARM ON THE WAY TO THE HOSPITAL 59

TEACH THE CAREGIVER TO TREAT LOCAL INFECTIONS AT HOME

TO TREAT SKIN PUSTULES OR UMBILICAL INFECTION 60
TO TREAT THRUSH (WHITE PATCHES IN MOUTH) OR MOUTH ULCERS 60
TREAT EYE INFECTION WITH TETRACYCLINE EYE OINTMENT 60

TEACH THE MOTHER/CAREGIVER TO GIVE ORAL DRUGS AT HOME

GIVE AN APPROPRIATE ORAL ANTIBIOTIC FOR LOCAL BACTERIAL INFECTION 60

GIVE THESE TREATMENTS IN A HEALTH FACILITY ONLY

NEONATAL RESUSCITATION 61
TREATMENT OF POSSIBLE SERIOUS BACTERIAL INFECTION OR VERY SEVERE DISEASE 61
TREAT AT RISK OF CONGENITAL SYPHILIS 61
TREAT DIARRHOEA 61
WHERE REFERRAL IS REFUSED OR IS NOT FEASIBLE - FURTHER CLASSIFY THE SICK YOUNG INFANT 62
WHERE REFERRAL IS REFUSED OR NOT FEASIBLE: FOR A SICK YOUNG INFANT WITH POSSIBLE SERIOUS BACTERIAL INFECTION OR VERY SEVERE DISEASE 63
TREAT OR PREVENT LOW BLOOD SUGAR IN THE YOUNG INFANT 63
TREAT FOR NEONATAL CONVULSIONS 64
EARLY INFANT DIAGNOSIS OF HIV IN INFANTS AND CHILDREN 64

COUNSEL THE MOTHER

TEACH CORRECT POSITIONING AND ATTACHMENT FOR BREASTFEEDING 65
TEACH THE MOTHER TO TREAT BREAST OR NIPPLE PROBLEMS 65
TEACH THE MOTHER HOW TO KEEP THE YOUNG INFANT WITH LOW WEIGHT FOR AGE OR LOW BODY TEMPERATURE WARM AT HOME 65
TEACH THE MOTHER ON UMBILICAL CORD CARE 65
ADVISE MOTHER TO GIVE HOME CARE FOR THE YOUNG INFANT 66
WHEN TO RETURN FOR A FOLLOWUP VISIT 66
WHEN TO RETURN IMMEDIATELY 66

GIVE FOLLOW-UP CARE FOR THE SICK YOUNG INFANT

WHERE REFERRAL IS NOT FEASIBLE: CLINICAL SEVERE INFECTION 67
PNEUMONIA 67
LOCAL BACTERIAL INFECTION 67
EYE INFECTION 67
DIARRHOEA: SOME DEHYDRATION OR NO DEHYDRATION 67
JAUNDICE 67
FEEDING PROBLEM 68
LOW WEIGHT FOR AGE OR LOW BIRTH WEIGHT 68
THRUSH OR MOUTH ULCERS 68

ANNEXES

RECORDING FORMS

MANAGEMENT OF THE SICK CHILD AGE 2 MONTHS UP TO 5 YEARS 69
MANAGEMENT OF THE SICK YOUNG INFANT AGE UP TO 2 MONTHS 71

GROWTH CHARTS 73

ABBREVIATIONS

Abbreviation	Meaning
3TC	Lamivudine
ABC	Abacavir
AL	Artemether-Lumefantrine
AVPU	Alert, Voice, Pain, Unresponsive
AZT	Zidovudine
BCG	Bacillus Calmette-Guerin vaccine
b-OPV	Bivalent Oral Polio Vaccine
CCC	Comprehensive Care Clinic
CLHIV	Children Living with HIV
CPAP	Continuous Positive Airway Pressure
CPT	Cotrimoxazole Preventive Therapy
CTX	Cotrimoxazole
DBS	Dried Blood Spot
DHA-PPQ	Dihydroartemisinin-Piperaquine
DRTB	Drug-Resistant Tuberculosis
DRV	Darunavir
DS	Double Strength tablet
DSTB	Drug-Susceptible Tuberculosis
DT	Dispersible Tablet
DTG	Dolutegravir
EID	Early Infant Diagnosis
EPTB	Extrapulmonary Tuberculosis
FDC	Fixed-Dose Combination
FNA	Fine Needle Aspiration
FQ	Fluoroquinolone
HEI	HIV-Exposed Infant
HIV	Human Immunodeficiency Virus
HTS	HIV Testing Services
ICS	Inhaled Corticosteroid
IGRA	Interferon-Gamma Release Assay
iKMC	Immediate Kangaroo Mother Care
INH	Isoniazid
IPV	Inactivated Polio Vaccine
KMC	Kangaroo Mother Care
LF-LAM	Lateral Flow Lipoarabinomannan
LFX	Levofloxacin
MCH	Maternal and Child Health
MDI	Metered-Dose Inhaler
MDR/RR	Multidrug-Resistant / Rifampicin-Resistant
MIYCN	Maternal, Infant and Young Child Nutrition
mRDT	Malaria Rapid Diagnostic Test
MTCT	Mother-to-Child Transmission
MUAC	Mid-Upper Arm Circumference
NRM	Non-Rebreather Mask
NVP	Nevirapine
ORS	Oral Rehydration Solution
pALD	Paediatric Antiretroviral Dispersible
PBFW	Pregnant and Breastfeeding Women
PCR	Polymerase Chain Reaction
PCV	Pneumococcal Conjugate Vaccine
PCV10	10-valent Pneumococcal Conjugate Vaccine
PDR	Poly-Drug Resistant
pDTG	Paediatric Dolutegravir
PLHIV	People Living with HIV
PMA	Postmenstrual Age
PMTCT	Prevention of Mother-to-Child Transmission
PNA	Postnatal Age
PTB	Pulmonary Tuberculosis
RBS	Random Blood Sugar
RHZE	Rifampicin + Isoniazid + Pyrazinamide + Ethambutol
ROTA	Rotavirus vaccine
RUTF	Ready-to-Use Therapeutic Food
SD	Standard Deviation
SS	Single Strength
STIs	Sexually Transmitted Infections
TAF	Tenofovir Alafenamide
TDF	Tenofovir Disoproxil Fumarate
TB	Tuberculosis
TCV	Typhoid Conjugate Vaccine
TPT	Tuberculosis Preventive Therapy
VDRL	Venereal Disease Research Laboratory test
VL	Viral Load
WAZ	Weight-for-Age Z-score
WFH/L	Weight-for-Height/Length
WLZ	Weight-for-Length Z-score
XDR	Extensively Drug-Resistant

INTEGRATED MANAGEMENT OF NEWBORN AND CHILDHOOD ILLNESS (IMNCI) CHART BOOKLET

Background

In Kenya, 41 out of every 1000 children born, do not live to be 5years of age with 3/4 of them occuring within the first year (Infant mortality rate of 32 deaths per 1000 live births) and half occurring within the neonatal period (NMR 21 deaths per 1000 live births) (KDHS 2022). Of note, 70% of all deaths in children can be attributed to easily preventable and treatable diseases namely: Acute respiratory infections (mostly pneumonia), Diarrhoea, Measles, Malaria, Malnutrition & Anaemia, HIV and Tuberculosis. Often children succumb to a combination of these conditions - with most children presenting in health facilities with the combined signs and symptoms of more than one of these diseases. Evidence has shown that many of these children are not comprehensively assessed, treated and given the appropriate advice. Recognizing the need to improve on the care of these children, WHO and UNICEF developed the Integrated Management of Newborn and Childhood Illness (IMNCI) strategy which emphasizes on integrated case management of the most common childhood diseases. In 2000, Kenya adopted the IMNCI strategy, which forms a critical part of the Kenya Essential Package for Health (KEPH).

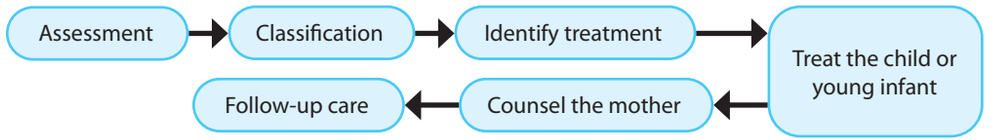
What is IMNCI?

The Integrated Management of Newborn and Childhood Illness (IMNCI) case management approach offers simple and effective methods to comprehensively prevent and manage the leading causes of serious illnesses and mortality in children below 5years. With IMNCI, sick children or young infants are not only treated for the signs and symptoms they present within a health facility, but are also assessed for other disease conditions they may be suffering from.

IMNCI is based on the following principles:

- All sick children aged up to 5 years are examined for general danger signs and all young infants are examined for signs of very severe disease. These signs indicate the need for immediate referral or admission to hospital.
- Children and infants are then assessed for main symptoms. For the older children, the symptoms include cough, difficulty in breathing, diarrhoea, fever, TB, HIV, ear infections, anaemia, measles and malnutrition.
- A combination of individual signs then lead to the child's or young infant's classification within one or more symptom groups.
- Essential drugs are then used to treat the children or young infants. Lastly counseling of caregivers regarding home care, appropriate feeding and when to return to facility - immediately or on follow-up - is done.

IMNCI 6 major steps



Where can IMNCI be applied?

The IMNCI approach is designed for use in clinical settings at all levels of health care in Kenya where children under 5 years are managed. These are dispensaries, health centres, sub-county hospitals, county hospitals and national referral hospitals including, faith based and private health facilities.

What is the IMNCI Chart booklet?

Kenya's IMNCI guidelines are packaged in this chart booklet. The chart booklet provides a step-by-step guide to a healthcare worker on case management of children below 5 years of age visiting health facilities. The chart booklet summarizes and describes the 6 step IMNCI process illustrated above.

By using the IMNCI chart booklet and implementing the guidelines the health workers will be able to implement on the key elements of IMNCI:

1. Assessment

The assess column in the chart booklet describes how to take history and do a physical exam

- Routinely assess for general danger signs (or possible serious bacterial infection in a young infant) - a general danger sign indicates that a child has a serious and life threatening condition that requires urgent attention
- Assess for common illnesses in children or young infant's by asking questions about common conditions, examining the child or young infant and checking the need for other routine services such as immunization and nutrition.
- Look for other health problems.

2. Classification

The classify (signs and classify) column of the chart means the health worker has to make a decision on the severity of the illness.

Healthcare workers will be able to classify children or young infants illnesses using the colour- coded triage system.

Colour	Classification
Pink	= Severe Classification needing admission or pre-referral treatment and referal
Yellow	= A classification needing specific medical treatment and advice
Green	= Not serious and in most cases no drugs are needed. Simple advice on home management is given

3. Identify Treatment

The identify treatment column helps the healthcare workers quickly and accurately identify treatment for the selected. If a child or young infant has more than one illness, the healthcare worker must look at more than one table to identify the appropriate treatment.

4. Treat

The treat column shows how to administer the treatment. Treat means giving the treatment in the facility, prescribing drugs or other treatments to be given at home and also teaching the mother/ caregiver how to administer treatment at home.

The following rules should be adhered to.

- If a child or young infant requires admission or referral (pink), it is important the essential treatment is offered to the child or young infant before admission or referral.
- If the child or young infant requires treatment (yellow), develop a treatment plan, administer drugs to be given at the facility, advise on treatment at home and counsel the mother/ caregiver accordingly.
- If no serious conditions have been found (green), advise the mother/caregiver on care of child at home.

5. Counsel

If follow up care is indicated, teach the mother/caregiver when to return to the clinic. Also teach the mother/ caregiver how to recognize signs indicating that the child or young infant should be brought back to the facility immediately.

6. Follow up

Some children or young infants need to be seen more than once for a current episode of illness. Identify such children or young infants and when they are brought back, offer appropriate follow up care as indicated in the IMNCI guideline including reassessing the child or young infant for any new problems.

The guidelines also aim to empower healthcare workers to:

- Correctly interview caregivers.
- Provide counseling for appropriate preventative and treatment measures.
- Correctly counsel the mother about her own health.

Who can apply the IMNCI approach?

- The IMNCI process can be applied to any healthcare worker working in settings where children or young infants below 5years are managed. These should include: Doctors, Clinical Officers and Nurses.

What should you do when you receive the IMNCI chart booklet at your facility?

- The Ministry of Health has distributed the IMNCI chart booklet to all health facilities in Kenya. Upon receiving a copy at your facility, all health workers should familiarize themselves with the guidelines and begin immediate implementation/use. **NB: the hard copy of the chart booklet should remain at the health facility at all times.**
- A mobile application of the IMNCI guidelines has been developed by the Ministry of Health (MOH) and is available for free to all healthcare workers. For instructions on how to download, please visit www.health.go.ke or contact the Newborn Child and Adolescent Health Unit (NCAHU) at the MOH.

ASSESS, CLASSIFY AND IDENTIFY TREATMENT FOR THE SICK CHILD (AGE 2 MONTHS UP TO 5 YEARS)

ASK THE MOTHER WHAT THE CHILD'S PROBLEMS ARE

- Determine if this is an initial or follow-up visit for this problem.
  - If follow-up visit, use the follow-up instructions on TREAT THE CHILD chart.
  - If initial visit, assess the child as follows:

CHECK FOR GENERAL DANGER SIGNS

ASSESS

ASK:

- Is the child able to drink or breastfeed?
- Does the child vomit everything?
- Has the child had convulsions in this illness?

LOOK:

- See if the child is lethargic or unconscious.
- Is the child convulsing now? If yes treat immediately.

URGENT ATTENTION

CLASSIFY

SIGNS	CLASSIFY
Any general danger sign	**VERY SEVERE DISEASE**

IDENTIFY TREATMENT

TREATMENT
(URGENT PRE-REFERRAL TREATMENTS ARE IN BOLD PRINT.)

- **If the child is unconscious or lethargic give oxygen** (Pg 30).
- **Give diazepam if convulsing now** (Pg 31) **– Do not give more than 2 doses of diazepam in 24hrs once phenobarbital is used.**
- **Quickly complete the assessment.**
- **Give any pre-referral treatment immediately.**
- **Test for and treat / prevent low blood sugar** (Pg 31).
- **Keep the child warm.**
- **Refer URGENTLY.**
- **Screen for possible TB disease** (Pg 21) **and check for HIV** (Pg 19 - 20).

A child with any general danger sign needs URGENT attention; complete the assessment and administer any pre-referral treatment immediately so referral is not delayed.

ACUTE MEDICAL PROBLEMS UNDER IMNCI SEVERE CLASSIFICATIONS

- Very severe disease
- Severe pneumonia or Very severe disease
- Hypovoalemic Shock from Diarrhoea/ Dehydration
- Severe dehydration
- Severe persistent diarrhoea
- Very severe febrile disease or severe malaria
- Severe complications of measles
- Mastoiditis
- Severe acute malnutrition with complications for children > 6 months
- Severe anaemia

THEN ASK ABOUT MAIN SYMPTOMS:

DOES THE CHILD HAVE COUGH OR DIFFICULTY IN BREATHING?

ASSESS

IF YES, ASK
For how long?

LOOK, LISTEN, FEEL (CHILD MUST BE CALM)

- Count the breaths in one minute. Use respiratory rate timers where available
- Check for central cyanosis
- Listen for stridor*
- Listen for grunting
- Look for lower chest wall indrawing**
- Listen for wheeze***
- Check for oxygen saturation using pulse oximetry
- Check AVPU****
- If wheezing with either chest in- drawing or fast breathing: Assess for possible Asthma
- If wheezing assess for possible TB disease

Fast breathing is:

If the child is:

- 2 months up to 12 months - 50 breaths per minute or more
- 12 months up to 5 years - 40 breaths per minute or more

Classify COUGH or DIFFICULTY IN BREATHING



CLASSIFY		IDENTIFY TREATMENT
SIGNS	**CLASSIFY**	**TREATMENT** (URGENT PRE-REFERRAL TREATMENTS ARE IN BOLD PRINT.)
• Any general danger sign **OR any of** • Oxygen saturation less than 90% • Grunting • Stridor in a calm child. • Central Cyanosis • AVPU = V, P or U	**SEVERE PNEUMONIA OR VERY SEVERE DISEASE**	• **If oxygen saturation is less than 90%, start oxygen therapy** (Pg 30) **and refer or admit.** • **Give first dose of Benzyl Penicillin and Gentamicin** (Pg 29). • **Test and Treat for / prevent low blood sugar** (Pg 31). • **Keep the child warm.** • **Treat wheeze if present** (Pg 30). • **Admit or refer urgently to hospital.** • **If severe viral croup, nebulize with 3-5mls of adrenaline 1:1000** (Pg 31). **Give oxygen and refer.** • **Screen for possible TB disease** (Pg 21) **and check for HIV** (Pg 19 - 20).
• Lower chest wall indrawing in calm child **OR** • Fast breathing	**PNEUMONIA (NON SEVERE PNEUMONIA)**	• **Give high dose Amoxicillin (Amoxicillin Dispersible Tablet (DT))** (Pg 26). • Treat wheeze if present (Pg 30) and follow-up in 2 days. • Soothe the throat and relieve the cough with a safe remedy (Pg 28). • Screen for possible TB disease (Pg 21) and check for HIV (Pg 19 - 20). • Advise caregiver/ mother when to return immediately. • Review in 2 days. **If not possible, admit OR refer children with chest indrawing.**
No signs of pneumonia or very severe disease.	**NO PNEUMONIA: COUGH OR COLD**	• Treat wheeze if present (Pg 30) and follow-up in 2 days. • Soothe the throat and relieve the cough with a safe remedy (Pg 28). • Screen for possible TB disease (Pg 21) and check for HIV (Pg 19 - 20). • Advise caregiver/ mother when to return immediately. • Follow-up in 5 days.

Note:
*Stridor - A harsh sound heard during inspiration.
**Chest indrawing - Present if the lower chest wall moves in during inspiration.
***Wheeze - A musical sound heard during expiration.
****AVPU - Alert, responsive to Voice, responsive to Pain, Unresponsive

DOES THE CHILD HAVE DIARRHOEA?

ASSESS

IF YES, ASK:
- For how long?
- Is there blood in the stool?

LOOK, LISTEN, FEEL:
- Check:
  - Weak/ absent pulse
  - Not alert; AVPU*<A
  - Cold hands +Temp gradient
  - illary refill time 3 secs or more
- Look at the child's general condition. **Is the child:**
  - Lethargic or unconscious?
  - Restless and irritable?
- Look for sunken eyes
- Offer the child fluids. **Is the child:**
  - Unable to drink or drinking poorly?
  - Drinking eagerly, thirsty?
- Pinch the skin of the abdomen. Does it go back:
  - Very slowly (≥2 secs)?
  - Slowly (1-2sec)?
  - Immediately?

Note: Classify all children with diarrhoea for dehydration.

Classify DIARRHOEA

for DEHYDRATION

1



CLASSIFY		IDENTIFY TREATMENT
SIGNS	**CLASSIFY**	**TREATMENT** (URGENT PRE-REFERRAL TREATMENTS ARE IN BOLD PRINT.)
ALL FOUR OF: • Weak/absent pulse • AVPU < A • Cold hands + Temp gradient • illary refill 3secs or more **PLUS** • Sunken eyes and slow skin pinch.	**HYPOVOLAEMIC SHOCK FROM DIARRHOEA/ DEHYDRATION**	• **TREAT FOR SHOCK. A second bolus may be given if required before proceeding to step 2 of PLAN C** (Pg 33). • **Treat for / prevent low blood sugar** (Pg 31). • **Assess for severe acute malnutrition** (Pg 16 - 17). • **Assess for severe anaemia** (Pg 18). • **NB: If HB<5g/dl transfuse urgently. Give maintenance fluids as you wait for blood BUT do not give bolus IV fluid.** • **Admit or refer urgently to hospital.** • **Screen for possible TB disease** (Pg 21) **and check for HIV** (Pg 19 - 20)**.**
• Unable to drink OR AVPU<A (lethargic or unconscious) **PLUS ANY OF THE FOLLOWING:** • Sunken eyes • Return of skin pinch ≥ 2 sec.	**SEVERE DEHYDRATION**	• **Admit or refer URGENTLY to hospital with mother giving frequent sips of Low osmolarity ORS on the way. Give fluids for severe dehydration. (Plan C)** (Pg 33)**.** **OR** • **If Child has no other severe classification:** • **Admit in your facility.** • **Give fluids for severe dehydration. (Plan C)** (Pg 33). • **Give Low osmolarity ORS 5 ml/kg/hr via NGT and Zinc Sulphate** (Pg 27). • **Assess for severe acute malnutrition** (Pg 16 - 17)**.** • **Assess for severe anaemia** (Pg 18)**.** • **If child is 2 years or older and there is cholera in your area, give Doxycycline 4mg/kg for cholera (See pg 26 for alternative treatment). Notify Subcounty Disease Surveillance Officer.** • **Screen for possible TB disease** (Pg 21) **and check for HIV** (Pg 19 - 20) • **Counsel on preventive measures such as proper sanitation and waste disposal, safe drinking water including purification (eg boiling), handwashing, exclusive breastfeeding**

Continue next page

NB: If referral is not possible, manage the child as described in Integrated Management of Newborn and Childhood Illness and Basic Paediatric Protocols.
*AVPU - Alert, responsive to voice, responsive to pain, unresponsive.

for DEHYDRATION 2

• Able to drink adequately or drinking eagerly **PLUS 2 OR MORE OF:** • Sunken eyes • Return skin pinch 1-2secs. • Restless/irritable	**SOME DEHYDRATION**	• **If Child has a severe classification:** • **Admit or refer URGENTLY to hospital with mother giving frequent sips of ORS on the way. Advise the mother to continue breast feeding.** **OR** • If the child has no severe classification: • **Give fluids (Low osmolarity ORS), Zinc sulphate and food for some dyhydration (Plan B)** (Pg 32). • Follow-up in 2 days. • Screen for possible TB disease (Pg 21) and check for HIV (Pg 19 - 20). • Advise mother when to return immediately. • Counsel on preventive measures such as proper sanitation and waste disposal, safe drinking water including purification (eg boiling), handwashing, exclusive breastfeeding.
Not enough signs to classify as some or severe dehydration.	**NO DEHYDRATION**	• **If child has a severe classification:** • **Admit or refer URGENTLY to hospital with mother giving frequent sips of Low osmolarity ORS on the way. Advise the mother to continue breast feeding.** **OR** • If the child has no severe classification: • **Give Low osmolarity ORS, fluids, Zinc suphate and food to treat diarrhoea at home (Plan A)** (Pg 32). • Follow-up in 5 days. • Screen for possible TB disease (Pg 21) and check for HIV (Pg 19 - 20). • Advise mother when to return immediately. • Counsel on preventive measures such as proper sanitation and waste disposal, safe drinking water including purification (eg boiling), handwashing, exclusive breastfeeding.

and if DIARRHOEA 14 DAYS OR MORE

Dehydration present: (hypovolemic shock, severe dehydration, some dehydration).	**SEVERE PERSISTENT DIARRHOEA**	• **Treat hypovolaemic shock and any other form of dehydration before referral unless the child has another severe classification.** • **Give Low osmolarity ORS and Zinc Sulphate** (Pg 27). • **Admit or refer URGENTLY to hospital with mother giving frequent sips of ORS on the way.** • **Screen for possible TB disease** (Pg 21) **and check for HIV** (Pg 19 - 20)**.**
No dehydration.	**PERSISTENT DIARRHOEA**	• Advise the mother on feeding a child who has PERSISTENT DIARRHOEA. • **Give Low osmolarity ORS and Zinc Sulphate** (Pg 27). • Follow-up in 5 days. • Screen for possible TB disease (Pg 21) and check for HIV (Pg 19 - 20).

and if BLOOD IN STOOL

Blood in the stool.	**DYSENTERY**	• **Treat with ciprofloxacin.** For dosage and second line treatment, see Pgs. 26 and 48 respectively. • **Give Low osmolarity ORS and Zinc Sulphate** (Pg 27). • Follow-up in 2 days. • Screen for possible TB disease (Pg 21) and check for HIV (Pg 19 - 20).

DOES THE CHILD HAVE FEVER?

(by history or feels hot or axillary temperature 37.5°C or above)

ASSESS

IF YES ASK:
- For how long?
- Has the child travelled to a high malaria burden area* in the last 1 month?

Decide Malaria burden: high or low burden.

THEN ASK:
- Has the child had signs of measles within the last 3 months? If yes, assess and classify for measles

LOOK AND FEEL:
- Look or feel for stiff neck.
- Look for any other cause of fever****

High Malaria burden area:
- Do a malaria test

Low malaria burden area:**
- Do a malaria test, if No obvious cause of fever

TEST
- Microscopy
- mRDT

NOTE: If you can't test, don't withhold treatment

Consider Measles if child has:
- Generalized rash with any of the following:
  - Cough,
  - Runny nose, or
  - Red eyes.

Refer to Fever: Measles chart on Pg 13.

CLASSIFY FEVER: HIGH OR LOW MALARIA BURDEN AREA

CLASSIFY		IDENTIFY TREATMENT
SIGNS	**CLASSIFY**	**TREATMENT** (URGENT PRE-REFERRAL TREATMENTS ARE IN BOLD PRINT.)
• Any general danger sign or • Stiff neck or • Hypoglycaemia (Glucose ≤ 2.5 mmol/l) or • Respiratory Distress **AND** • Confirm malaria with a test.	**VERY SEVERE FEBRILE DISEASE OR SEVERE MALARIA**	• **Give first dose of Artesunate for severe malaria** (Pg 29). • **Give first dose of Ceftriaxone** (Pg 29). • **Test, Treat/prevent low blood sugar** (Pg 31). • **Give one dose of paracetamol in clinic for high fever (≥38.5 °C)** (Pg 27). • **Admit or Refer URGENTLY to hospital.** • **Screen for possible TB disease** (Pg 21) **and check for HIV** (Pg 19 - 20)
• Malaria test POSITIVE***	**UNCOMPLICATED MALARIA**	• **Give Artemether + Lumefantrine (AL)** (Pg 27). • **Give one dose of paracetamol in clinic for high fever (≥38.5 °C)** (Pg 27). • If fever is present every day for more than 7 days assess further or refer. • Screen for possible TB disease (Pg 21) and check for HIV (Pg 19 - 20). • Advise when to return immediately. • Follow up in 3 days if fever persists. • Counsel the caregiver on preventive measures such as proper use of LLIN, indoor residual spraying, Malaria vaccine and environmental control.
• Malaria test NEGATIVE	**FEVER: NO MALARIA**	• Give one dose of paracetamol in clinic for high fever (≥38.5 °C) (Pg 27). • Assess for other possible causes of fever. • If fever is present every day for more than 7 days assess further or refer. • Screen for possible TB disease (Pg 21) and check for HIV (Pg 19 - 20). • Advise mother when to return immediately. • Follow up in 3 days if fever persists. • Counsel the caregiver on preventive measures such as proper use of LLIN, indoor residual spraying, Malaria vaccine and environmental control.

* High malaria burden area (Very high, High and moderate burden)
**Low malaria burden area (Low and very low burden)
***If no malaria test available: High burden - classify as UNCOMPLICATED MALARIA
**** Look for local tenderness, refusal to use a limb, hot tender swelling, red tender skin or boils, lower abdominal pain or pain on passing urine

DOES THE CHILD HAVE FEVER? (CHECK FOR MEASLES)

(by history or feels hot or axillary temperature 37.5°Cor above).

ASSESS

ASK:

- For how long?
- Has the child been in contact with anyone with measles?
- Has the child had signs of measles within the last 3 months?

Look for signs of MEASLES:

- Generalized rash and any of the following:
  - Cough,
  - Runny nose, or
  - Red eyes.

Check for Complications of MEASLES

- If the child has signs of measles now or within the last 3 months
  - Look for mouth ulcers, are they deep or extensive?
  - Look for pus draining from the eye
  - Look for clouding of the cornea

CLASSIFY MEASLES

IF SUSPECTED MEASLES NOW OF WITHIN THE LAST 3 MONTHS, CLASSIFY

CLASSIFY		IDENTIFY TREATMENT
SIGNS	**CLASSIFY**	**TREATMENT** (URGENT PRE-REFERRAL TREATMENTS ARE IN BOLD PRINT.)
• Generalized rash of measles **AND ONE OF:** • Cough, • Runny nose or • Red eyes	**SUSPECTED MEASLES**	• **Give Vitamin A** (Pg 27). • **Give one dose of paracetamol in clinic for high fever (≥39.0 °C)** (Pg 27). • Notify subcounty disease surveillance officer, take blood sample for confirmation of measles. • Screen for possible TB disease (Pg 21) and check for HIV (Pg 19 - 20). • Advise mother when to return immediately.
• Any general danger sign **OR** • Clouding of cornea **OR** • Deep or extensive mouth ulcers.	**SEVERE COMPLICATIONS OF MEASLES***	• **Give Vitamin A** (Pg 27). • **If clouding of the cornea or pus draining from the eye, apply tetracycline eye ointment** (Pg 28)**.** • **Admit or refer URGENTLY to hospital.** • **Screen for possible TB disease** (Pg 21) **and check for HIV** (Pg 19 - 20)**.**
• Pus draining from the eye **OR** • Mouth ulcers.	**EYE OR MOUTH COMPLICATIONS OF MEASLES**	• **Give Vitamin A** (Pg 27). • **If pus draining from the eye, treat eye infection with tetracycline eye ointment** (Pg 28)**.** • **If mouth ulcers, treat with Gentian Violet** (Pg 28). • Follow-up in 3 days. • If child has no indication for referral, notify and draw blood sample for confirmation of measles. • Screen for possible TB disease (Pg 21) and check for HIV (Pg 19 - 20).
• No pus draining from the eye or mouth ulcers.	**NO EYE OR MOUTH COMPLICATIONS OF MEASLES**	• **Give Vitamin A** (Pg 27) **if not received in the last 1 month.** • If child has no indication for referral, draw blood and send for confirmation of measles. • Screen for possible TB disease (Pg 21) immediately after the measles infection and check for HIV (Pg 19 - 20).

*Other important complications of measles - pneumonia, stridor, diarrhoea, ear infection and malnutrition are classified in other tables.

CHECK FOR AN EAR PROBLEM

ASSESS

ASK:
- Is there ear pain?
- Is the child rubbing or pulling the ear?
- Is there ear discharge? If yes, for how long?

LOOK AND FEEL:
- Look for pus draining from the ear.
- Feel for tender swelling behind the ear.

CLASSIFY EAR PROBLEM

CLASSIFY

IDENTIFY TREATMENT

SIGNS	CLASSIFY	TREATMENT (URGENT PRE-REFERRAL TREATMENTS ARE IN BOLD PRINT.)
• Tender swelling behind the ear.	**MASTOIDITIS**	• **Give first dose of Cefriaxone** (Pg 29)**.** • **Give first dose of Paracetamol for pain** (Pg 27). • **Refer URGENTLY to hospital or admit.** • **Check for HIV** (Pg 19 - 20)**.**
• Pus is seen draining from the ear or discharge is reported for **less than 14 days** or • Ear pain.	**ACUTE EAR INFECTION**	• **Give high dose Amoxicillin (Amoxicillin Dispersible Tablet (DT)) for 10 days** (Pg 26)**.** • Give paracetamol for pain (Pg 27). • Dry the ear by wicking (Pg 28). • Check for HIV infection (Pg 19 - 20). • Follow-up in 5 days.
• Pus is seen draining from the ear or discharge is reported **for 14 days or more**	**CHRONIC EAR INFECTION**	• Dry the ear by wicking (Pg 28). • Ciprofloxacin ear drops 0.2%- 3 drops twice a day for 2 weeks. • Check for HIV infection (Pg 19 - 20) • Follow-up in 2 weeks. If ear discharge is still present, refer for ENT review.
• No ear pain and No pus seen or reported draining from the ear.	**NO EAR INFECTION**	• No treatment.

CHECK INFANTS UNDER 6 MONTHS FOR RISK OF POOR GROWTH AND DEVELOPMENT

ASSESS

ASK:

- Was infant born too early (preterm) or too small (Low birthweight)? (Reported or documented)
- Has infant failed to gain weight including neonate who has not regained birthweight? (reported or documented)*
- Does the infant have difficulties feeding?**
- Does infant usually receive any foods or drinks other than breastmilk?
- Does mother have feeding concern(s) or breast problem(s)? (Reported or observed)
- Does mother have illness that requires further assessment? (reported or observed)
- Has the mother had any difficulties taking care of the infant or herself recently?

CHECK FOR:

IMNCI Danger Signs

- Unable to drink or breastfeed
- Vomits everything
- Convulsions
- Lethargic or unconscious

Other Danger Signs

- Difficulty in breathing
- Bilateral oedema (+, ++, or +++)
- Temperature (High >38ºC or Low <35.5ºC)
- Has infant recently lost weight?
- Acute medical complications as per IMNCI (Pg 8)

Measure child's:

- WAZ/WLZ and /or MUAC

Measure mother's MUAC

CLASSIFY ALL MOTHERS AND INFANTS

CLASSIFY

SIGNS	CLASSIFY	IDENTIFY TREATMENT TREATMENT (URGENT PRE-REFERRAL TREATMENTS ARE IN BOLD PRINT.)
Any one or more of the following signs: • Not able to feed at all • Vomits everything • Convulsions • Severe chest indrawing • Fast breathing • High or low body temperature • Movement only when stimulated or no movement at all • Bilateral oedema (+, ++, or +++) • Recent weight loss	**VERY SEVERE DISEASE**	• **Provide pre- referral treatment according to IMNCI.** • **Refer URGENTLY to health facility for admission.**
Any one or more of the following signs: • Infant born preterm • Low birthweight • Failure to gain weight • Neonate has not regained birthweight by two weeks of age • MUAC less than 11.5cm • WLZ less than -2SD **ANY one or more of the following signs:** • Infant has difficulties feeding • Infant usually receives food or drinks other than breastmilk • Mother has feeding concerns or breast problem **Any one or more of the following signs:** • Mother has illness requiring further assessment • Mother indicates she has had difficulties taking care of her infant or herself recently • Mother has MUAC less than 21cm	**POTENTIAL RISK**	• Refer mother and infant to nearest health facility for further assessment.
• No signs of severe disease or potential risk	**LOW RISK**	• Praise & reassure. • Provide or refer for routine health care and maternal and IYCF counseling.

Note: *If there is documented weight loss or failure to gain adequate weight (Less than 5g/kg/day) or if mother reports that infant has lost weight or failed to gain weight, then refer for indepth assessment.
**This refers to any mode of feeding: Breast fed, non-breastfed or mixed feeding.

CHECK FOR ACUTE MALNUTRITION (6 - 59 MONTHS)

ASSESS

ASK:
- Was the baby born preterm?

LOOK AND FEEL:
- Look for oedema of both feet.
- Determine the child's weight for Height/length (WFH/L*) and plot on the IMNCI Chart booklet (see pages 73 & 76) to determine the Z-score.**
- Determine the growth pattern; Is the growth faltering? (Weight curve is flattening or dropping for at least 2 consecutive months?)
- Measure MUAC***

IF WFH/L Z-score less than -3SD or MUAC less than 11.5cm then:

- Check for any **medical** complication present:
  - Any general danger signs.
  - Any acute medical problem under severe classification (Pg 8).

- If no medical complication present:
  - Conduct appetite test (Pg 41).
    - Has the child passed or failed the appetite test?

If child has acute malnutrition and is receiving RUTF****, **DO NOT** give iron because there is already adequate amount of Iron in RUTF.



CLASSIFY NUTRITIONAL STATUS

CLASSIFY

SIGNS	CLASSIFY	IDENTIFY TREATMENT TREATMENT (URGENT PRE-REFERRAL TREATMENTS ARE IN BOLD PRINT.)
• Bilateral pitting Oedema grade ++/+++, **OR** • Any grade of bilateral pitting oedema with severe wasting (WFH/L Z-score less than -3SD OR MUAC less than 11.5 cm) **OR** • Severe wasting **AND** Any one of the following: • One or more IMNCI danger sign • Any acute medical problem as per IMNCI severe classification • Failed appetite test	**SEVERE ACUTE MALNUTRITION WITH COMPLICATIONS**	• **Check blood glucose and treat if RBS< 3mmol/l (Give 5ml/kg 10% Dextrose). If glucose test unavailable, treat for hypoglycemia if not alert** (Pg 31)**.** • **Check for hypothermia, (Axillary temperature <35ºC). Keep the child warm.** • **Check for dehydration if child has diarrhoea. If in shock, give IV fluids Ringer's Lactate in 5% Dextrose and refer urgently** (Pg 33)**.** • **Give first dose of Benzyl Penicillin + Gentamicin** (Pg 29). • **Give Vitamin A if eye signs present or child has measles now or in the past 3 months** (Pg 27). • **Screen for possible TB disease** (Pg 21) **and check for HIV** (Pg 19 - 20). • **Refer URGENTLY to hospital for admission in a paediatric ward.**
• Bilateral pitting Oedema grade + **OR** • Severe wasting (WFH/L Z-score less than -3SD OR MUAC less than 11.5 cm) **AND** • DOES NOT have any IMCI danger signs or acute medical problems (as per IMCI classification) **AND** • Has PASSED THE APPETITE TEST	**SEVERE ACUTE MALNUTRITION WITHOUT COMPLICATIONS**	• **Prevent and treat for hypoglycemia if RBS <3mmol/l** (Pg 31). • **Keep warm.** • **Give high dose Amoxicillin (Amoxicillin Dispersible Tablet (DT)) for 5 days** (Pg 26)**.** • **Refer to Outpatient therapeutic care for nutrition treatment with ready to use therapeutic food (RUTF).** • **Screen for possible TB disease** (Pg 21) **and check for HIV** (Pg 19 - 20). • Assess the child's feeding and counsel the mother on the feeding recommendations (Pg 41 - 43). • Immunize as per schedule (Pg 25). • Advise mother when to return immediately. • Follow up in 7 days.

Continue next page

*WFH/L is Weight-for-Height or Weight-for-Length determined by using the WHO growth standards charts.
** If the child was born preterm, use the preterm growth chart on pg 73
***MUAC is Mid-Upper-Arm-Circumference measured using MUAC tape in all children 6 weeks or older.
****RUTF is Ready-to-use Therapeutic Food for conducting the appetite test and feeding children with severe acute malnutrition.

• WFH/L Z-score ≥ -3 to < -2SD **OR** • MUAC 11.5 to 12.4cm	**MODERATE ACUTE MALNUTRITION**	• Conduct comprehensive medical and psychosocial assessment and record (medical history, immunization status, physical examination, mental health and psychosocial assessment and follow up of medical condition needing long term medical care). • Refer infants and children identified with medical and/or psychosocial problems to the appropriate care available. • Assess the child's feeding and counsel the mother/caregiver on use of locally available nutrient-dense home diet (Pg 41 - 43). • Counsel the caregiver on age-appropriate stimulation, play and communication activities (Pg 44). • Check the child's developmental milestones (Pg 22) to note any notable developmental delays for further follow-up. • Immunize as per schedule (Pg 25). • Give Vitamin A to children who are on schedule for routine supplementation (Pg 27). • Give Albendazole if child is 1 year or older and has not had a dose in the previous 6 months (Pg 26). • Screen for possible TB disease (Pg 21) and check for HIV (Pg 19 - 20). • Advise mother when to return immediately. • Advise the mother/caregiver to return for follow-up in 14 days.
• Static weight or losing weight • WFH/L Z-score > -2 to ≤-1SD **OR** • MUAC 12.5 to 13.4cm	**AT RISK OF ACUTE MALNUTRITION**	• Assess the child's feeding and counsel the mother on feeding according to the feeding recommendations (Pg 41 - 43). • If feeding problem, follow-up in 5 days. • Immunize as per schedule (Pg 25). • Give Albendazole if child is 1 year or older and has not had a dose in the previous 6 months (Pg 26). • Screen for possible TB disease (Pg 21) and check for HIV (Pg 19 - 20). • Advise mother when to return immediately. • Follow up in 14 days.
• WFH/L Z-score > -1SD **OR** • MUAC >13.5 cm	**NO ACUTE MALNUTRITION**	• Assess the child's feeding and counsel the mother on feeding according to the feeding recommendations (Pg 41 - 43). • If feeding problem, follow-up in 5 days. • Give Albendazole if child is 1 year or older and has not had a dose in the previous 6 months (Pg 26). • Immunize as per schedule (Pg 25). • Screen for possible TB disease (Pg 21) and check for HIV (Pg 19 - 20).

NOTE:

Infants and children 6–59 months old with SAM who do not meet any of the above criteria but have any of the following characteristics should undergo an in-depth assessment by a healthcare worker to determine if a referral to inpatient care is necessary:

i. Medical problems that do not need immediate inpatient care, but do need further examination and investigation (eg., bloody diarrhoea, hypoglycaemia, HIV-related problems);
ii. Medical problems that need mid- or long-term follow-up care and have a significant association with nutritional status (eg., congenital heart disease, cerebral palsy or other disability, HIV, tuberculosis);
iii. Failure to gain weight (weight gain less than 5 g/kg/day observed over 2 consecutive weeks);
iv. Failure to improve clinically in outpatient care;
v. Previous episode(s) of severe wasting and/or nutritional oedema

CHECK FOR ANAEMIA

ASSESS

Look:

- Look for palmar pallor. Is there:
  - Severe palmar pallor?
  - Some palmar pallor?
  - No palmar pallor?
- If any pallor, check haemoglobin (Hb) level.

CLASSIFY ANAEMIA

CLASSIFY

SIGNS	CLASSIFY	IDENTIFY TREATMENT TREATMENT (URGENT PRE-REFERRAL TREATMENTS ARE IN BOLD PRINT.)
• Severe palmar pallor **OR** • Hb < 5 g/dl	**SEVERE ANAEMIA**	**Refer URGENTLY.**
• Some palmar pallor **OR** • Hb 5 g/dl up to 11 g/dl.	**ANAEMIA**	• Give iron (Pg 27) and counsel on iron-rich foods. • Assess feeding and counsel regarding any feeding problems (Pg 41 - 43). • Give Albendazole if child is 1 year or older and has not had a dose in the last 6 months (Pg 26). • Investigate for other possible causes of anaemia. • Advise caregiver when to return immediately. • Follow-up in 14 days.
• No pallor.	**NO ANAEMIA**	• Assess child's feeding and counsel the mother according to the feeding recommendations (Pg 41 - 43).

NOTE:

- Assess for sickle cell anaemia if common in your area.
- DO NOT give iron if the child is receiving RUTF. Small amounts of iron are available in RUTF.
- DO NOT give iron in acutely ill children since free iron promotes bacterial growth and can make some infections worse
- Iron is extremely toxic in overdose, particularly in children. All medication should be stored out of reach of children.

CHECK FOR HIV EXPOSURE AND INFECTION

ASSESS

ASK

- Ask for mother's HIV status to establish child's HIV exposure
  - Is it Positive, Negative or Unknown?
- Inquire for a positive contact history with person with TB or a chronic cough

LOOK, FEEL AND DIAGNOSE:

<u>Child <18 months</u>

- If mother is HIV positive*, Conduct DNA PCR at birth and or first contact, 6 weeks, 6months and 12 months (Or earlier, if child develops symptoms sugestive of HIV as per WHO staging criteria)
- If mother's HIV status is unknown, conduct an antibody test (rapid test) on mother to determine HIV exposure.If exposed, start ARV prophylaxis and test with DNA PCR for the baby

<u>Child ≥18 months</u>

- If mother's antibody test is POSITIVE, the child is exposed. Conduct an antibody test on the child.
- If breastfeeding, do HIV antibody test every 6months while breastfeeding and 6weeks after complete cessation of breastfeeding.

Child whose mother is NOT available:

- **Child < 18 months**
  - Do an antibody test on the child. If positive, do a DNA PCR test.
- **Child ≥ 18 months**
  - Do an antibody test to determine the HIV status of the child

NB: See Early Infant Diagnosis (EID) algorithm on Pg 64.

CLASSIFY HIV STATUS

CLASSIFY

SIGNS	CLASSIFY	IDENTIFY TREATMENT TREATMENT (URGENT PRE-REFERRAL TREATMENTS ARE IN BOLD PRINT.)
Child < 18 months and DNA PCR test POSITIVE	**PRESUMED HIV INFECTION**	• **Initiate ART immediately** (Pg 34) **and take confirmatory HIV DNA PCR and baseline VL.** • **Retest at 6 weeks with DNA PCR.** • **Educate caregivers on adherence and its importance.** • **Screen for possible TB disease** (Pg 21) **at every visit.** • **For those who do not have TB disease, start TPT. Screen for possible TB throughout TPT.** • **Treat, counsel and follow up existing infections.** • **Initiate or continue cotrimoxazole prophylaxis** (Pg 35)**.** • **Assess child's feeding and provide appropriate counseling to the mother/caregiver** (Pg 41 - 43). • **Offer routine follow up for growth, nutrition and development and HIV services.** • **Immunize for measles at 6 months, 9 months and boost at 18 months** (Pg 25). • **Follow up and offer comprehensive management of HIV as per the national HIV guidelines.**
Presumptive diagnosis of children < 18 months • Positive HIV antibody test **AND** • Symptomatic with 2 or more of the following: • Oral candidiasis • Severe pneumonia • Severe sepsis **OR** Any of the following: • Any WHO clinical stage 4 condition • Recent maternal death(if likely to have been HIV related) or advanced HIV disease • Child's CD4% <25%		
Child ≥ 18 months and Antibody test POSITIVE		

Continue next page

* All HIV positive mothers should be initiated on ARVs and linked to psychosocial support. Refer to National PMTCT guidelines.

Child<18 months • If mother test is positive and child's DNA PCR is negative **OR** • If mother is unavailable; child's antibody test is positive and DNA PCR is negative	**HIV EXPOSED**	• Treat, counsel and follow up existing infections. • Initiate or continue Cotrimoxazole prophylaxis (Pg 35). • Categorize the mother and child based on MTCT risk (High/Low) (Pg 35): • High risk infant/child should receive triple infant prophylaxis with ABC+3TC+DTG for 14 weeks, after which they should continue with NVP prophylaxis until 6 weeks after complete cessation of breastfeeding • Low risk infant/child should receive NVP and AZT for 6 weeks, after which they should continue with NVP prophylaxis until 6 weeks after complete cessation of breastfeeding • Repeat DNA PCR test at 6weeks, and if negative at 6 months and at 12 months. If negative, repeat antibody test at 18 months (see EID algorithm Pg 64). • Assess child's feeding and provide appropriate counseling to the mother/caregiver (Pg 41 - 43). • Offer routine follow up for growth, nutrition and development. • Continue with routine care for under 5 clinics. • Screen for possible TB at every visit (Pg 21). • For those who do not have TB disease, start TPT. Screen for possible TB throughout TPT. • Immunize for measles at 6 months, 9 months and boost at 18 months (Pg 25).
• Mother's HIV status is NEGATIVE **OR** • Mother's HIV status is POSITIVE and child is ≥18 months with antibody test NEGATIVE 6 weeks after completion of breastfeeding	**HIV NEGATIVE**	• Manage presenting conditions according to IMNCI and other recommended national guidelines. • Advise the mother about feeding and about her own health (Pg 42 & 46 respectively).

SCREEN FOR TUBERCULOSIS

ASSESS

ASK

- Cough for >2 weeks.
- Persistent fever and/or night sweats >2 weeks
- Weight loss or poor weight gain/failure to thrive
- Fatigue (lethargy)/ reduced playfulness/ less active >2 weeks

*Enquire for a positive contact history with a person with TB or person with a Chronic cough.

LOOK, LISTEN AND FEEL

- Fast breathing (Respiratory rate: 0 - 2 months ≥60b/min, 2 -12months ≥50b/min, 1- 5yr ≥40b/min)
- Respiratory distress (flaring of alae nasi, chest wall indrawing, grunting or noisy breathing)
- Wheeze or crepitations or asymmetry of findings
- Oxygen saturation <90% or central cyanosis
- Temperature >37.5°
- Check weight on growth curve, compare to the previous weight
- Check for EPTB signs - Look for abnormal signs arising from other affected organs other than the lungs (refer to the simplified diagnostic algorithm for EPTB) (Pg 38)

CLASSIFY TB

PULMONARY TB

CLASSIFY / IDENTIFY TREATMENT

SIGNS	CLASSIFY	TREATMENT (URGENT PRE-REFERRAL TREATMENTS ARE IN BOLD PRINT.)
2 or more symptoms and any abnormal respiratory exam finding	**PROBABLE (PRESUMPTIVE) PULMONARY TB**	• Investigate and evaluate for TB disease as per the simplified algorithm for diagnosis of TB in children (Pg 37). • If bacteriologically confirmed with TB start treatment as per the TB guidelines (Pg 39 - 40). • If clinically diagnosed start treatment as per the TB guidelines (Pg 39 - 40). • Screen for malnutrition (Pg 16 - 17). • Test for HIV (Pg 19 - 20). • If the child is a contact of bacteriologically confirmed PTB and active TB has been ruled out, start them on TPT as per the TB guidelines.
Less than 2 symptoms and a normal respiratory exam finding.	**NOT PRESUMPTIVE FOR PULMONARY TB**	• Consider other diagnosis & manage child appropriately. • If there is history of TB contact - evaluate to offer TPT. • Test for HIV (Pg 19 - 20). • Screen for malnutrition (Pg 16 -17). • Ask parent/guardian to bring the child back after 3 days if there is no improvement for the current illness. • If there is improvement reassess the child after 1 week.

EXTRAPULMONARY

SIGNS	CLASSIFY	TREATMENT (URGENT PRE-REFERRAL TREATMENTS ARE IN BOLD PRINT.)
Presence of two or more of the symptoms apart from cough plus presence of one of: • Positive contact history • Abnormal sign arising from the AFFECTED ORGAN (see EPTB algorithm) • Investigation relevant to AFFECTED ORGAN is positive (where available) • Positive mantoux or IGRA test (where available)	**EXTRAPULMONARY TB**	• Refer for investigations, further evaluation and management as per the EPTB algorithm (Pg 38). • If clinical diagnoses confirmed then start treatment as per the TB guidelines (Pg 39 - 40). • Test for HIV (Pg 19 - 20). • Screen for Malnutrition (Pg 16 - 17).

Note:
The following categories of children with TB should be treated as inpatients:

- Severe forms of PTB and EPTB (e.g. Spinal TB) for further investigation and initial management
- TB meningitis
- Severe malnutrition for nutritional rehabilitation
- Signs of severe pneumonia (i.e. chest wall indrawing, oxygen saturation <90%, central cyanosis, grunting, inability to breastfeed/drink)
- Other co-morbidities e.g. severe anaemia
- Social or logistic reasons to ensure adherence
- Severe adverse reactions such as hepatotoxicity

CHECK FOR CHILD'S DEVELOPMENTAL MILESTONES

Ask if the child was born preterm. For preterm babies, use corrected gestational age to assess their developmental milestones.

ASSESS

UNDER 1 MONTH
- Moro reflex
- Blinking reflex
- Sucking reflex
- Stepping reflex
- Grasping reflex

0 - 2 MONTHS
- Vocalizing
- Alternate kicking
- Opening of the hands
- Social smile (baby smiles back)
- Baby follows a colourful object dangled before their eyes
- Startles to loud sound

2 - 4 MONTHS
- Holds the head upright
- Follows the object or face with their eyes
- Turns the head, rolls over, responds to sound
- Smiles when you speak
- Looks at mother's face

4 - 6 MONTHS
- Reaches for and grasps objects with hand
- Takes objects to her mouth
- Babbles (makes sounds)
- Enjoys noisy toys

6 - 9 MONTHS
- Sits without support
- Moves object from one hand to the other
- Repeats syllables (bababa, mamama)

9 - 12 MONTHS
- Takes steps with support
- Picks up small object or string with 2 fingers
- Listens when spoken to
- Says 2-3 words
- Imitates simple gestures (claps hands, bye)

12 - 18 MONTHS
- Walks without support
- Drinks from a cup
- Says 7-10 words
- Points to some body parts on request

18 - 24 MONTHS
- Kicks a ball
- Builds tower with 3 blocks or small boxes
- Points at pictures on request
- Speaks in short sentences

24 – 30 MONTHS
- Turns door knobs and lids
- Can name some body parts
- Knows if he or she is a boy or a girl

30 - 36 months
- Plays next to and with other children
- Follow simple instructions
- Able to take off some clothes by self
- Speech is understood by family and friends

36 - 48 months
- Says first name, when asked
- Draws a circle, when shown
- Puts on some clothes by him/herself

4 - 5 years
- Names a few colors of items
- Draws a person with 3 or more body parts
- Unbuttons some buttons
- Serves food and water

5 years and older
- Follows rules or takes turns when playing games with other children
- Does simple chores at home
- Counts to 10

CLASSIFY FOR DEVELOPMENTAL MILESTONES

CLASSIFY

IDENTIFY TREATMENT

SIGNS	CLASSIFY	TREATMENT (URGENT PRE-REFERRAL TREATMENTS ARE IN BOLD PRINT.)
• Absence of one or more milestones from current age group **AND** • Absence of one or more milestones from earlier age group **OR** • Regression of milestones	**DEVELOPMENTAL MILESTONES DELAY**	• **Counsel the caregiver appropriately.** • **Refer to appropriate specialty.** • **Screen for mother's health needs and risk factors and other possible causes including Malnutrition, TB disease and hyperthyroidism** (Pg 25).
• Absence of one or more milestones from current age group	**DEVELOPMENTAL MILESTONES ALERT**	• Praise caregiver on milestones achieved. • Counsel caregiver on play & communication activities to do at home (Pg 44 - 45). • Screen for possible TB disease (Pg 21) and other causes. • Advise to return for follow up in 30 days.
• All milestones for the current age group are present	**DEVELOPMENTAL MILESTONES NORMAL**	• Praise caregiver on milestones achieved. • Encourage caregiver to give more challenging activities for the next age group (Pg 44). • Advise to continue with follow up consultations.

ASSESS FOR INTERACTION, COMMUNICATION AND RESPONSIVENESS

ASSESS

ASK

- How do you play with your child? (Ask the caregiver to demonstrate)
- How do you talk with your child? (Ask the caregiver to demonstrate)
- How do you get your child to smile? (Ask the caregiver to demonstrate)
- How does your child express their needs and how do you respond to the need?

LOOK, LISTEN

(Health care provider should observe how the caregiver interacts, communicates with the child and how they respond to the child)

- Does the caregiver play with child?
- Does the caregiver talk to child?
- Does the caregiver smile with the child?

CLASSIFY FOR INTERACTION, COMMUNICATION AND RESPONSIVENESS

CLASSIFY

SIGNS	CLASSIFY
General signs for all children • Does not move/play with the child, or controls child's movements • Is not able to comfort child, and child does not look to caregiver for comfort • Scolds the child **Age < 6 Months** • Does not play with baby • Does not talk to baby • Tries to force smile or is not responsive to baby **Age 6 Months & Older** • Does not play with child • Does not talk, or talks harshly to child • Says child is slow to learn	**POOR INTERACTION AND/OR COMMUNICATION AND RESPONSIVENESS**

IDENTIFY TREATMENT

TREATMENT

For All Children:

- Ask caregiver to copy child's movements and to follow child's lead.
- Help caregiver look into child's eyes while gently talking and holding child.
- Help caregiver distract child from unwanted actions by giving alternative toys or activities.

AGE < 6 Months

- Discuss ways in which to help baby see, hear, feel and move appropriate for their age.
- Ask the caregiver to look into baby's eyes while talking to the baby.
- Ask caregiver to make large gestures and cooing sounds and/or copy baby's sounds and gestures, and see baby's response.

AGE: 6 Months & Older

- Ask caregiver to do play or communication activity, appropriate for age.
- Help caregiver interpret what child is doing and thinking, while observing child's response and smile.
- Counsel the caregiver on appropriate activity to do together with the child (Pg 44 - 45).
- Encourage more activity with the child, check hearing and seeing.
- Refer child with difficulties appropriately.

(See the counsel the mother card on recommendation for care for Child's Development (Pg 44 - 45)**).**

Follow up in 14 days.

Continue next page

General signs for all children • Caregiver moves towards the child, talks to or makes sounds with child • Caregiver smiles or is responsive to child • Caregiver looks into child' s eyes while talking softly to them, gently touches child or holds them closely. • Caregiver distracts child from unwanted actions with appropriate toys. **Age <6 Months** Caregiver: • Moves the baby 's arms and legs while gently stroking the baby. • Gets baby's attention with shaker or other • ways • Looks into baby's eyes while talking softly to baby • Responds to baby's sounds and gestures to get baby to smile **Age 6 Months & older** Caregiver: • Plays word games or with toys • Looks into child 's eyes and talks softly to child • Draws smile out from the child • Says the child is learning well	**GOOD INTERACTION AND /OR COMMUNICATION AND RESPONSIVENESS**	Praise caregiver and encourage to continue age appropriate play and communication (See the counsel the mother card on recommendation for care for Child's Development (Pg 44 - 45)).

CHECK THE CHILD'S IMMUNIZATION, VITAMIN A & DEWORMING STATUS

IMMUNIZATION *(The schedule below also applies to Preterm babies)*

VACCINES										
Age	**BCG**	**Polio Vaccine (Bivalent Oral Polio Vaccine)**	**IPV**	**Diphtheria/ Pertussis/ Tetanus/ Hepatitis B/ Haemophilus type B (Pentavalent)**	**Pneumococcal PCV10**	**Rotavirus Vaccine**	**Measles Rubella**	**Yellow Fever**	**Typhoid Conjugate Vaccine (TCV)**	**Malaria vaccine**
Birth	BCG*	bOPV 0 (birth –2wks)								
6 weeks		bOPV 1		Pentavalent 1	PCV10 1	ROTA 1				
10 weeks		bOPV 2		Pentavalent 2	PCV10 2	ROTA 2				
14 weeks		bOPV 3	IPV1	Pentavalent 3	PCV 10 3	ROTA 3				
6 months							Measles, rubella**			****Malaria vaccine
7 months										****Malaria vaccine
9 months			IPV2				Measles, rubella	Yellow Fever***	TCV	****Malaria vaccine
18 months							Measles, rubella			
24 months										****Malaria vaccine

* Do not give BCG to a child with symptomatic HIV/AIDS. HIV infected, receiving ART and clinically well and immunologically stable, should be vaccinated. In child exposed to TB disease at birth, do not give BCG, instead give child TPT for 6 months then administer BCG 2 weeks after completion of TPT. No repeat dose of BCG after primary dose is given, even if the scar does not develop.

**Measles Rubella vaccine at 6 months for HIV exposed/infected and immunocompromised children. Repeat at 9 months and 18 months as per the immunization schedule.

***Yellow fever should not be administered to children with symptomatic HIV/AIDS and is only offered in the following counties: Turkana, West Pokot, Baringo and Elgeyo Marakwet.

**** Malaria Vaccine is only offered in the following counties: Busia, Bungoma, Kakamega, Vihiga, Kisumu, Siaya, Migori and Homabay.

VITAMIN A & DEWORMING

Age	Vitamin A	De-worming
6 months	Vitamin A	
12 months	Vitamin A	Deworming
18 months	Vitamin A	Deworming
24 months	Vitamin A	Deworming
30 months	Vitamin A	Deworming
36 months	Vitamin A	Deworming
42 months	Vitamin A	Deworming
48 months	Vitamin A	Deworming
54 months	Vitamin A	Deworming
60 months	Vitamin A	Deworming

ASSESS FOR OTHER PROBLEMS THAT THE CHILD MAY HAVE and HAVE NOT BEEN DISCUSSED IN IMNCI

- Examples: Burns, fracture, skin conditions etc. and manage them accordingly.

ASSESS THE MOTHER'S / CAREGIVER'S HEALTH NEEDS

- Nutritional status and anaemia, contraception.
- Check the mother's/ caregiver's HIV status.
- Screen for cancer e.g: breast, cervical, prostate if caregiver is male.
- Check the mother's/ caregiver's psychosocial support needs.
- Check hygienic practices.
- Check/assess mother's/ caregiver's mental status and SGBV (Sexual Gender Based Violence).

Mental health assessment - Patient Health Questionnaire-2 (PHQ-2)				
Over the last 2 weeks, how often have you been bothered by any of the following problems?	Not at all	Several days	More than half the days	Nearly every day
1. Little interest or pleasure in doing things	0	1	2	3
2. Feeling down, depressed, or hopeless	0	1	2	3
For office coding:	0	+_____	+_____	+_____
= Total Score _____				
Recommended actions for persons scoring 3 or higher: Refer to a Psychologist				

TREAT THE CHILD

(CARRY OUT THE TREATMENT STEPS IDENTIFIED ON THE ASSESS AND CLASSIFY CHART)

TEACH THE MOTHER/CAREGIVER TO GIVE ORAL DRUGS AT HOME

Follow the instructions below for every oral drug to be given at home. Also follow the instructions listed with each drug's dosage table.

- Determine the appropriate drugs and dosage for the child's age or weight.
- Tell the mother/caregiver the reason for giving the drug to the child.
- Demonstrate how to measure a dose.
- Watch the mother/caregiver practise measuring a dose by themselves.

GIVE AN APPROPRIATE ORAL ANTIBIOTIC

Oral antibiotic doses

Weight (kg)	**High dose Amoxicillin** for pneumonia & severe infections 40 - 45mg/kg/dose			**Amoxicillin** (for mild infections) 25mg/kg/dose		**Flucloxacillin** 15mg/kg/dose		**Ciprofloxacin** 15mg/kg/dose (for 3 days)
	12 hrly			**12hrly**		**8hrly**		**12hrly**
	Syrup		**D. Tab**					
	125mg /5mls	**250mg /5mls**	**250mg tabs**	**mls** 125mg/5ml	**250mg** tabs	**mls** 125mg/5ml	**250mg** s or tabs	**250mg** tabs
3.0	5mls	2.5	½ tab	3		2.5	¼	
4.0	7.5mls	3.75		4		2.5	¼	¼
5.0	10mls	5	1 tab	6		5	¼	¼
6.0	10mls	5		6		5	½	¼
7.0		7.5		8		5	½	½
8.0		7.5		8		5	½	½
9.0		7.5		8		5	½	½
10.0		10	2 tabs	12	1	5	1	½
11.0		10		12	1	10	1	1
12.0		10		12	1	10	1	1
13.0		12.5		12	1	10	1	1
14.0		12.5	3 tabs	12	1	10	1	1
15.0		12.5		15	1	10	1	1
16.0				15	1	10	1	1
17.0				15	1	10	1	1
18.0				15	1	10	1	1
19.0				15	1	10	1	1
20.0				15	2	10	1	1

Antibiotic Therapy for cholera

Category	**Antibiotic**	**Children**
First line	Doxycycline PO	4 mg/kg single dose
Alternative	Azithromycin PO	20 mg/kg single dose
	Ciprofloxacin PO	20 mg/kg single dose

DEWORMING USING ALBENDAZOLE

Give Albendazole 200mg as a single dose for children 1 to 2 years and 400mg if child is 2 years or older.

TREATMENT FOR UNCOMPLICATED MALARIA

FIRST- LINE ANTIMALARIAL: ARTEMETHER + LUMEFANTRINE (AL)

AGE or WEIGHT	ARTEMETHER + LUMEFANTRINE TABLETS (Each tablet contains Artemether 20mg + Lumefantrine 120mg)	
WEIGHT	**Age in years**	**Dose of AL to be administered at 0hrs, 8hrs, 24 hrs, 36hrs, 48hrs and 60hrs.**
5 to <15Kg	Below 3 years	1 tablet
15 to < 25Kg	3 - 7 years	2 Tablets
25 to <35kg	8 - 11 years	3 Tablets
>35kg	≥12 years	4 tablets

In a child below 5kg, evaluation of other causes of fever including malaria should be undertaken. Where malaria is confirmed, the current recommended treatment is one tablet of AL given according to the schedule above, under close supervision.

Counsel the mother or caregiver on managing uncomplicated malaria:

- Show all caregivers of young children how to prepare the dispersible tablet prior to administration. Ensure she/he understands how to administer the same to the child prior to leaving the facility.
- If vomiting occurs within 30 minutes after drug administration, the dose should be repeated. And if vomiting persists, the patient should return to the facility for review.
- Explain the dosing schedule, use probing questions to determine the patient's understanding.
- Emphasize that all 6 doses must be taken over 3 days even if the patient feels better after a few doses. Follow up after 3 days of treatment.
- Advise patients to return immediately to the nearest health facility if the condition deteriorates at any time or if symptoms have not resolved after 3 days.

SECOND-LINE ANTIMALARIAL: DIHYDROARTEMISININ-PIPERAQUINE(DHA-PPQ)

Body Weight (Kg)	Dihydroartemisinin (20mg) + piperaquine (160mg) given once daily for 3 days
5 to < 8	1 tablet
8 to <11	1.5 tablets
11 to < 17	2 tablets

GIVE VITAMIN A

- Therapeutic Vitamin A: Give two doses for treatment of Measles. Give the first dose in clinic and give mother another dose to give at home the next day.
- Vitamin A supplementation: Give first dose at 6 months of age and thereafter every 6 months up to 5 years to **ALL CHILDREN** as per Vitamin A schedule for prevention (Pg 25).
- To give Vitamin A, cut open the sule and give drops.

AGE	VITAMIN A SULES		
	200 000 IU	100 000 IU	50 000 IU
Up to 6 months			1 sule
6 months up to 12 months		1 sule	2 sules
12 months up to 5 years	1 sule	2 sules	4 sules

GIVE IRON

Give one dose at 3-6 mg/Kg of iron daily for 3 months.
NB: Avoid iron in a child known to suffer from Sickle Cell Anaemia.

AGE or WEIGHT	IRON TABLET (200mg)
2 months up to 4 months (4 - 6 kg)	1/4
4 months up to 12 months (6 - 10 kg)	1/4
12 months up to 3 years (10 - 14 kg)	1/2
3 years up to 5 years (14 - 19 kg)	1/2

GIVE ZINC SULPHATE

For a sick young infant or child with diarrhoea:

AGE	Zinc Sulphate Give once daily for 10 days
	DISPERSIBLE Tablet (20 mg)
0 months up to 6 months	1/2
6 months up to 5 years	1

GIVE PARACETAMOL FOR FEVER OR EAR PAIN

Give PARACETAMOL (10-15mg/kg body weight 6 to 8 hourly) until fever or ear pain is gone (3 days for fever and 5 days for ear pain).

TEACH THE MOTHER/CAREGIVER TO TREAT LOCAL INFECTIONS AT HOME

- Explain to the mother/caregiver what the treatment is and why it should be given.
- Describe the treatment steps listed in the appropriate sections.
- Watch the mother/caregiver as they give the first treatment in the clinic (Except remedy for cough or sore throat).
- Tell them how often to give the treatment at home.
- If needed for treatment at home, give mother/caregiver the tube of tetracycline eye ointment or a small bottle of gentian violet.
- Check the mother's/caregiver's understanding before they leave the clinic.

TREAT EYE INFECTION WITH TETRACYCLINE EYE OINTMENT

- **Clean both eyes 4 times daily.**
  - Wash hands.
  - Ask the child to close their eyes.
  - Use a clean cloth and water to gently wipe away pus.

- **Then apply tetracycline eye ointment in both eyes 4 times daily.**
  - Ask the child to look up.
  - Squeeze a small amount of ointment on the inside of the lower lid.
  - Wash hands again.

- Treat until there is no pus discharge (minimum 5 days).
- Do not use other eye ointments or drops, or put anything else in the eye.

CLEAR THE EAR BY DRY WICKING

- **Dry the ear at least 3 times daily.**
  - Roll clean absorbent cloth or soft, strong tissue paper into a wick.
  - Place the wick in the child's ear.
  - Remove the wick when wet.
  - Replace the wick with a clean one and repeat these steps until the ear is dry.
  - Tell the mother/caregiver not to place anything in the ear between wicking treatments. For chronic ear infection, instill Ciprofloxacin ear drops twice daily for 2 weeks after dry wicking.
  - Do not allow the child to go swimming or get any water in their ear

TREAT MOUTH ULCERS WITH GENTIAN VIOLET

- **Treat the mouth ulcers twice daily.**
  - Wash hands.
  - Clean the child's mouth with clean soft cloth wrapped around the fingers and wet with salt water.
  - Paint the mouth with half-strength gentian violet.
  - Wash hands again.

TREAT FOR THRUSH WITH NYSTATIN

- **Treat thrush 4 times daily for 7 days**
  - Clean the mouth with clean soft cloth wrapped around the finger and wet with salt water.
  - Instill Nystatin 1 ml 4 times a day (After feeds).
  - If breastfed, advise mother to wash her breasts after feeds and apply the same medicine on the areola and nipples.
  - If severe, recurrent or pharyngeal thrush, refer for HIV testing.

SOOTHE THE THROAT, RELIEVE THE COUGH WITH A SAFE REMEDY

- **Safe remedies to recommend:**
  - Breast milk for exclusively breast-fed infant
  - Warm water with Lemon tea and honey (children >1year)

- **Harmful remedies to discourage:**
  - Codeine and ephedrine containing cough mixtures
  - Herbal smoke inhalation
  - Vicks drops by mouth
  - Any mixture containing vinegar

GIVE THESE TREATMENTS IN A HEALTH FACILITY ONLY

TREATMENT OF VERY SEVERE DISEASE:

- Explain to the mother/caregiver why the drug is given.
- Determine the appropriate dose for the child weight (or age).
- Use a sterile needle and sterile syringe. Measure the dose accurately.
- Give the drug as an Intravenous/intramuscular injection.

GIVE AN INTRAVENOUS/ INTRAMUSCULAR ANTIBIOTIC

FOR CHILDREN BEING REFERRED URGENTLY:

- Give dose of Ceftriaxone and REFER IMMEDIATELY.

WHERE REFERRAL IS NOT POSSIBLE OR DELAYED:

- Treat to prevent hypoglycemia.
- Continue CEFTRIAXONE FOR 10 DAYS.

Weight (kg)	Penicillin (50,000 iu/kg)	Flucloxacillin (50mg/kg)	Gentamicin (7.5mg/kg)	Ceftriaxone Max 50mg/kg 24hrly for neonates** Meningitis/ Severe Sepsis, 50mg/kg BD not to exceed 4g/day	Metronidazole (7.5mg/kg) not to exceed 4g/day
	IV/IM	IV / IM	IV / IM	IV/ IM	IV
	6hrly	8hrly	24hrly	50mg/kg	Age <1m: 12hrly Age ≥ 1m: 8hrly
3.0	150,000	150	20	150	20
4.0	200,000	200	30	200	30
5.0	250,000	250	35	250	35
6.0	300,000	300	45	300	45
7.0	350,000	350	50	350	50
8.0	400,000	400	60	400	60
9.0	450,000	450	65	450	65
10.0	500,000	500	75	500	75
11.0	550,000	550	80	550	80
12.0	600,000	600	90	600	90
13.0	650,000	650	95	650	95
14.0	700,000	700	105	700	105
15.0	750,000	750	110	750	110
16.0	800,000	800	120	800	120
17.0	850,000	850	125	850	125
18.0	900,000	900	135	900	135
19.0	950,000	950	140	950	140
20.0	1,000,000	1000	150	1000	150

TREATMENT OF SEVERE MALARIA

FIRST LINE TREATMENT FOR SEVERE MALARIA: ARTESUNATE

Artesunate typically comes as a powder together with a 1ml vial of 5% bicarbonate that then needs to be further diluted with either normal saline or 5% dextrose- the amount to use depends on whether the drug is to be given IV or IM (see table below). If IV administration is intended, ensure you secure IV access before drug reconstitution.

- DO NOT use water for injection to prepare artesunate for injection.
- DO NOT give artesunate if the solution in the syringe is cloudy.
- DO NOT give artesunate as a slow IV drip (infusion).
- YOU MUST use artesunate within 1 hour after it is prepared for injection.

Preparing IV / IM Artesunate	IV	IM
Artesunate powder (mg)	60mg	60mg
Sodium Bicarbonate (mls,5 %)	1ml	1ml
Normal Saline or 5% Dextrose (mls)	5 mls	2mls
Artesunate concentration (mg/ml)	10mg/ml	20mg/ml

ARTESUNATE is given IV/IM for a minimum of 24 hours. As soon as the child can eat drink (After 24 hours for artesunate) then change to a full course of artemisinin combination therapy (ACT) typically the 1st line oral anti-malarial, Artemether Lumefantrine.

ARTEMETHER FOR SEVERE MALARIA

- Administer a loading dose of 3.2mg/kg IM stat then 1.6mg/kg IM daily until the patient is able to tolerate oral medications.
- Thereafter a complete course of AL is given.

NOTE:

- **Admit or refer patients with the following:**
  - Severe anaemia (HB level of <5g/dl or haematocrit of <15%)
  - Two or more convulsions within a 24-hr period
  - Hyperparasitaemia and are stable without prostration, respiratory distress or altered consciousness

These patients can be treated with AL, DHA-PPQ or oral quinine where ACT is not available. However, they need to be closely monitored in an inpatient setting.

Weight (kg)	Artesunate, 3mg/kg At 0,12 and 24h then daily for max 7 days		
	IV mls of 60mg in 6mls	Dose in mg	IM mls of 60mg in 3mls
3.0	0.9	9	0.45
4.0	1.2	12	0.6
5.0	1.5	15	0.8
6.0	1.8	18	0.9
7.0	2.1	21	1.1
8.0	2.4	24	1.2
9.0	2.7	27	1.4
10.0	3	30	1.5
11.0	3.3	33	1.6
12.0	3.6	36	1.8
13.0	3.9	39	12
14.0	4.2	42	2.1
15.0	4.5	45	2.3
16.0	4.8	48	2.4
17.0	5.1	51	2.6
18.0	5.4	54	2.7
19.0	5.7	57	2.9
20.0	6.0	60	3

PRESCRIBING OXYGEN

Oxygen Administration Device	Flow rate	Fraction of inspired Oxygen (FiO_2)
Nasal prongs 	Standard Flow Rate: **Neonates:** 0.5 - 1 L/min **Infants:** 1 - 2 L min **Child:** 1 - 4 L min	Delivers **35%** O_2 to the patient.
	High Flow rate **Preterm Neonates:** 1L/min **Term neonates:** 2L/min **Infants:** 4 L/min **Child:** 4 - 8L/min	Delivers **50%** O_2 to patient
Nasal Catheter 	**Neonates:** Not recommended **Infants/child:** 1 - 2L/min	Delivers **40%** O_2 to patient
Oxygen face mask with reservoir bag (non-rebreather mask) 	**All groups:** 10 - 15L/min (The bag should not deflate so as not to dilute the O_2 concentration)	Delivers **80-95%** O_2 to patient

*Humidification is needed for all patients on high flow rates.
*Check for abdominal distension regularly.

Always DOCUMENT the flow rate, delivery device, monitoring frequency and target oxygen saturation as part of the oxygen prescription.

TREAT FOR POSSIBLE ASTHMA

Wheeze + History of cough or difficulty breathing,
(Likelihood of asthma much higher if age >12m and recurrent wheeze)

Yes

Severe Asthma
Any one of these:
- Oxygen saturation < 90%
- Central Cyanosis
- Inability to drink / breast feed
- AVPU= 'V', 'P' or 'U',or
- Inability to talk/complete sentences
- Pulse rate >200 bpm (0-3yrs)and >180 bpm (4-5yrs)
- Silent chest on auscultation

Yes →

Immediate Management

Admit/Refer
- Oxygen
- Nebulize 2.5mg salbutamol or 6 puffs of inhaler with spacer and mask, repeat every 20 minutes upto 3 doses if needed
- Prednisolone 1-2mg/kg*
- Consider ipratropium bromide 250mcg when nebulized or 2 puffs of 80 mcg every 20 min if poor response**
- Antibiotics as for severe pneumonia

No

Mild or Moderate Asthma
Wheeze PLUS
- Lower chest wall indrawing **OR**
- Fast breathing
  RR ≥50/min (Age 2 - 11mo)
  RR ≥40/min (Age 12 - 59mo)

Yes →

Salbutamol 2 puffs of inhaler (or 2.5mg nebulized) every 20 minutes upto 3 doses if needed

Monitor closely for 1-2 hours

If mild symptoms allow home on salbutamol MDI give 2 puffs every 6 hours.
Counsel caregiver on signs of deterioration and schedule review within 48 hours.

If lack of response to salbutamol, increasing respiratory rate, worsening saturation, any signs of severe asthma, refer to Immediate Management above.

- Recurrence of asthma symptoms:
  - Consider Inhaled corticosteroid (ICS) therapy or adjust the doses if already on ICS. (Look out for other comorbidities).
  - Demonstrate MDI and spacer use to the caregiver before discharge from the health facility.
  - Preferably use spacer with face masks for <3 years and for 4 - 5 years use facemask or mouthpiece.
  - Advise on regular follow up.

*Prednisolone administered for 3 - 5 days. Max dose of 20mg/day for <2 years and 30mg/day for 2-5 years.
** Repeat every 20 minutes for one hour if needed.

MANAGE THE CHILD FOR LOW BLOOD SUGAR

PREVENT

- **If the child is able to breastfeed:**
  - Ask the mother to breastfeed the child.

- **If the child is not able to breastfeed but is able to swallow:**
  - Give expressed breast milk or if unavailable, a breast milk substitute. If neither of these is available, give sugar water.
  - Give 30-50 ml of milk or sugar water before departure.
  - To make sugar water: Dissolve 4 level teaspoons of sugar (20 grams) in a 200 ml cup of clean water.

- **If the child is not able to swallow:**
  - Insert nasogastric tube, check position and give 20-50 ml of milk or sugar water.

TREAT

- **For Suspected / confirmed low blood sugar***
  - Give 10% glucose** 5 ml/kg by nasogastric tube OR
  - Give same amount slowly intravenously if a line is available over 2-3min. Keep warm.
  - Admit or refer urgently to hospital.

RAPID ACTING BRONCHODILATOR	
Salbutamol for asthma	
Salbutamol inhaler using a spacer and mask	Give 2 puffs (mild or moderate asthma) or 6 puffs (severe asthma) • Allow 5 -10 breaths per puff. • If still wheezing repeat every 20minutes in first hour and 2 - 4 hourly thereafter.
Nebulized salbutamol (5mg/ml preparation).	2.5mg (0.5ml) salbutamol in 2ml saline
Epinephrine for stridor	
Nebulized epinephrine (adrenaline)(1: 1000=0.1 %)	3 - 5mls of adrenaline 1mg in 1ml preparation (1:1000): • Add 3 - 5 ml of 1:1000 adrenaline to 1 ml of saline and administer using a nebulizer. • Always use oxygen at flow-rate of 6 - 8 litres. • Repeat every 15 minutes, until the child is transferred (or the stridor disappears). • Give one dose of prednisone as part of pre-referral treatment for stridor.

TREATMENT OF CONVULSIONS

Age > 1 month.

Child convulsing →

Ensure safety and check ABCD
A - Place in lateral position, suction if indicated.
B - Start on oxygen via NRM.
C - Check for temp gradient, severe pallor.
D - Check RBS or give 5mls/kg of 10% Dextrose.

Convulsion lasting > 5min?

If children have up to 2 fits lasting <5mins, they **DO NOT** require emergency drug treatment

Yes

- Give IV Diazepam 0.3mg/kg slowly over 1minute OR rectal diazepam 0.5mg/kg. *Alternatives include IV Lorazepam or buccal midazolam*
- Check ABCD when convulsion stops, observe and investigate cause / refer appropriately.

Convulsion continues 5mins after first dose of diazepam

- Give the second dose of IV diazepam 0.3 mg/kg slowly over 1 minute, OR rectal diazepam 0.5 mg/kg.
- Continue oxygen.
- Check airway and breathing when convulsion stops, investigate & treat cause / refer appropriately.

Convulsion continues 5mins after second dose of diazepam

No

Child having 3rd convulsion lasting <5 mins in <2 hrs (short multiple convulsions).

Yes

- Give IM phenobarbitone 15mg/kg (loading dose).
- Initiate maintenance therapy with phenobarbitone 2.5 mg/kg OD for 48 hrs then review.
- Continue oxygen during active seizure.
- Check ABC when convulsion stops, investigate and treat cause / refer appropriately.

- DO NOT give more than 2 doses of diazepam in 24hrs once phenobarbitone is used.
- DO NOT give a phenobarbitone-loading dose to an epileptic on maintenance phenobarbitone.
- Phenytoin, levetiracetam and IV sodium valproate are alternatives to phenobarbitone.

* Low blood sugar (hypoglycemia) may be suspected in any infant or child who is convulsing or has loss of consciousness for which there is no obvious cause; has a rectal temperature of below 35.5°C, is drowsy or sweating, is lethargic, floppy or jittery - Particularly when less than 2 months old.
**To constitute 10% glucose from 50% glucose: Mix 1 part of 50% glucose with 4 parts of water for injection.

REHYDRATION THERAPY AND FEEDING FOR DIARRHOEA

PLAN A: TREAT DIARRHOEA WITH NO DEHYDRATION

Counsel the mother/caregiver on the 4 Rules of Home Treatment:

- **Give Extra Fluids, Give Zinc Sulphate, Continue Feeding, Advise when to Return**

1. **GIVE EXTRA FLUID** (Give ORS and other Fluids- as much as the child will take)

- **ADVISE THE MOTHER:**
  - **For infants aged 0-6 months:**
    - Breastfeed frequently and for longer at each feed.
    - If the child is exclusively breast-fed, give ORS in addition to breast milk.
    - If the child is not breastfed, give ORS solution and safe breastmilk substitute.
  - **For infants aged 6-59 months:**
    - Breastfeed frequently and for longer at each feed.
    - Counsel on locally available nutrient-dense diet including food-based fluids(such as soup, enriched uji, and yoghurt drinks e.g. Mala), or safe water.
    - Give diluted fresh fruit juice in ration of 1:1, fruits or bananas to provide potassium.
    - Advise mothers/caregivers to continue giving ORS as instructed.

- **TEACH THE MOTHER/CAREGIVER HOW TO MIX AND GIVE ORS. GIVE THE MOTHER/CAREGIVER 4 PACKETS OF ORS TO USE AT HOME.**

- **SHOW THE MOTHER /CAREGIVER HOW MUCH FLUID TO GIVE IN ADDITION TO THE USUAL FLUID INTAKE:**
  - Up to 2 years: 50 to 100 ml after each loose stool
  - 2 years or more: 100 to 200 ml after each loose stool

- **ADVISE THE MOTHER/CAREGIVER TO:**
  - Give frequent small sips from a cup.
  - If the child vomits, wait 10 minutes. Then continue, but more slowly.
  - Continue giving extra fluids until the diarrhoea stops.

2. **GIVE ZINC SULPHATE**

- **TELL THE MOTHER/CAREGIVER HOW MUCH ZINC SULPHATE TO GIVE:**
  - Up to 6 months: 10mg (1/2 tablet) per day for 10 days
  - 6 months or more: 20mg (1 tablet) per day for 10 days

- **SHOW THE MOTHER HOW TO GIVE ZINC SULPHATE:**
  - Infants: Dissolve tablet in a small amount of expressed breastmilk, ORS or clean water in a cup.
  - Older children: Tablets can be chewed or dissolved in a small amount of water.

3. **CONTINUE FEEDING -** (See Counsel the mother chart Pg 43.

4. **WHEN TO RETURN -** (See Counsel the mother chart Pg 47.

PLAN B: TREAT DIARRHOEA AT FACILITY WITH ORS (SOME DEHYDRATION)

Give recommended amount of ORS over 4-hour period in clinic.

1. **DETERMINE AMOUNT OF ORS TO GIVE DURING FIRST 4 HOURS.**
   - The approximate amount of ORS required (in ml) should be calculated by multiplying the child's weight (in kg) with 75 i.e. ORS 75ml/kg.
   - If the child wants more ORS than prescribed, give more.

2. **SHOW THE MOTHER/CAREGIVER HOW TO GIVE ORS SOLUTION.**
   - Give frequent small sips from a cup (or a spoon every 1-2 minutes for a child under 2 years).
   - Check from time to time to see if there are any problems. If the child vomits, wait 10 minutes then continue, but more slowly.
   - Continue breast feeding whenever the child wants.
   - If the child's eyelids become puffy, it is a sign of overhydration and does need any more ORS at this time:
     - Stop ORS and give plain water or breastmilk.
     - Give ORS according to plan A when the puffiness is gone.

3. **AFTER 4 HOURS:**
   - Reassess the child and classify the child for dehydration.
   - Select the appropriate plan to continue treatment.
   - Begin feeding the child in the clinic.

4. **IF THE MOTHER MUST LEAVE BEFORE COMPLETING TREATMENT:**
   - Show her how to prepare ORS solution at home.
   - Show her how much ORS to give to finish 4-hour treatment at home.
   - Give her enough ORS sachets to complete rehydration under this plan. Also give her 4 ORS sachets to continue with **Plan A**. Explain the 4 Rules of Home Treatment:
     - Give extra fluid
     - Give zinc sulphate
     - Continue feeding
     - When to return

TREAT HYPOVOLAEMIC SHOCK

Give 20ml/kg bolus of IV/IO Ringer's Lactate (over 15 minutes).

→ Reassess the child.

→ Signs of shock persist → Repeat bolus of Ringer's Lactate (over 15 minutes). Reassess*.

→ Signs of shock not present → Treat child for severe dehydration - Plan C, STEP 2.

* If signs of shock still persist, refer urgently.

PLAN C: TREAT SEVERE DEHYDRATION QUICKLY

FOLLOW THE ARROWS. IF ANSWER IS "YES", GO ACROSS. IF "NO", GO DOWN.

START HERE

Can you give intravenous (IV) fluid immediately?

YES → Start IV immediately. Give ORS by NGT while the drip is set up. Give 100 ml/ kg Ringer's Lactate solution or if not available, use Normal Saline as follows:

AGE	STEP 1 First give 30 ml/kg in:	STEP 2 Then give 70 ml/kg in:
Infants (under 12 months)	1 hour*	5 hours
Children (12 months up to 5 years)	30 minutes*	2 hours 30 minutes

* Repeat once if radial pulse is still very weak or not detectable.

- Reassess the child every 1- 2 hours. If hydration status is not improving, give the IV drip more rapidly.
- Also give ORS (about 5 ml/kg/hour) as soon as the child can drink: usually after 3-4 hours (infants) or 1-2 hours (children).
- Reassess an infant after 6 hours and a child after 3 hours. Classify dehydration. Then choose the appropriate plan (A, B, or C) to continue treatment.

NO

- Fix a nasogastric tube.
- Start rehydration with ORS solution: give 20 ml/kg/hour for 6 hours (total of 120 ml/kg). Reassess the child every 1-2 hours:
  - If there is repeated vomiting or increasing abdominal distension, give the fluid more slowly.
  - If hydration status is not improving after 3 hours, send the child for IV therapy.
- After 6 hours, reassess the child. Classify dehydration. Then choose the appropriate plan (A, B, or C) to continue treatment.

NOTE: If possible, observe the child at least 6 hours after rehydration to be sure the mother can maintain hydration giving the child ORS solution by mouth.

REHYDRATION THERAPY FOR DIARRHOEA IN CHILDREN WITH SEVERE ACUTE MALNUTRITION

MANAGEMENT OF A CHILD WITH SEVERE ACUTE MALNUTRITION AND DIARRHEA/ VOMITING

History of diarrhea/vomiting

Yes

On examination, does the child have ALL FOUR of the following signs?

i. **Weak/absent pulses**
ii. **AVPU < A**
iii. **Cold hands + temperature gradient**
iv. **Capillary refill 3 seconds or more**

No → Child with dehydration from ongoing losses → Give ReSoMal at 5ml/kg every 30 minutes for the first 2 hours (10ml/kg/hour over 2 hours).

Yes → Child in hypovolaemic shock → Give IV/IO 10 ml/kg Ringer's Lactate in 5% Dextrose over 1 hour. Reassess after 1 hour.

Is there clinical improvement and no signs of fluid overload?

Yes → Give another 10 ml/kg 5% Dextrose in Ringer's Lactate over 1 hour to have given a total of 20mls/kg over 2 hours.

No → Investigate for other causes of shock e.g septic shock.

Refer Urgently

HIV TREATMENT

Refer to the current National HIV guidelines

INITIAL ART REGIMEN

Age	Weight	Preferred Regimen	Dosing (correct weight-based dosing must be confirmed at every visit)
Birth to 4 weeks Term infants (≥37 weeks gestation)	≥3kg	ABC+3TC+DTG	Refer to the National HIV guidelines for weight and age-based dosing (ART FOR INFANTS 0 – 4 WEEKS AND >37 WEEKS table)
Pre-term Infants <37 weeks	Any	AZT+3TC+NVP	Refer to the National HIV guidelines for weight and age-based dosing (ART FOR INFANTS <37 WEEKS table)
>4 weeks to <15 years	<25 kg	ABC + 3TC + DTG (pALD)	Refer to the National HIV guidelines for weight-based dosing (ARVS DOSING CHARTS table)
	>25 kg	TAF/3TC/DTG	TDF/3TC/DTG (300/300/50mg): 1 tab once daily
≥15 years	Any	TAF/TDF + 3TC + DTG	TAF/TDF/3TC/DTG (300/300/50mg): 1 tab once daily

ARVS DOSING CHARTS

Drug	Strength of tablet	Number of tablets or sules by weight band once daily					Strength of adult tablet	Number of tablets or sules by weight band once daily
		3–5.9 kg	6–9.9 kg	10–13.9 kg	14–19.9 kg	20–24.9 kg		25–34.9 kg
pALD	Tablet (dispersible) 60/30/5mg	1	3	4	5	6	NA	NA
ABC/3TC	Tablet (dispersible) 120/60 mg	1	1.5	2	2.5	3	600 mg/300 mg	1
DTG	Tablet (dispersible) 10 mg	0.5	1.5	2	2.5	3	NA	
DTG	Tablet 50 mg	-	-	-	-	1	50 mg	1
DTG/TDF/3TC		-	-	-	-	-	50/300/300	1
TAF/3TC/DTG	Tablet 25/300/50mg	-	-	-	-	-	25/300/50	1
DRV 75mg				3	5	5		
DRV 600mg							600mg	1

ART FOR INFANTS 0 – 4 WEEKS AND >37 WEEKS

Age band	Drug and dosing schedule	Formulation	Dose (Tab)
0 - <2weeks	DTG* 5mg given every 48hours	pDTG 10mg scored dispersible tablet	½ tab
	ABC/3TC 30/15mg given every 48hours	ABC/3TC 120/60mg double scored dispersible tablet	¼ tab
2 - <4weeks	DTG 5mg given every 24 hours	pDTG 10mg scored dispersible tablet	½ tab
	ABC/3TC 30/15mg given every 24hours	ABC/3TC 120/60mg double scored dispersible tablet	¼ tab

*DTG cannot be used in infants <37 weeks gestation
Infants > 4 weeks and weighing ≥3kg can be transitioned to pALD as per the dosing chart above (ARVS DOSING CHARTS)

ART FOR INFANTS <37 WEEKS

Drug	Age of Infant	Gestational age		
		<30 weeks	**30 – 35 weeks**	**>35 weeks**
AZT	**0 – 4 weeks**	2mg/kg BD	2mg/kg BD	4mg/kg BD
	4 – 8 weeks	3 mg/kg BD	3 mg/kg BD	12mg/kg BD
	8 – 10 weeks	12mg/kg BD	12mg/kg BD	12mg/kg BD
		<30 weeks	**30 – 35 weeks**	**>35 weeks**
3TC	**<4 weeks**	2mg/kg BD	2mg/kg BD	2mg/kg BD
	>4 weeks	4mg/kg BD	4mg/kg BD	4mg/kg BD
		32 - 34 weeks	**34 – 37 weeks**	**>37 weeks**
NVP	**0 – 2 weeks**	2 mg/kg BD	4 mg/kg BD for 1 week then 6mg/kg BD	6mg/kg BD
	2 – 4 weeks	4mg/kg BD		

COTRIMOXAZOLE PROPHYLAXIS

WEIGHT (KG)	SUSPENSION 240MG PER 5ML	SINGLE STRENGTH TABLET 480MG (SS)	DOUBLE STRENGTH TABLET 960MG (DS)
1 - 4	2.5ml	¼ SS tab	-
5 - 8	5ml	½ SS tab	¼ DS tab
9 - 16	10ml	1 SS tab	½ DS tab
17 - 30	15ml	2 SS tab	1 DS
>30 (Adults and adolescents)	-	2SS	1DS

Note:

- If creatinine clearance (CrCl) 15-30 ml/min then use 50% of normal recommended dose; if CrCl <15 ml/min, then CTX should be avoided.
- Cotrimoxazole can cause anaemia and neutropenia in some patients, as well as a skin rash.

ART FOR CHILDREN WITH TB CO-INFECTION

Age	Weight	Initial ART Regimen if TB/HIV Co-infection
Birth to 4 weeks	Any	Start anti-TB treatment immediately; start ART after 4 weeks of age, once tolerating anti-TB drugs (follow the regimen recommendations for children ≥ 4 weeks old).
>4 weeks to <15 years	<30 kg	ABC + 3TC + DTG Increase DTG dosing frequency to twice daily for duration of rifampicin-containing TB treatment and for an additional 2 weeks after TB treatment is completed, then revert to once daily dosing.
	≥30 kg	Give TDF/3TC/DTG FDC morning + DTG 50mg evening for duration of rifampicin-containing TB treatment and for an additional 2 weeks after TB treatment is completed, then revert to TDF/3TC/DTG FDC once daily.
≥15 years	Any	Give TDF/3TC/DTG FDC morning + DTG 50mg evening for duration of rifampicin-containing TB treatment and for an additional 2 weeks after TB treatment is completed, then revert to TDF/3TC/DTG FDC once daily.
Any		TAF/3TC/DTG (TAF-LD) Do not initiate TAF-LD in ART-naïve patients (Never used ART before) who are on Anti TB treatment. Use other options which include; ABC/3TC/DTG plus DTG 50mg until 2 weeks after completion of Rifampicin based TB treatment; re-enforce adherence interventions.

RISK CATEGORIZATION FOR MTCT CLIENTS TO DETERMINE INFANT PROPHYLAXIS

High Risk MTCT Clients	Low Risk MTCT Clients
• Adolescent girls and young women <19 yrs (new or known positive) • Pregnant and breastfeeding women (PBFW) with high viral load >50copies/ml • Lack of an established family/social support system; non-disclosure • PMTCT client with previous HIV infected client • Multiple sexual partners • Partners with unknown HIV status • Recent STIs (syphilis and Hepatitis B virus) • Poor treatment adherence and/default from care, stigma, refusal of ART • Sexual and partner violence survivor • New or established Mental health conditions or behavioural risk • Presence of uncontrolled comorbidities (DM, opportunistic infections, malnourished – low MUAC etc) • PBFW who are female sexual workers, people who inject drugs, alcohol use and brewers/sellers	• HIV positive client on treatment and virally suppressed at baseline (<50 copies/ml) • Good adherence to treatment and follow up • Has an established social support system • Controlled comorbidities

ABBREVIATIONS for HIV medication

3TC	Lamivudine
ABC	Abacavir
DRV	Darunavir
DTG	Dolutegravir
FDC	Fixed drug combinations
NVP	Nevirapine
pALD	Paediatric Abacavir/Lamivudine/Dolutegravir fixed-dose combination
TAF	Tenofovir alafenamide
TAF-LD	TAF/Lamivudine/ Dolutegravir fixed-drug combination
TDF	Tenofovir

INFANT HIV PROPHYLAXIS

Refer to the current National HIV guidelines

- Categorize the child/infant based on MTCT risk into High or Low Risk of Vertical Transmission of HIV (Pg 35).
- Infants at high risk of HIV acquisition should receive triple infant prophylaxis with ABC+3TC+DTG for 14 weeks, after which they should continue with NVP prophylaxis until 6 weeks after complete cessation of breastfeeding
- Infants at low risk of HIV acquisition should receive NVP and AZT for 6 weeks, after which they should continue with NVP prophylaxis until 6 weeks after complete cessation of breastfeeding
- Ensure correct formulation and tablet scoring before administration

INFANT PROPHYLAXIS DOSING: ABC+3TC+ DTG REGIMEN FOR INFANTS AT HIGH RISK OF HIV ACQUISITION

Dosing for Term Infants <4 weeks of age and >2kg*			
Age in Weeks	**Drug Strength of Paediatric Formulation**		
	ABC/3TC Double-scored tablet (dispersible) 120mg/60mg	**DTG tablet** (dispersible) 10mg**	
< 2 weeks	1/4 tablet (On alternate days)	1/2 tablet (On alternate days)	
2 - < 4 weeks	1/4 tablet (Daily)	1/2 tablet (Daily)	
For infants aged 4 weeks and above, and ≥ 3kg – Use Paediatric FDC ABC/3TC/DTG (60mg/30mg/5mg)			
Age in Weeks	**Weight Band**		
> 4 weeks***	3 – 5.9kg	6 – 9.9kg	10 – 13.9kg
	1 tablet Daily	3 tablets Daily	4 tablets Daily

*For infants <4 weeks of age and >2kg, use ABC/3TC (120mg/60mg) and pDTG (10mg) as per the dosages and schedule of administration as shown above
**DTG should not be administered to infants <37 weeks of gestation. For pre-term infants <37 weeks gestation, use AZT/3TC and NVP (ART FOR INFANTS <37 WEEKS, Pg 34)
***For infants age 4 weeks and above, and ≥ 3kg, use paediatric FDC ABC/3TC/DTG (60mg/30mg/5mg) as per the dosing chart above

NEVIRAPINE AND ZIDOVUDINE (BIRTH TO 6 WEEKS) AND EXTENDED NEVIRAPINE PROPHYLAXIS FOR INFANTS AT LOW RISK OF VERTICAL TRANSMISSION OF HIV, AND INFANTS CONTINUING INFANT PROPHYLAXIS

Infant Age/Weight (kg)	**Nevirapine Once Daily Dose (10 mg/ml)**	**Zidovudine Twice Daily Dosing (10mg/ml)**
Birth to 6 weeks (Low risk of vertical transmission)		
≤ 2.0 kg	2mg/kg/dose	4mg/kg/dose
2.0 – 2.49	10mg (1 ml)	10mg (1ml)
>2.5	15mg (1.5 ml)	15mg (1.5ml)
From 6 weeks onwards (For all high- and low-risk infants, continue prophylaxis until 6 weeks after complete cessation of breastfeeding)		
6 – 12 weeks	20mg (2 ml)	-
12 weeks – 6 months	20mg (2ml)	-
6 – 9 months	30mg (3ml)	-
9 – 24 months	40mg (4ml)	-

MINISTRY OF HEALTH
REPUBLIC OF KENYA

MINISTRY OF HEALTH

SIMPLIFIED ALGORITHM FOR THE DIAGNOSIS OF PULMONARY TUBERCULOSIS IN CHILDREN (AGED BELOW 15 YEARS)

NATIONAL TUBERCULOSIS, LEPROSY AND LUNG DISEASE PROGRAM

Is this the first visit to any health facility for the current illness?
(This should raise index of suspicion and avoid more delays)

Does the child have presence of TWO OR MORE of the following symptoms suggestive of TB?
- Cough for >2 weeks[a]
- Persistent fever and/or night sweats >2 weeks
- Weight loss or poor weight gain
- Fatigue (lethargy), reduced playfulness, less active >2 weeks

Enquire for positive contact history with a person with TB or person with a chronic cough?[b]

Box 1

On physical examination does the child have any of the following;
- Abnormal exam (any of the following signs)
  - Increased respiratory rate (0-2 months>60/min, 2 -12months >50/min. 1-5yr >40/min., 6 -10yr >30/min. 11-14yr>25/min)
  - Respiratory distress (flairing of alae nasi, chest indrawing)
  - Wheeze or crepitations or asymmetry of findings
  - Oxygen saturation <90% or cyanosis
- Temperature >37.5˚
- Check growth curve, weight, compare to previous weight[c]

Box2

Less than two symptoms (Box 1)
Normal exam findings (Box 2)
Then this is not presumptive for TB.

- Consider other diagnosis & manage child appropriately per ETAT/IMCI ('IMNCI) guidelines.
- Ask parent/guardian to bring the child back after three (3) days if there is no improvement for the current illness.
- However, if there is improvement reassess the child after 1 week.
- If there is history of TB contact - evaluate to offer TPT.

On reassessment of the child after 1 week, if symptoms persist, and/or new symptoms develop, do a thorough clinical review as per box 1& 2

Two or more symptoms (Box 1)
Abnormal respiratory exam findings (Box 2)
This is **probable (presumptive) TB**

No

Investigations for TB available?

Where investigations are not available or delay in obtaining result

Yes

- **Chest X-ray**
- **Tuberculin Skin Test (TST/mantoux) or IGRA**

Bacteriological testing
- Sputum (expectorated or induced), gastric aspirate, nasopharyngeal aspirate (Xpert/MtbRif/Truenat/ZN Microscopy/TB culture)
- Stool (for Xpert)
- Urine (TB LAM) for HIV positive only[d] as an add on test

If test is Negative or not done

Bacteriologic Test Positive

CLINICAL DIAGNOSIS OF PTB

Does the child have presence of TWO OR MORE of the following symptoms?
- Cough for >2 weeks[a]
- Persistent fever and/or night sweats >2 weeks
- Weight loss or poor weight gain
- Fatigue (lethargy), reduced playfulness, less active >2 weeks

PLUS
Presence of ANY ONE OF the following ..
- Positive history of contact with a person with TB in the last 12 months[b]
- Abnormal Respiratory exam
- Abnormal CXR suggestive of PTB[e] (where available)
- Positive Mantoux or IGRA test (where available)

Child does not meet criteria for Clinical diagnosis

Give the child antibiotics for 1 week and ask them to return for reassessment after a week

On reassessment of the child after 1 week, if symptoms persist, re-evaluate and send for CXR if not previously done. If CXR is not accessible and symptoms have persisted consider clinical diagnosis of TB

MAKE A CLINICAL DIAGNOSIS OF TB. START TB TREATMENT

THIS IS BACTERIOLOGICALLY CONFIRMED TB. START TB TREATMENT

Additional Information

[a] Respiratory symptoms suggestive of TB include

All ages: Cough > 2 weeks

Younger children:
- Cough
- Or breathing faster than usual
- Or wheeze not responding to salbutamol

NB/all children started on TB treatment should have their HIV status known

[b] History of contact – ask if there is an individual with known TB or chronic cough who:
- Currently lives in the same household as child ('80% of children have no household contact) or
- Who has visited the home and spent time in the home in contact with child in the preceding 12 months (e.g. relative, maid, driver, worker) or
- For school going children – individual in the same school classroom or dormitory with TB or chronic cough or
- Pre-school children – if they spend time in a day care centre within community or
- All children - if they have spent significant time in a hospital / health facility setting in the preceding months (high risk of exposure to patients with TB in hospital months setting)

Diagnosing DR TB - Clinical considerations
- Diagnosis in children is largely based on clinical and radiological diagnosis
- Consider DR TB diagnosis in a child with a TB diagnosis and contact history as above
- Bacteriologic diagnosis should always be attempted, however, this should not delay treatment initiation

[c] Check weight / poor weight gain includes:
- Weight loss or
- Weight for age Z-score (WAZ) or weight for height z-score (WHZ) below -2. or
- Poor weight gain:-flattening of growth curve (chart) or documented crossing of percentile preceding months or
- Older child (or parent) report they have noticed that they (their child) have lost weight /become thinner in recent weeks to months.

[e] Abnormalities on Chest X-ray that are suggestive of PTB include:
- Enlarged perihilar or paratracheal lymph nodes (+/- narrowing of airways due to compression by lymph nodes)
- Miliary opacification (diffuse micronodules in lung fields)
- Pleural effusion
- Opacification of a lobe or segment of a lobe
- Cavitation (more frequent in teenagers).

[d] TB LF-LAM Indications

A rapid point-of-care urine dip-stick test
LF-LAM SHOULD NOT be used as an alternative test to mWRD but as an adjunct test to help diagnose TB while awaiting for mWRD results.

Indications include;
- PLHIV adults, adolescents and Children ≥5 years with advanced HIV disease (CD4 cell count <200 cells/mm3 or WHO stage 3 and 4 disease.
- All CLHIV Less than 5 Years
- PLHIV that have danger signs or severe illness (RR,30BPM, Temp>39˚c, Heart rate >120BPM, confusion or disorientation, unable to walk unaided)
- PLHIV currently admitted in Hospital and have signs and symptoms of PTB
- PLHIV with signs and symptoms of TB (presumptive TB) in outpatient setting

Drug Sensitive Tuberculosis (DSTB) Treatment Regimens for Kenya for Children 10 Years and Below

Note: All children less than 1 year of age will require the 6 month regimen

	Eligibility for 4 month regimen	Eligibility for 6 month regimen	Eligibility for 12 months
Age	>1 year to 10 years	All infants (<12 months) Above 10 years (11 to <15 years)	TB Meningitis & Ostoearticular TB
Type of TB **Indicators of severity**	Non-severe Pulmonary TB Stable enough to be managed as an outpatient at the point of diagnosis No danger signs TB cervical lymph nodes	• Severe Pulmonary TB • All hospitalised patients at point of diagnosis • Severely ill at diagnosis with any danger sign such as; Difficulty in breathing with associated central cyanosis, Grunting, Oxygen saturation <90%, ; Increased respiratory rate (age 0-2months >60/min; 2-12months >50/min; 1-5yr >40/min; 6-10yr >30/min) ; unable to drink/breastfeed; Weak or absent pulse; Coma/convulsing /confusion; Not responding to pain or unresponsive • Extra Pulmonary TB	
Immune status	• HIV negative. No severe acute malnutrition • No other immunosuppressive condition	• HIV positive • Severe acute malnutrition (SAM) • Any immunosuppressed child	
Bacteriologic status	• Bacteriologically negative • Clinically diagnosed	Bacteriologically confirmed	
Treatment regimen	4 months - 2RHZE / 2RH	6 months - 2RHZE / 4RH	12 months- 2RHZE / 10RH

* If the child has known contact with a person with drug-resistant TB , this table does not apply, start the child on treatment as per the drug-resistant TB guidelines

Childhood Drug Resistant TB (DRTB) Treatment Regimen

Resistance pattern	Target group	Regimen (Child friendly formulations)
1. Isoniazid Mono-resistant TB	Children <25 kgs	6RHZE/LFX
2. Poly-drug resistance (PDR) TB	Children <25 kgs	9RHZE/LFX
3. MDR/RR TB	<15 years Non severe dx	6Bdq/Lzd/Lfx/Cs/Cfz 3Lfx/Cs/Cfz
	<15 years Severe dx	6Bdq/Lzd/Lfx/Cs/Cfz 6Lfx/Cs/Cfz
4. Pre-XDR (FQ Res.) TB	<15 years (Children)	6Bdq/Dlm/Cfz/Cs/Lzd 3Dlm/Cfz/Cs
5. XDR TB		Consult national clinical review team
Any Difficult to manage child (DS or DR TB)		Consult sub county and county clinical review teams

*Children (15 years) weighing above 25 Kgs with INH mono resistance or poly drug resistance should be put on the adult dose

MINISTRY OF HEALTH
REPUBLIC OF KENYA

MINISTRY OF HEALTH

NATIONAL TUBERCULOSIS, LEPROSY AND LUNG DISEASE PROGRAM

SIMPLIFIED ALGORITHM FOR THE DIAGNOSIS OF EXTRA PULMONARY TUBERCULOSIS IN CHILDREN (AGED BELOW 15 YEARS)

Presence of TWO OR MORE of the following symptoms lasting > 2 weeks

- SYMPTOM ARISING FROM AFFECTED ORGAN (see table)
- Weight loss or poor weight gain
- Persistent fever and/or night sweats
- Fatigue, reduced playfulness, less active

PLUS

Presence of ONE OR MORE of the following:

- Positive contact history
- Abnormal sign arising from the AFFECTED ORGAN (see table)
- Investigation relevant to AFFECTED ORGAN is positive (where available)
- Positive mantoux or IGRA test (where available)

MAKE A CLINICAL DIAGNOSIS OF EXTRA PULMONARY TB
START TB TREATMENT

EXTRA PULMONARY TUBERCULOSIS IN CHILDREN – SYMPTOMS, SIGNS & INVESTIGATIONS

(In addition to the clinical features due to EPTB listed below, the child may have one or more of the classic non-specific signs: persistent fever, poor weight gain/weight loss and lethargy/ reduced activity, lasting >2 weeks)

Site of EPTB	Symptoms and Signs for EPTB specific to the affected site	Investigation
TB lymphadenitis (cervical, axillary or inguinal LN)	• Lymph node enlargement for more than one month. • Painless, non-tender • Often asymmetrical • +/- caseous (cheese like) discharge • Most commonly in neck area	**Fine needle aspiration (FNA) for:** • Xpert, culture • Microscopy – predominance of lymphocytes, AFB **Lymph node biopsy - Histology**
Pleural TB	• +/- Chest pain often one-sided. • +/- Cough • Large effusion - fast breathing, breathlessness • Dullness on percussion, reduced breath sounds on affected side	• CXR • Chest ultrasound • Pleural tap1
TB meningitis	**Persistent CNS symptoms:** • Early signs – Persistent headache, irritability/abnormal behaviour, one sided weakness, changing gait, blurred vision, squint (signs progressively worsening over weeks) • Late (severe) signs - reduced level of consciousness, convulsions, neck stiffness bulging fontanelle, cranial nerve palsies.	• Lumbar puncture – CSF1 • CT scan brain with contrast • Cranial ultrasound in infants <6 months with open anterior fontanelle
Miliary TB	• Non-specific signs: persisting fever, weight loss/poor weight gain, lethargy. • Often have respiratory symptoms & signs (fast breathing, +/- cough, wheeze, respiratory distress)	• CXR – diffuse miliary opacities (micronodules) • Fundoscopy – see micro-nodules on retina • High risk for other extra pulmonary sites. Look for other lesions.
Abdominal TB	• Abdominal pain >2 weeks • Progressive swelling of abdomen over several weeks. • Exam: Abdominal swelling, ascites (shifting dullness, fluid thrill.	• Ascitic tap[1] • Abdominal ultra-sound[2]
Spinal TB	• Persisting pain in focal point in the back. • Early sign: Tender / pain when apply pressure at part of spine, loss of lordosis / reduced curvature in lower back if located in lumbar vertebrae. • Advanced disease: Deformity of spine • Progressive lower limb weakness	• X-ray of affected spine – lateral and antero-posterior views.
Pericardial TB	• Breathless with minimal exertion, palpitations (feeling of rapid heart beat),cough may be present • Cardiac failure (tachycardia, pedal oedema, infants – periorbital puffiness • Distant heart sounds • Apex beat difficult to palpate	• CXR – global enlargement of heart. • Echocardiogram (Cardiac ultrasound). Pericardial tap

1. Cerebrospinal fluid (CSF), pleural fluid, ascitic fluid specimens, joint fluid - the following findings are suggestive of TB: Colour – clear or light yellow colour. Biochemistry – high protein and low glucose Microscopy - increased white cell counts, predominantly lymphocytes. (note that bacteriologic tests rarely detect MTb from these body fluids).

2. Abdominal ultra-sound shows ascites +/- septation, enlarged abdominal lymph nodes

All specimens (FNA, CSF, aspirates etc) may be sent for bacteriologic tests such as GeneXpert, AFB microscopy or TB culture as appropriate, however detection rate is lower than sputum

Childhood Drug-sensitive TB (DSTB) treatment regimen for Kenya			
Criteria	**Eligibility for 4 months**	**Eligibility for 6 months**	**Eligibility for 12 months**
Age	• 1yr to 10 yrs (Non-severe TB)	• <1yrs, or 11 to <15 yrs • 1yr to 10 yrs (severe TB)	• All ages
Type of TB	• PTB/ Uncomplicated TB cervical lymphadenitis	• Extra Pulmonary TB including complicated TB cervical lymphadenitis (Except TBM or Osteoarticular TB)	• TB Meningitis or Osteoarticular TB
Clinically status (stable/unstable)	• Non-severe TB • Stable enough to be managed as an outpatient at the point of diagnosis • No danger signs	• Severe TB • All hospitalized patients at the point of diagnosis • Severely ill at diagnosis with any danger signs such as; ◦ Difficulty in breathing with associated central cyanosis, Grunting, ◦ Oxygen saturation <90%; Increased respiratory rate (age 0-2months >60/min; 2-12months >50/min; 1-5yr >40/min; 6-10yr >30/min); ◦ unable to drink/breastfeed; Weak or absent pulse; Coma/convulsing /Confusion; Not responding to pain or unresponsive	
HIV status	• HIV negative/HEI	HIV positive	
Nutrition status	• Normal/MAM/ Overweight/Obese	Severe acute malnutrition (SAM)	
Immunosuppressive conditions (cancer, Diabetes, etc)	• Condition absent	• Condition present	
Bacteriologic status	• Clinically diagnosed • Bacteriologically negative	• Bacteriologically confirmed TB	
Treatment regimen	**2RHZE / 2RH**	**2RHZE / 4RH**	**2RHZE / 10RH**
* If the child has known contact with a person with drug-resistant TB , this table does not apply, start the child on treatment as per the drug-resistant TB guidelines			

Childhood drug-resistant TB (DRTB) treatment regimen		
Resistance pattern	**Target group**	**Regimen**
1. Isoniazid Mono-resistant TB*	Children <25 kgs	6RHZ(FDC)/E/LFX (Child-friendly formulations)
2. Poly-drug resistance (PDR) TB*	Children <25 kgs	9RHZ(FDC)/E/LFX (Child-friendly formulations)
3. MDR/RR	<15 years non-severe disease	2Bdq/Lzd/Lfx/MFx/Cfz/Z/E/Hh 2 or 4**Bdq/Lfx/MFx/Cfz/Z/E/Hh 5Lfx/MFx/Cfz/Z/E/Hh
	<15 years severe disease	6Bdq/Lfx/Lzd/Cs/Cfz 12Lfx/ Cs/Cfz
4. Pre-XDR (FQ Res.)	<15 years non-severe disease	6Bdq/Dlm/L/Cs/Cfz 3Dlm/Cs/Cfz
	<15 years Severe disease	6Bdq/Dlm/L/Cs/Cfz 12Dlm/ Cs/Cfz
5. XDR	All ages	Individualized Regimen
*Children weighing above 25 kgs with INH monoresistance or PDR should be put on the adult dose ** Extend the regimen for 2 months if the patient has not converted at month 2		

TB TREATMENT

ELIGIBILITY CRITERIA FOR PAEDIATRIC TB REGIMENS

	Eligibility for 4 month regimen (Non-severe TB)	Eligibility for 6 month regimen (Severe TB)	Eligibility for 12 months
Age	1 year to 10 years	• All infants (<12 months) • Above 10 years (11 to <15years)	Child of any age
Type of TB	• Non severe Pulmonary TB(PTB) • Uncomplicated TB Lymphadenitis	• Severe Pulmonary TB • Extra Pulmonary TB including complicated TB lymphadenitis (EXCEPT TB Meningitis or Osteoarticular TB)	TB Meningitis & Osteoarticular TB
Clinical Status (stable/unstable) at the point of diagnosis	• Stable enough to be managed as an outpatient **OR** • Has no danger signs	• Hospitalized patients **OR** • Requires admission **OR** • Danger signs present: • Difficulty in breathing with associated: • Central cyanosis, • Grunting, • Oxygen saturation <90%; • Increased respiratory rate (age 0-2months ≥60b/min; 2-12months ≥50b/min; 1-5yr ≥40b/min) • Unable to drink/breastfeed • Weak or absent pulse • Coma/convulsing/confusion • Not responding to pain or unresponsive	
Immunosuppressive conditions (cancer, DM etc.)	Condition absent	Condition present	
Nutrition status	Normal, Moderate acute Malnutrition, overweight, Obese	Severe Acute Malnutrition(SAM)	
HIV Status	HIV Negative	HIV Positive	
Bacteriologic status	Clinically diagnosed i.e. Bacteriologic tests are negative.	Bacteriologically confirmed	
Treatment regimen	If **ALL** of the above criteria have been met	If **ANY** of the above criteria have been met	
	4 months - 2RHZE / 2RH	**6 months - 2RHZE / 4RH**	**12 months- 2RHZE / 10RH**
***If the child has known contact with a person with drug-resistant TB , this table does not apply, start the child on treatment as per the drug-resistant TB guidelines.**			

DRUG REGIMEN FOR TREATMENT OF TB IN CHILDREN

DOSAGES FOR PAEDIATRIC TB TREATMENT (IMPROVED FORMULATIONS) DOSAGES FOR A CHILD UP TO 3.9KGS

After giving the child their dose for the day, discard the rest of the solution. Prepare a fresh solution every day

Weight band (Kgs)	Number of Tablets				
	Intensive Phase			Continuation Phase	
	RHZ (75/50/150mg)	E* (100mg)	How to reconstitute the medicines	RH(75/50mg)	How to reconstitute the medicines
Less than 2kg	1/4	1/4	Dissolve one (1) tablet of RHZ in 20 ml of safe drinking water. Once fully dissolved, add the completely crushed one (1) tablet of Ethambutol and give 5ml (1/4) of this solution.	1/4	Dissolve one (1) tablet of RH in 20 ml of safe drinking water. Once fully dissolved, give 5ml (1/4) of this solution.
2-2.9	1/2	1/2	Dissolve one (1) tablet of RHZ in 20 ml of safe drinking water. Once fully dissolved, add the completely crushed one (1) tablet of Ethambutol and give 10ml (1/2) of this solution.	1/2	Dissolve one (1) tablet of RH in 20 ml of safe drinking water. Once fully dissolved, give 10ml (1/2) of this solution.
3-3.9	3/4	3/4	Dissolve one (1) tablet of RHZ in 20 ml of safe drinking water. Once fully dissolved, add the completely crushed one (1) tablet of Ethambutol and give 15ml (3/4) of this solution.	3/4	Dissolve one (1) tablet of RH in 20 ml of safe drinking water. Once fully dissolved, give 15ml (3/4) of this solution.

*Ethambutol is not dispersible. Crush it completely before adding to be prepared solution of RHZ during the intensive phase.

DOSAGES FOR A CHILD BETWEEN 4 - 25KGS

Weight band (Kgs)	Number of Tablets				
	Intensive Phase			Continuation Phase	
	RHZ (75/50/150mg)	E(100mg)	How to reconstitute the medicines	RH(75/50mg	How to reconstitute the medicines
4 - 7.9	1	1	Dissolve the tablet(s) of RHZ in 20 ml of safe drinking water. Once fully dissolved, add the completely crushed tablet(s) of Ethambutol and give ALL of this solution to the child.	1	Dissolve the tablet(s) of RH in 20 ml of safe drinking water. Once fully dissolved give ALL of this solution to the child.
8 - 11.9	2	2		2	
12 - 15.9	3	3		3	
16 - 24.9	4	4		4	
25kg and above	Use adult dosages and preparations.			Use adult dosages and preparations.	

PYRIDOXINE (VITAMIN B6) DOSING FOR CHILDREN ON TB TREATMENT

Weight band (Kgs)	Dose in mg	Number of 25mg tablets	Number of 50mg tablets
Less than 5	6.25	Half a tablet 3 TIMES PER WEEK	Not suitable for young infants
5.0 - 7.9	12.5	Half a tablet daily	Half of 50mg tablet 3 TIMES PER WEEK
8.0 - 14.9	25	One tablet daily	Half of 50mg tablet daily
15kg and above	50	Two tablets daily	One 50 mg tablet daily

COUNSEL THE MOTHER

ASSESS CHILD'S FEEDING

ASSESS FEEDING IF CHILD IS LESS THAN 2 YEARS OLD, HAS MODERATE ACUTE MALNUTRITION, AT RISK OF ACUTE MALNUTRITION, ANAEMIA, CONFIRMED HIV INFECTION, OR IS HIV EXPOSED

- **Ask** questions about the child's usual feeding and feeding during this illness. Compare the mother/caregiver's answers to the Feeding Recommendations for the child's age (Pg 42).

- **Ask** - How are you feeding your child?
  - If the child is receiving any breast milk, **Ask**:
    - How many times during the day?
    - Do you also breastfeed during the night? (Y/N)
    - Does the child take any other food or fluids? (Y/N)
    - How many times per day?
    - What do you use to feed the child?

- **IF MODERATE ACUTE MALNUTRITION or if a child with CONFIRMED HIV INFECTION** fails to gain weight or loses weight between monthly measurements, **Ask:**
  - How large are the servings?
  - Is the child served on their own plate/cup? (Y/N)
  - Who feeds the child?
  - How is the child fed? (Probe for responsive feeding)
  - What foods are available in the home?
  - During this illness, has the child's feeding changed? (Y/N). If yes, how?

In addition, for **HIV EXPOSED** child:

- **If mother and child are on ARV treatment or prophylaxis and child breastfeeding, Ask:**
  - **Ask** the mother**:**
    - Do you take ARV drugs? (Y/N)
      - If yes, do you take all doses?/have you miss any doses?/have you stopped taking medication? (Y/N)
      - If no, counsel and initiate treatment.

  - **Ask** about the child:
    - Is the child on HIV prophylaxis? (Y/N)
      - If yes, does he/she take all doses?/has he/she missed any doses?/ Has he/she stopped taking medication? (Y/N)
      - If no, counsel and initiate prophylaxis.

CONDUCT THE APPETITE TEST

- All children aged 6 months or more with **SEVERE ACUTE MALNUTRITION** (oedema of both feet or WFH/L Z-score less than -3SD or MUAC less than 11.5 cm) **AND NO MEDICAL COMPLICATION** should be assessed for appetite.
- By far the most important criterion to decide if a patient should be sent to in- or out- patient management is the Appetite Test.
- A poor appetite means that the child has an infection or a major metabolic abnormality such as liver dysfunction, electrolyte imbalance, cell membrane damage or damaged biochemical pathways. These are the patients at immediate risk of death.

How to do the appetite test?

- The appetite test should be conducted in a separate quiet area.
- Explain to the mother/caregiver the purpose of the appetite test and how it will be carried out.
- The mother/caregiver, where possible, should wash her/his hands.
- The mother/caregiver should sit comfortably with the child on her/his lap and either offer the RUTF from the packet or put a small amount on her/his finger and give it to the child.
- The mother/caregiver should offer the child the RUTF gently, encouraging the child all the time. If the child refuses then the mother/caregiver should continue to quietly encourage the child and take time over the test.
- The test usually takes a short time but may take up to one hour. The child must not be forced to take the RUTF.
- The child needs to be offered plenty of water to drink from a cup as he/she is taking the RUTF.

The result of the appetite test

- **PASS:**
  - A child that takes at least the amount shown in the table below passes the appetite test.

- **FAIL:**
  - A child that does not take at least the amount of RUTF shown in the table below should be referred for inpatient care.
  - Even if the caregiver/health worker thinks the child is not taking the RUTF because s/he doesn't like the taste or is frightened, the child still needs to be referred to inpatient care for at least a short time. If it is later found that the child actually takes RUTF to pass the test then they can be immediately transferred to the outpatient treatment.

Minimum amount of Plumpy'nut / RUTF per kg of body weight required to pass the Appetite Test

Plumpy'nut / RUTF	
Body Weight (kg)	**Sachets**
Less than 4 kg	1/8 to ¼
4 - 6.9	¼ to 1/3
7 - 9.9	1/3 to ½
10 - 14.9	½ to ¾
15 - 29	¾ to 1
Over 30kg	>1

FEEDING RECOMMENDATIONS FOR ALL CHILDREN DURING SICKNESS AND HEALTH, INCLUDING HIV EXPOSED CHILDREN ON ARV PROPHYLAXIS



NEWBORN, BIRTH UP TO 1 WEEK

- Immediately after birth, put your baby in skin-to-skin contact with you.
- Allow your baby to take the breast within the first hour. Give your baby colostrum, the yellowish, thick milk. It protects the baby from many illnesses.
- Breastfeed day and night, as often as your baby wants, at least 8 times in 24 hours. Frequent feeding produces more milk.
- If your baby is small (low birth weight), feed at least every 2 to 3 hours. Wake the baby for feeding after 3 hours, if baby does not wake self.
- DO NOT give other foods or fluids. Breast milk is all your baby needs. This is especially important for infants of HIV-positive mothers. Mixed feeding increases the risk of HIV mother-to-child transmission when compared to exclusive breastfeeding.



1 WEEK UP TO 6 MONTHS

- Breastfeed as often as your child wants. Look for signs of hunger, such as beginning to fuss, sucking or moving lips.
- Breastfeed day and night whenever your baby wants, at least 8 times in 24 hours. Frequent feeding produces more milk.
- Do not give other foods or fluids
- Breast milk is all your baby needs.



6 UP TO 9 MONTHS

- Breastfeed as often as your child wants.
- Also give thick porridge or well mashed foods, including animal-source foods and Vitamin A-rich fruits and vegetables.
- Start by giving 2 to 3 tablespoons of food. Gradually increase to 1/2 cups (1 cup = 250ml).
- Give 2 to 3 meals each day.
- Offer 1 or 2 snacks each day between meals when the child seems hungry.



9 UP TO 12 MONTHS

- Breastfeed as often as your baby wants.
- Also give a variety of mashed or chopped family food, including animal-source foods and Vitamin A- rich fruits and vegetables.
- Give 1/2 cup at each meal (1 cup = 250ml).
- Give 3 to 4 meals each day.
- Offer 1 or 2 snacks between meals. The child will eat if hungry.
- For snacks, give small chewable items that the child can hold. Let your child try to eat the snack, but provide help if needed.



12 MONTHS UP TO 2 YEARS

- Breastfeed as often as your child wants.
- Also give a variety of mashed or chopped family food, including animal-source foods and Vitamin A-rich fruits and vegetables.
- Give 3/4 cup at each meal (1 cup = 250ml).
- Give 3 to 4 meals each day.
- Offer 1 or 2 snacks between meals.
- Continue to feed your child slowly, patiently. Encourage - but do not force - your child to eat.



2 YEARS & OLDER

- Give a variety of family foods to your child, including animal-source foods and Vitamin A-rich fruits and vegetables.
- Give at least 1 cup at each meal (1 cup = 250ml).
- Give 3 to 4 meals each day.
- Offer 1 or 2 snacks between meals.
- If your child refuses food, offer "tastes" several times. Show that you like the food. Be patient.
- Talk with your child during a meal, and keep eye contact.

NB:

COUNSEL ON FEEDING PROBLEMS INCLUDING GUIDANCE ON INFANT FEEDING IN HIV CONTEXT

If the child is not being fed as described in the above recommendations, counsel the mother accordingly. In addition:

Ensure hygiene during food preparation: Clean hands, Clean food, Clean utensils and Clean storage.







- **If the mother reports difficulty with breast feeding, assess breastfeeding** (Pg 65).
  - As needed, show the mother correct positioning and attachment for breast feeding.
- **If the child is less than 6 months old and is taking other milk or foods:**
  - Build mother's confidence that she can produce all the breast milk that the child needs.
  - Suggest breastfeeding more frequently and for longer periods day and night, and gradually reducing other milk or foods.
- **If other milk needs to be continued, counsel the mother to:**
  - Breastfeed as much as possible, including at night.
  - Make sure that other milk is a safe appropriate breast milk substitute.
  - Make sure other milk is correctly and hygienically prepared and given in adequate amounts.
  - Make sure that left over feeds are not fed to the baby.
- **If the mother is using a bottle to feed the child:**
  - Recommend reinstating breastfeeding if possible.
  - Recommend using a cup instead of a bottle.
  - Show the mother how to feed the child with a cup.
  - Re-emphasize hand washing & hygiene.
- **If a child > 6months is not being fed actively, counsel the mother to:**
  - Sit with the child and encourage eating.
  - Give the child an adequate serving in a separate plate or bowl.
- **If the child is not feeding well during illness, counsel the mother to:**
  - Clear a blocked nose if it interferes with feeding.
  - Breastfeed more frequently and for longer if possible.
  - For the child under 6 months who is not breastfed, increase the number of milk feeds.
  - For a child >6months already on complementary feeds, give soft, varied, appetizing, enriched favorite foods including milk to encourage the child to eat as much as possible.
  - Offer small frequent feeds.
  - Give snacks between meals and check regularly for oral thrush or ulcers.
  - Expect that appetite will improve as child gets better.
- **If the child has sore mouth or mouth ulcers:**
  - Recommend soft foods that will not burn the mouth such as thick enriched porridge, eggs, mashed potatoes, bananas, pumpkin or avocado.
  - Chop foods.
  - Avoid salty, spicy, acidic foods that may irritate the child.
  - Give cold nutritious drinks - they'll not make the child sick or sicker.
- **Follow-up any feeding problem in 5 days.**

GUIDANCE ON INFANT FEEDING IN HIV CONTEXT

Provide the mother with ART to achieve optimal viral suppression while the infant receives ARV prophylaxis or ART in line with the current national guidelines.

- All HIV positive mothers should be given information on the government guidance on breastfeeding in the context of HIV and counseled on the benefits and challenges of breastfeeding.
- All HIV positive mothers should be encouraged and supported to exclusively breastfeed for the first 6 months of life, introducing appropriate complementary foods at six months and continue breastfeeding for 2 years and beyond.
- Give information on prevention and management of breast conditions.

Feeding HIV Infected Infants

- Exclusive breastfeeding is recommended for the first 6 months of life with continued breastfeeding for 2 years and beyond.
- Maternal and infant ARVs should be administered as per recommendations in the Kenya HIV guidelines.
- Counsel the mother on risks associated with mixed feeding. For infants who cannot access breast milk, give information on safe preparation, storage and appropriate feeding techniques chosen for replacement feeds. Ensure adherence to the Breast Milk Substitutes (Regulation and Control) Act, 2021.
- Appropriate complementary feeding should be introduced at six months with consideration of the energy needs as per the MIYCN policy and guidelines.
- Stoppage of breast feeding for children should happen gradually within one month.

FEEDING INFANTS UNDER SPECIAL CIRCUMSTANCES

- Special circumstances include maternal death, abandoned children, severe maternal breast disease as determined by the clinician.
- Refer to current MIYCN policy 2019 and Breast Milk Substitutes (General) Regulations 2021 Cap 34(1).

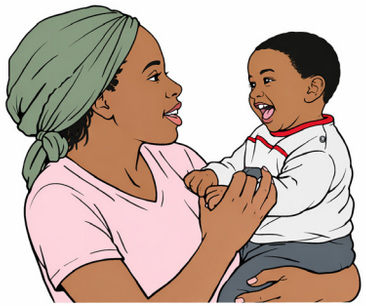
RECOMMENDATIONS FOR CARE FOR CHILD DEVELOPMENT

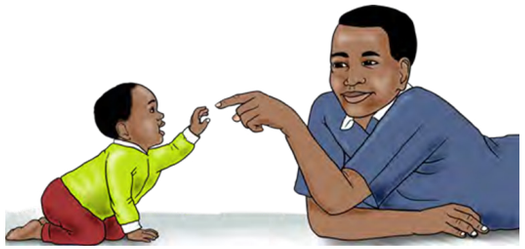
CONCEPTION TO BIRTH	NEWBORN, BIRTH TO 1 WEEK	1 WEEK UP TO 6 MONTHS	6 MONTHS UP TO 9 MONTHS	9 MONTHS UP TO 12 MONTHS	12 MONTHS UP TO 2 YEARS	2 YEARS AND OLDER
PLAY						
Your baby begins learning in the womb. Gently rub your belly. The baby enjoys when you gently rub your belly. Pat the belly when your child kicks. 	Your baby can see and hear you at birth — start using the child's name. Skin to skin contact is good. Provide ways for your child to see, hear, feel, move arms and legs freely, and touch you. Gently soothe, stroke and hold your baby. 	Provide ways for your child to see, hear, feel, move freely and touch you. Slowly move colourful things for your child to see and reach for. Example of play items: Shaker rattle, ring, big ring on a string. 	Give your child clean, safe household things to handle, bang, and drop. Examples of play items: containers with lids, metal pot and spoon. 	Hide an attractive item for the child under a cloth or box. See if the child can look for it. Play peek-a-boo. 	Give your child things to stack and put into containers and take out. Examples of play items: Nesting and stacking objects, containers and pegs. 	Help your child count, name and compare things. Help your child make simple play items. Examples of play items: balls, dolls and cars. 
COMMUNICATE						
Gently rub your belly and talk to the child. Take time for intentional relaxed breathing. Sing soothing songs as you rub your belly. 	Look into baby's eyes and talk to your baby. When you are breastfeeding is a good time. 	Smile, laugh and talk with your child. Respond to your child by copying their sounds and gestures. 	Respond to your child's sounds and interests. Call the child's name and see your child respond. 	Tell your child name things in their environment, such as Items, people, animals. Show your child how to say things with their hand, like bye-bye. Example of play items: A doll. 	Ask your child simple questions. Respond to your child's attempts to talk. Show and talk about nature, people and things. 	Encourage your child to talk and answer your child's questions. Tell your child or create stories, songs and games. Create stories, songs and games. Examples of play items: Simple books with pictures, dolls, balls and cars. 

• Give your child affection and show love • Be interested in your child's interests and respond to them • Praise your child for trying to learn skills
• Avoid screentime (TV, phone, tablet) before 2 years of age • Plan the child's daily activities

ABOUT CARE FOR DEVELOPMENT PROBLEMS











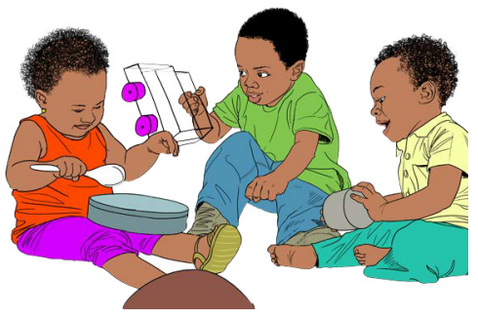






IF THE CHILD IS NOT BEING CARED FOR AS DESCRIBED IN THE ABOVE RECOMMENDATIONS, COUNSEL THE CAREGIVER ACCORDINGLY. IN ADDITION:

- **Discuss ways to have the baby see, hear, feel and move appropriately for age.**
- **If the child cannot be breastfed, counsel the Caregiver to:**
  - Hold the child close when feeding, look at the child, and talk or sing to her/him.
- **If the Caregiver does not know what her/his child does to play or communicate:**
  - Inform her/him that children play and communicate from birth.
  - Demonstrate for her/him how the child responds to activities.
- **If the Caregiver feels she/he does not have enough time to provide care for development, encourage her/him to:**
  - Combine care for development with other child care activities (feeding, bathing, dressing).
  - Ask other family members to help provide care for development or help her/him with other tasks.
- **If the Caregiver has no toys for the child to play with, counsel the Caregiver to:**
  - Use any household objects that are clean and safe.
  - Make simple toys.
  - Play with the child, as the child will learn by playing with her/him and other people.
- **If the child is not responding, or seems "slow":**
  - Encourage the Caregiver to do extra care for development activities.
  - Check to see whether the child is able to see and to hear.
  - Encourage the Caregiver and other family members to play and communicate with the child through touch and movement.
- **If the child is being raised by someone other than the mother, help the caretaker to:**
  - Identify at least one person who can care for the child regularly and give the child love and attention.
  - Expect that, with love and special attention, the child can recover from the loss of a parent.
- **Advise the mother to return for follow-up in 14 days:**
  - Assess as per the recommendations for care for development (Pg 44).





GIVE GOOD HOME CARE FOR YOUR CHILD

FOR ANY SICK CHILD

- Breastfeed frequently and for longer at each feed.
- Increase fluid intake if on complementary feeding. Give soup, porridge, yoghurt drinks or clean and safe water.
- If the child vomits, wait 10 minutes then continue - but more slowly.
- Continue providing extra feeds for up to 2 weeks.



CHILD WITH DIARRHOEA
Giving more fluids can be life saving. Give Zinc Sulphate.

1) For children not on exclusive breastfeeding:

- Give extra fluids as much as the child will take:
  - ORS
  - Food based fluids such as; Soup, Porridge, Yoghurt, Clean and safe water
- Breastfeed more frequently and longer at each feeding.
- Continue giving extra fluids until diarrhoea stops.

2) For babies on exclusive breastfeeding:

- Breastfeed more frequently and longer at each breastfeed.
- Give ORS.

If the child vomits, wait for 10 minutes then give small frequent sips.



EXCLUSIVELY BREASTFEED THE YOUNG INFANT

- Give only breast milk to the young infant.
- Breastfeed frequently, as often and for as long as the infant wants.

MAKE SURE THAT THE YOUNG INFANT IS KEPT WARM AT ALL TIMES

- In cool weather, cover the infant's head and feet, and dress the infant with extra clothing.

ABOUT MOTHER'S OWN HEALTH

If the mother is sick, provide care for her, or refer her for help.

- If she has a breast problem (such as engorgement, sore nipples, breast infection), provide care for her or refer her for help (Refer to Pg 65 on breast problems).
- Advise her to express milk from breasts if separated from the baby for more than one day to maintain lactation and prevent breast engorgement.
- Advise her to eat well to keep up her own strength and health.
- Advise her on rest and self care for her mental well being(Refer to pg 25 on the mental health assessment - Patient Health Questionnaire -2 (PHQ-2)).
- Encourage male involvement in reproductive and child health care.
- Check her immunization status including the 5TTs.
- Check vitamin A status to be given if within 6 weeks of delivery.

Make sure she has access to:

- Breast examination and Cervical Cancer Screening
- Family planning
- Counseling on STI and Reproductive Tract Infection (RTI) prevention

If the mother is HIV Positive:

- Emphasize on importance of early testing for the child and enrollment into HIV care if child is positive.
- Emphasize on the importance of adhering to HIV treatment for both mother and child.
- Refer mother for HIV care and support if services not in facility.
- Encourage partner to seek Voluntary Counseling and Testing (VCT).
- Re-emphasize the importance of safe sex and early treatment of STI/RTI and counsel regarding future pregnancies.

If baby is less than 2 months:

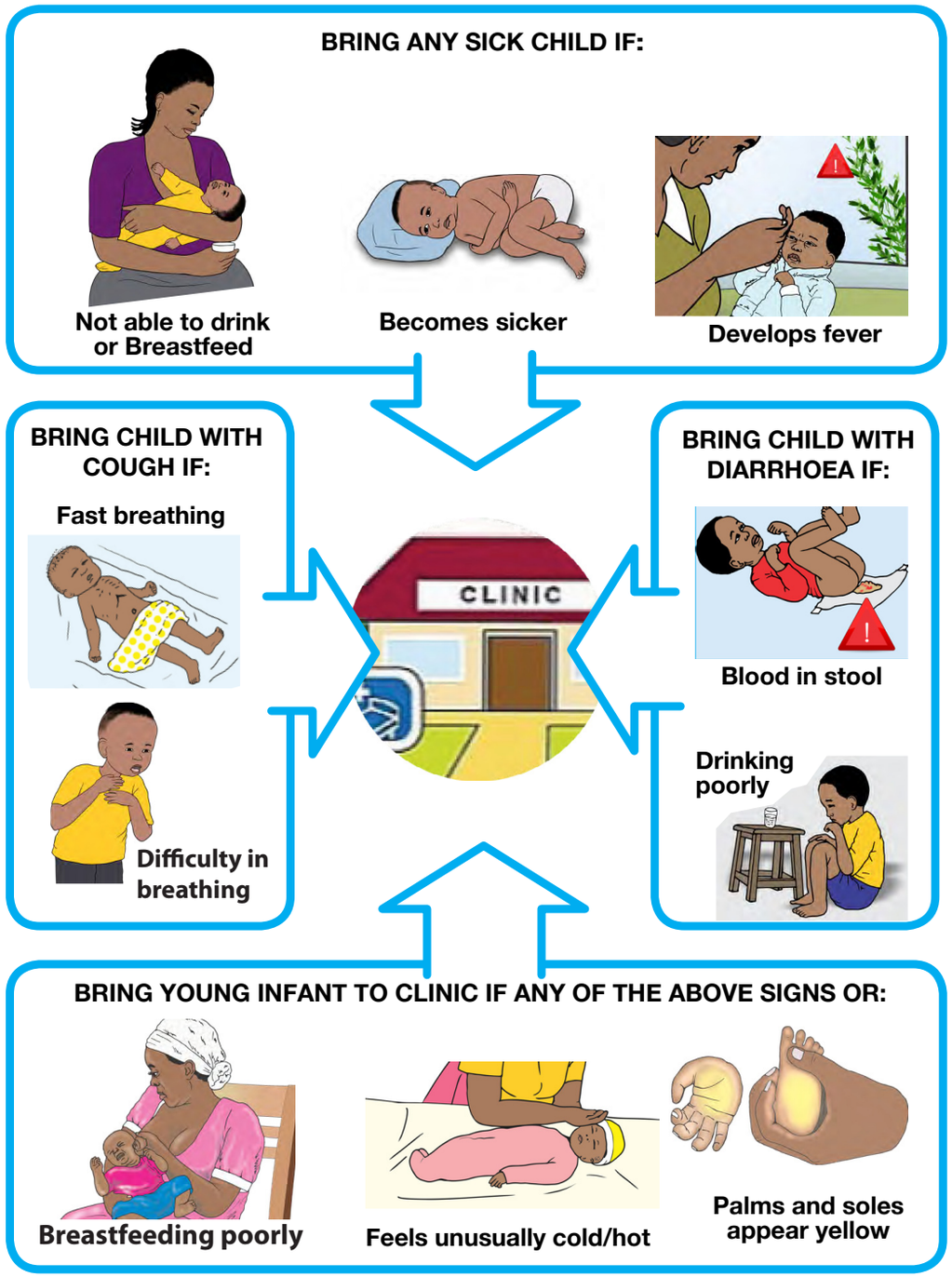
- Ask mother about her lochia.
- Ask about perineal care.
- Ask about Caesarean Section scar where applicable.
- If PMTCT not done, counsel and test.
- If on ART or other medications, counsel on compliance.
- Remind her of postpartum schedule (Within 48hrs, 2-4weeks, 4-6 weeks).

WHEN TO RETURN FOR SCHEDULED VISITS

If the child has:	Return for follow- up in:
• Pneumonia (Non-Severe Pneumonia) • Some dehydration • Dysentery	2 Days
• Uncomplicated Malaria • Fever - no Malaria, if fever persists • Eye or mouth complications of Measles • Not presumptive for Pulmonary TB, if not improving	3 Days
• Persistent Diarrhoea • No Pneumonia: Cough or cold • No dehydration • Acute ear infection • Feeding problems • Any other illness, if not improving	5 Days
• Severe Acute Malnutrition without complications • Not Presumptive for Pulmonary TB, if improving	7 Days
• Chronic ear infection • Moderate Acute malnutrition • At Risk of Acute Malnutrition • Anaemia • Poor interaction and/or communication and responsiveness	14 Days
• Developmental milestone alert	30 Days
• HIV Infected • HIV Exposed	Monthly



WHEN TO RETURN IMMEDIATELY



GIVE FOLLOW - UP CARE

- **Care for the child who returns for follow-up using all the boxes that match the child's previous classifications.**
- **For every follow-up visit, a full assessment of the child or young infant should be done.**

PNEUMONIA

- **After 2 days:**
  - Check the child for general danger signs. Assess and classify the child for cough or difficulty in breathing (Pg 9).
  - **Ask:**
    - Is the child breathing slower?
    - Is there less fever?
    - Is the child eating better?

  - **Treatment:**
    - If any **general danger sign**, give the first dose of IV Benzyl Penicillin + Gentamicin, then admit or refer URGENTLY to hospital.
    - If **lower chest wall indrawing, breathing rate, fever and eating have not improved, or if the child was on amoxicillin,** change to Benzyl Penicillin + Gentamicin and ADMIT or REFER (If this child had measles within the last 3 months or is known or confirmed HIV infection, refer).
    - If **breathing slower, less fever, or eating better,** complete the 5 days of antibiotics.

WHEEZING

- **After 2 days:**
  - Check the child for general danger signs or lower chest wall indrawing. Assess and classsify the child for cough or difficulty in breathing (Pg 8).
  - **Ask:**
    - Is the child breathing slower?
    - Is the child still wheezing?
    - Is the child eating better?

  - **Child under 1 year:**
    - If **wheezing and any of the following**;
      - General danger sign, stridor in a calm child, lower chest wall indrawing, fast breathing or poor feeding: Give the first dose of IV Benzyl Penicillin + Gentamicin, then admit or refer URGENTLY to hospital.
    - If **no wheezing, breathing slower and eating better:** continue the treatment for 5 days.

  - **Child over 1 year;**
    - If **wheezing and any of the following**;
      - General danger sign, stridor in a calm child, lower chest wall indrawing, fast breathing or poor feeding: Give the first dose of IV Benzyl Penicillin + Gentamicin, then admit or refer URGENTLY to hospital.
      - If breathing rate and eating have not improved: Give the first dose of IV Benzyl Penicillin + Gentamicin, then admit or refer URGENTLY to hospital.
      - If still wheezing: continue inhaled bronchodilator.
      - If child has unilateral wheeze and not responding to bronchodilators, consider alternative diagnosis and refer for evaluation.
    - If **breathing slower, no wheezing and eating better,** continue the treatment for 5 days.

PERSISTENT DIARRHOEA

- **After 5 days:**
  - **Ask:**
    - Has the diarrhoea stopped?
    - How many loose stools is the child having per day?

  - **Treatment:**
    - If the **diarrhoea has not stopped** (child is still having 3 or more loose stools per day), do a full reassessment of the child (Pg 8 - 25). Give any treatment needed. Then refer to hospital.
    - If the **diarrhoea has stopped** (child having less than 3 loose stools per day), tell the mother to follow the usual feeding recommendations for the child's age, but give one extra meal every day for 1 month. Ask her to continue giving Zinc Sulphate for a total of 10 days.
    - **NB:** Attention to diet is an essential part of the management of a child with persistent diarrhoea.

DYSENTERY

- **After 2 days:**
  - Assess and classify the child for diarrhoea (Pg 10 - 11).
  - Look for signs of improvement.

  - **Ask:**
    - Are there fewer episodes of loose stool?
    - Is there less blood in the stool?
    - Is there less fever?
    - Is there less abdominal pain?
    - Is the child eating better?

  - **Treatment:**
    - If **there is dehydration**, treat as per classification (Pg 10 - 11).
    - If **there is no improvement after 2 full days of antibiotic treatment:**,
      - Stop the first antibiotic.
      - Investigate for possible cause through stool analysis.
      - If unable to do the lab investigation, give IM/IV Ceftriaxone 50-80mg/kg as the second-line antibiotic for 3 days.
      - If the two antibiotics have each been given for 2 days, with no sign of clinical improvement, check for other conditions (eg amoebiasis).
      - If amoebiasis, give metronidazole at 10 mg/kg three times a day for 5 days.
      - Admit the child if there is an indication requiring hospital treatment.
    - If **there is improvement**, complete the antibiotics.

UNCOMPLICATED MALARIA

- **If fever persists after 3 days, or recurs within 14 days:**
  - Do a full reassessment of the child (Pg 8 – 25) and assess for other causes of fever.
  - **DO NOT REPEAT** the Rapid Diagnostic Test if it was positive on the initial visit.

  - **Treatment:**
    - If the child has **any general danger sign or stiff neck,** treat as VERY SEVERE FEBRILE DISEASE.
    - If the child has **any other cause of fever other than malaria,** provide appropriate treatment.
    - If there is **no other apparent cause of fever:**
      - Do microscopy to look for malaria parasites. If parasites are present and the child has finished a full course of AL, give oral DIHYDROARTEMISININ-PIPERAQUINE (DHA-PPQ) , if available, or refer the child to a hospital. Give paracetamol. If child is under 5kgs and was given DHA-PPQ, assess further.
      - If there is no other apparent cause of fever and you do not have a microscopy to check for parasites, refer the child to a hospital.
      - If fever has been present for 7 days, refer for further evaluation.

FEVER - NO MALARIA

- **If fever persists after 3 days:**
  - Do a full reassessment of the child (Pg 8 – 25), assess for other causes of fever and repeat the malaria test.

  - **Treatment:**
    - If the child has **any general danger sign or stiff neck,** treat as VERY SEVERE FEBRILE DISEASE.
    - If a child has **a positive malaria test,** give first-line oral antimalarial and paracetamol. Advise the mother to return in 3 days if the fever persists.
    - If the child has any other **cause of fever other than malaria,** provide appropriate treatment.
    - If there is **no other apparent cause** of fever:
      - If the fever has been present for 7 days, refer for further evaluation.

EYE OR MOUTH COMPLICATIONS OF MEASLES

- **After 3 days:**
  - Look for red eyes and pus draining from the eyes.
  - Look at mouth ulcers.
  - Smell the mouth.

  - **Treatment for Eye Infection:**
    - If **pus is draining from the eye,** ask the mother/caregiver to describe how she has treated the eye infection.
      - If treatment has been correct, refer to hospital. If treatment has not been correct, teach mother/ caregiver correct treatment.
    - If **the pus is gone but redness remains,** continue the treatment.
    - If **no pus or redness,** stop the treatment.

  - **Treatment for Mouth Ulcers:**
    - If mouth ulcers are **worse, or there is a very foul smell from the mouth,** refer to hospital.
    - If mouth ulcers are the **same or better,** continue using half-strength Gentian Violet for a total of 5 days.

  - **Treatment for thrush:**
    - If **thrush is worse,** check that treatment is being given correctly.
    - If **the child has problems with swallowing,** refer to hospital.
    - If thrush is the **same or better,** and the child is feeding well, continue Nystatin for a total of 7 days.
    - If thrush is **no better or is worse,** consider symptomatic HIV infection.

EAR INFECTION

- **After 5 days:**
  - Reassess for ear problem (Pg 14).
  - Measure the child's temperature.

  - **Treatment:**
    - If **there is tender swelling behind the ear or high fever (38.5°C or above),** give the first dose of Ceftriaxone, then admit or refer URGENTLY to hospital.
    - **Acute ear infection:**
      - If **ear pain continues or discharge persists,** treat with 5 more days of the same antibiotic. Continue wicking to dry the ear. Follow-up in 5 days.
      - If **no ear pain or discharge**, praise the mother on her careful treatment and encourage her to complete 10 days of antibiotic.

- **After 2 weeks:**
  - **Chronic ear infection:** Check that the mother is wicking the ear correctly. Encourage her to continue. Review in 2 weeks.
    - If **ear discharge still present**, refer for ENT review.
    - If **no ear pain or discharge,** praise the mother for her careful treatment.

FEEDING PROBLEM

- **After 5 days:**
  - Reassess feeding (Pg 41).
  - Ask about any feeding problems found on the initial visit.
    - Counsel the mother/caregiver about any new or continuing feeding problems.
    - If you **counsel the mother/ caregiver to make significant changes in feeding**, ask her to bring the child back again after 5 days.
    - If the **child is moderately wasted**, ask the mother to return 14 days after the initial visit to measure the child's weight gain.

ANAEMIA

- **After 14 days:**
  - Give / continue iron. Advise mother to return in 14 days for more iron.
  - Continue giving iron every day for maximum of 3 months.
  - If the **child is not improving after 1month or develops other signs, refer for further evaluation.**

MALNUTRITION

- **After 14 days:**
  - If the **child is gaining weight,** encourage the mother to continue with feeding. Counsel the mother about any feeding problem.

MODERATE ACUTE MALNUTRITION

- **After 14 days:**
  - Measure weight and MUAC, and record in the register and in the MCH Hand Booklet at each visit (every 2 weeks).
  - Assess for nutritional oedema.
  - Height/Length measurements will be taken monthly.
  - Conduct medical assessment (including screening for TB), and provide treatment, if necessary.
  - Check if the mother/caregiver is implementing age-appropriate stimulation, play, and communication activities at home.
  - Schedule a follow-up appointment every 2 weeks with the caregiver and encourage them to attend.
  - Remind the caregivers to come with their MCH Hand Booklet on each visit.
  - Assess if the child is improving or not.

- **Treatment**
  - **If the child is deteriorating:**
    - Develops signs of SAM or you do not think feeding will improve, refer appropriately.
    - Not improving (not gaining weight or increasing MUAC), counsel on any feeding problem found.
  - **If the child is improving:**
    - Praise the mother and encourage her to continue use of locally available nutrient dense diet.

SEVERE ACUTE MALNUTRITION WITHOUT COMPLICATIONS

- **After 7 days:**
  - Measure weight and MUAC, and record in the register and in the MCH Hand Booklet at each visit (every week).
  - Assess for nutritional oedema.
  - Height/Length measurements will be taken monthly.
  - Check vital signs: temperature, respiratory rate, pulse rate, SpO_2.
  - Conduct appetite test.
  - Take medical history and conducts physical examination, including evaluation of health and nutrition status progress.
  - Check if the mother/caregiver is implementing age-appropriate stimulation, play, and communication activities at home.

- **Treatment**
  - **If the child is making good progress** – gaining weight, increasing MUAC, decreasing oedema, has a good appetite, without severe medical complications, and regularly attending weekly follow-up visits – the child continues the treatment through outpatient care weekly until he/she achieves the criteria for discharge.
  - **If the child is not responding to the treatment or if IMNCI danger signs are observed**, refer the child to inpatient care for treatment.

HIV EXPOSED & INFECTED CHILDREN

- **FOR EVERY VISIT:**
  - Assess the child's general condition. Do a full assessment (Pg 8 - 25).
  - Treat the child for any condition found.
  - Ask for any feeding problems and counsel the mother about any new or continuing feeding problems.
  - Screen for TB on every visit and evaluate for TPT eligibility.
  - Continue with routine follow-up for growth and development, nutrition, immunization, vitamin A, and deworming.

- **HIV INFECTED CHILD**
  - Advise the mother/caregiver to bring the child back if any new illness develops or she is worried.
  - Counsel the mother/caregiver on any other problems and ensure community support is being given. Attach to a CHP. Refer for further psychosocial support/ counseling if necessary.
  - Assess adherence to ART and Cotrimoxazole, and advice accordingly.
  - Offer or refer the child for comprehensive HIV management and care (including ART) as per the national ART guidelines.
  - Plan for defaulter tracking system, indentification and tracking of children.
  - Follow up monthly.

HIV EXPOSED CHILD: For children tested DNA PCR Negative

- **After 2 weeks:**
  - Give Cotrimoxazole prophylaxis from 6 weeks and emphasize the importance of compliance. Start or continue with ARV prophylaxis until 6 weeks after complete cessation of breast feeding.
  - Retest at 6 weeks, 6 months,12 months and 18 months (or earlier if child develops symptoms suggestive of HIV-as per WHO staging criteria).
  - If after 18 months and the infant continues breastfeeding, do HIV antibody test every 6 months. Repeat the HIV antibody test 6 weeks after complete cessation of breastfeeding.
  - Follow up HIV Exposed infant monthly up to 24 months.
  - Refer to early infant diagnosis (EID) algorithm for confirmation of HIV status (Pg 64).
  - Refer to the HIV exposed infant follow-up card and register for further follow-up instructions.

IF ANY MORE FOLLOW -UP VISITS ARE NEEDED BASED ON THE INITIAL VISIT OR THIS VISIT, ADVISE THE MOTHER OF THE NEXT FOLLOW-UP VISIT.

ALSO ADVISE THE MOTHER WHEN TO RETURN IMMEDIATELY.

IMNCI PROCESS FOR ALL YOUNG INFANTS (BIRTH UP TO TWO MONTHS)

HAS THE INFANT JUST BEEN DELIVERED?

YES

- **Does the young infant require resuscitation?**

YES

Resuscitate using the Newborn Resuscitation algorithm (Pg 61).

NO

- Provide essential newborn care (Pg 59).
- **Proceed to ASSESS, CLASSIFY AND IDENTIFY TREATMENT FOR THE SICK YOUNG INFANT** (Pg 52 - 58).
- Support mother to initiate breastfeeding (Pg 65).

NO

- **ASK:** Why the caregiver has brought the child to the health facility today?
- **DETERMINE IF THIS IS AN INITIAL, FOLLOW UP** or **ROUTINE VISIT.**
- Ensure that an infant who has come for an initial visit (i.e. because they are sick) is **fast-tracked**.
- **Proceed to ASSESS, CLASSIFY AND IDENTIFY TREATMENT FOR THE SICK YOUNG INFANT** (Pg 52 - 58).
- **TREAT (if necessary)** (Pg 59 - 64)
- **COUNSEL THE CAREGIVER** on home care for the sick young infant and when to return if necessary (Pg 65 - 66).
- Provide **FOLLOW-UP CARE** (if necessary) (Pg 67 - 68).
- **Assess breastfeeding** and support the mother to successfully breastfeed the infant (Pg 65).

ASSESS, AND CLASSIFY AND IDENTIFY TREATMENT

THE SICK YOUNG INFANT FROM BIRTH AGE UP TO 2 MONTHS

ASK THE MOTHER WHAT THE YOUNG INFANT'S PROBLEMS ARE

- Assess the young infant as follows:

CHECK FOR POSSIBLE SERIOUS BACTERIAL INFECTION, VERY SEVERE DISEASE, PNEUMONIA OR LOCAL BACTERIAL INFECTION

ASSESS

ASK

- Has the infant had convulsions? Fits/ twitching?

ASK and LOOK:

- Is the infant able to feed or breastfeed?

LOOK, LISTEN AND FEEL

- Count the breaths in 1 minute. (Young infant must be calm)
  - Repeat the count if it is 60 breaths or more per minute.
- Listen for grunting (noisy breathing)
- Look for severe lower chest wall indrawing.
- Check for oxygen saturation using pulse oximetry where available.
- Look for severe palmar pallor
- Measure axillary temperature.
- **Look at the young infant's movements:**
  - If the infant is sleeping, ask the mother to wake him/her.
  - Does the infant move on his/ her own?
  - If the infant is not moving, gently stimulate him or her.
    - Does the infant move only when stimulated but then stops?
    - Does the infant not move at all?
- **Look at the umbilicus:**
  - Is it red or draining pus?
- Look for skin pustules

CLASSIFY ALL YOUNG INFANTS

CLASSIFY AND IDENTIFY TREATMENT

SIGNS	CLASSIFY	TREATMENT (URGENT PRE-REFERRAL TREATMENTS ARE IN BOLD PRINT.)
Any of the following signs: • Not able to feed or breastfeed • Convulsions • Movement only when stimulated or no movement at all • Fast breathing (60 breaths per minute or more) **in infants less than 7 days old** • Severe lower chest wall indrawing • High body temperature (38°C* or above) • Low body temperature (less than 35.5°C*)	**POSSIBLE SERIOUS BACTERIAL INFECTION OR VERY SEVERE DISEASE**	• **If convulsing now, give Phenobarbitone as per the neonatal convulsions algorithm** (Pg 64). • **If oxygen saturation is less than 90%, start oxygen therapy and refer or admit** (Pg 30). • **Give first doses of Benzyl Penicillin and Gentamicin** (Pg 61). • **Test for and treat / prevent low blood sugar** (Pg 63). • **If severe palmar pallor, Refer for URGENT blood transfusion.** • **Advise mother how to keep the infant warm on the way to the hospital** (Pg 59). • **Screen for possible TB disease** (Pg 21) **and check for HIV exposure** (Pg 56). • **Admit or refer URGENTLY to hospital.** • **If REFERRAL IS REFUSED OR IS NOT FEASIBLE, further classify and treat in the clinic until referral is feasible.** (Pg 63).
• Fast breathing (60 breaths per minute or more) **in infants 7–59 days old**	**PNEUMONIA**	• **Give oral amoxicillin for 7 days** (Pg 60)**.** • Advise the mother to give home care (Pg 66). • Follow up in 3 days.
• Red umbilicus or draining pus **OR** • Skin pustules.	**LOCAL BACTERIAL INFECTION**	• Treat skin pustules and red umbilicus with Flucloxacillin Syrup for 7 days (Pg 60). • Teach the mother to treat local infections at home. (Pg 60). • Advise mother to give home care for the young infant (Pg 66). • Screen for possible TB disease (Pg 21) and check for HIV exposure (Pg 56). • Advise mother when to return immediately. • Follow-up in 2 days.
None of the signs of Very Severe Disease or Local Bacterial Infection	**VERY SEVERE DISEASE OR LOCAL BACTERIAL INFECTION UNLIKELY**	• Advise mother to give home care for the young infant (Pg 66). • Screen for possible TB disease (Pg 21) and check for HIV exposure (Pg 56). • Advise mother when to return immediately.

*These thresholds are based on axillary temperature. The thresholds for rectal temperature readings are approximately 0.5°C higher.

THEN, CHECK FOR JAUNDICE

ASSESS

ASK
- Does the young infant have yellow discoloration of eyes, palms or soles?
- If yes, for how long?

LOOK
- Look for jaundice (yellow eyes or skin)
- Look at the young infants' palms and soles. Are they yellow?

CLASSIFY JAUNDICE

CLASSIFY		IDENTIFY TREATMENT
SIGNS	**CLASSIFY**	**IDENTIFY TREATMENT**
• Any jaundice if age less than 24 hours **OR** • Yellow eyes, palms and soles at any age **OR** • Any visible yellowness in a preterm baby regardless of when it appears	**SEVERE JAUNDICE**	• **Test for and treat / prevent low blood sugar** (Pg 62). • **Refer URGENTLY to hospital.** • **Advise mother how to keep the infant warm on the way to the hospital** (Pg 59).
• Yellowness appearing after 24 hours of age • Yellowness in the eyes & skin • Palms and Soles **NOT** yellow	**JAUNDICE**	• If the young infant is less than 14 days: • Advise the mother to give home care for the young infant (Pg 66). • Advise mother to return immediately if palms and soles appear yellow (Pg 67). • If young infant older than 14 days, refer to hospital for further evaluation. • Follow up in 1 day.
No jaundice	**NO JAUNDICE**	• Advise mother to give home care for the young infant (Pg 66).

THEN CHECK FOR EYE CONDITIONS

ASSESS

ASK
- Is there an eye problem?

LOOK
- Crossed eyes
- Excessive tearing
- Fear of light
- Clouding of the cornea
- Red eyes
- Whiteness in the pupil
- Eye discharge, if yes for how long?
- Eyes draining pus
- Swollen eyes

CLASSIFY EYE CONDITION

CLASSIFY		IDENTIFY TREATMENT
SIGNS	**CLASSIFY**	**IDENTIFY TREATMENT**
• Crossed Eyes	**SQUINT**	• Admit or refer immediately to the hospital or appropriate unit within the facility.
• White spot on the pupil and crossed eyes	**CONGENITAL CANCER OF THE EYE**	
• Clouding of the cornea and no signs of measles, fear of light and exessive tearing	**CONGENITAL GLAUCOMA**	
• White spot on the pupil area.	**CONGENITAL CATARACT**	
• Eyes swollen or draining pus	**SEVERE EYE INFECTION**	• **If suspected gonococcal conjunctivitis, treat with single dose of IM Ceftriaxone 50mg/kg** (pg 61). • **Apply dose of Tetracycline Eye Ointment** (Pg 60). • **Test for and treat / prevent low blood sugar** (pg 63). • **Admit or Refer URGENTLY to hospital/specialized unit.** • Advise mother how to keep infant warm on the way to hospital (Pg 59). • Test and treat the mother and partner.
• Eyes are red or have a discharge	**EYE INFECTION**	• **Apply first dose of Tetracycline Eye Ointment** (pg 60). • Teach mother to treat eye infection at home (Pg 60) • Follow up in 2 days.
• No signs to classify for congenital condition or eye infection	**CONGENITAL CONDITION or EYE INFECTION UNLIKELY**	• Advise mother to give home care for the child (Pg 66).

DOES THE YOUNG INFANT HAVE DIARRHOEA?

ASSESS

LOOK, LISTEN AND FEEL

- **General condition:**
  - Is the young infant restless and irritable?
- **Infant's movements:**
  - Does the infant move on his/her own?
  - Does the infant move only when stimulated but then stops?
  - Does the infant not move at all?
- **Look for sunken eyes**
- **Pinch the skin of the abdomen.** Does it go back:
  - Very slowly (>2 secs)?
  - Slowly (1-2 secs)?
  - Immediately?

Note: A young infant has diarrhoea if the stools have changed from the usual pattern, and are many and watery (More water than faecal matter)

CLASSIFY DIARRHOEA FOR DEHYDRATION

CLASSIFY

IDENTIFY TREATMENT

SIGNS	CLASSIFY AS	IDENTIFY TREATMENT
Two of the following signs: • Movement only when stimulated or no movement at all? • Sunken eyes. • Skin pinch goes back very slowly.	**SEVERE DEHYDRATION**	• **If infant has no other severe classifi cation:** • **Give fluid for severe dehydration (Plan C,** Pg 33**).** **OR** • **If infant also has another severe classification:** • **Refer URGENTLY to hospital with the mother giving frequent sips of oral rehydration salts (ORS) on the way.** • **Advise the mother to continue breastfeeding.** • **Advise the mother how to keep the infant warm on the way to the hospital** (Pg 59)**.**
Two of the following signs: • Restless, irritable • Sunken eyes • Skin pinch goes back slowly.	**SOME DEHYDRATION**	• Give fluid and breast milk for some dehydration (Plan B, Pg 32) **OR** • If the infant also has another severe classification: • **Refer URGENTLY to hospital with the mother giving frequent sips of ORS on the way.** • Give zinc for 14 days (Pg 27). • Advise the mother to continue breastfeeding. • Advise the mother when to return immediately (Pg 67). • Follow-up in 2 days.
Not enough signs to classify as some or severe dehydration	**NO DEHYDRATION**	• Give fluids to treat for diarrhoea at home and continue breastfeeding (Plan A, Pg 32). • Give zinc for 14 days (Pg 27). • Counsel the caregiver on home care for the young infant (Pg 66). • Follow-up in 2 days.

THEN CHECK FOR SPECIAL TREATMENT NEEDS

ASSESS

If infant is less than 72 hours:

ASK, CHECK, RECORD

- Has the mother had any of the following during labour or within 24 hours before delivery:
  - Fever ≥ 38°C?
  - Infection treated with antibiotics?
  - Untreated maternal infections?
  - Membranes ruptured >18 hours before delivery?
  - Foul smelling amniotic fluid?

For all young infants:

ASK CHECK AND RECORD

- Mother tested VDRL positive?
- Mother tested HBsAg positive?
- Does the mother/caregiver/ close household contact have TB disease?
- Is the infant receiving any other foods or drinks?

CLASSIFY SPECIAL TREATMENT NEEDS

CLASSIFY

IDENTIFY TREATMENT

FOR INFANT IS LESS THAN 72 HOURS:

SIGNS	CLASSIFY	TREATMENT (URGENT PRE-REFERRAL TREATMENTS ARE IN BOLD PRINT.)
• Membranes ruptured >18 hours before delivery **OR** • Foul smelling amniotic fluid **OR** • Mother being treated with antibiotics for infection **OR** • Untreated maternal infections **OR** • Mother has fever ≥38°C	**RISK OF BACTERIAL INFECTION**	• **Give baby 1st dose of Benzyl Penicillin and Gentamicin** (Pg 61)**.** • **Test for and treat / prevent low blood sugar** (Pg 63). • **Keep warm** (Pg 59). • **Admit or refer urgently to hospital.**

FOR ALL YOUNG INFANTS:

• Mother tested positive for Hepatitis B surface antigen (HBsAg Positive)	**RISK OF HEPATITIS B**	• Give Hepatitis B vaccine at birth (within 24 hrs) and repeat at 6,10 and 14 weeks postnatal. • Infant should also receive Hepatitis B immunoglobulin within 24 hrs of birth. Refer for immunoglobulin. • If the infant presents after 24 hours of birth, give the Hepatitis B vaccine as per NVIP (Pentavalent) and do an antigen test (HBsAg) at 9months of age. DO NOT give Hepatitis B immunoglobulin.
• Mother tested VDRL positive **OR** • Mother's VDRL is not known, and it is not possible to get the result now	**RISK OF CONGENITAL SYPHILIS**	• Check for signs of congenital syphilis* and if present, refer to hospital. • If no signs of congenital syphilis, give IM Benzathine Penicillin (Pg 61). • Ask about the caregiver's health, and treat as necessary (Pg 25). • Ensure that the mother and partner receive full treatment for positive VDRL.
• History of mother /caregiver / close household contact started TB treatment <2 months before delivery of the baby	**RISK OF TUBERCULOSIS**	• If child has not been given BCG, withhold BCG and Screen for possible TB disease**. • If TB is ruled out, give TPT as per current guidelines. Screen for possible TB during treatment. • Upon completion of TPT, wait for 2 weeks then give BCG. • If child is found to have TB disease, give full course of TB treatment as per the current National TB guidelines (Pg 37- 40).
• Infants who are not breast fed due to the death or sickness of the mother e.g. Orphans	**SPECIAL FEEDING CIRCUMSTANCES**	• Counsel about feeding (Pg 41- 43). • Identify concerns of caregiver and family about feeding. • Advise the caregiver/s how to feed and keep the low weight infant warm at home (Pg 59). • If thrush, teach the caregiver/s to treat thrush at home (Pg 60). • Advise caregiver/s to give home care for the young infant (Pg 66). • Once the sick mother recovers, resume breast feeding. • Follow-up any feeding problem or thrush in 2 days. • Follow-up low weight for age in 14 days.

*SIGNS OF CONGENITAL SYPHILIS: Oedema, pallor, jaundice, reduced movements or irregular, jerky movements, full fontanelle, large lymph nodes, hepatomegaly, splenomegaly, respiratory distress, small red or purple spots in the skin (petechiae), blisters on hands and feet.

** NEONATAL TB: Neonatal sepsis with poor response to treatment, low birth weight, poor feeding, fever, lethargy, fast breathing, enlarged lymph nodes, enlarged liver and/or spleen, poor weight gain.

CHECK FOR HIV EXPOSURE AND INFECTION

ASSESS

ASK
- Has the mother and/or young infant had a HIV test?

IF YES:
- What is the mother's HIV status?:
  - Antibody test is POSITIVE
  - Antibody test is NEGATIVE

If mother is HIV positive and negative DNA PCR test in child

- **ASK:**
  - Is the mother on ART and young infant on ARV prophylaxis?(Risk categorize mother and child as high/low)

IF NO TEST: Mother and young infant status unknown Perform HIV Antibody test for the mother: If positive, perform DNA PCR test for the young infant

IF the mother is NOT available, do an antibody test on the child. If positive, do a DNA PCR test

CLASSIFY

CLASSIFY

IDENTIFY TREATMENT

SIGNS	CLASSIFY	TREATMENT (URGENT PRE-REFERRAL TREATMENTS ARE IN BOLD PRINT.)
POSITIVE DNA PCR test in young infant **Presumptive diagnosis of children < 18 months** • Positive HIV antibody test **AND** • Symptomatic with 2 or more of the following: • Oral candidiasis • Severe pneumonia • Severe sepsis **OR** Any of the following: • Any WHO clinical stage 4 condition • Recent maternal death(if likely to have been HIV related) or advanced HIV disease • Child's CD4% <25%	**PRESUMED HIV INFECTION**	• Initiate ART immediately (Pg 34), and take confirmatory HIV DNA PCR and baseline VL. • Retest at 6 weeks with DNA PCR. • Educate caregivers on adherence and its importance. • Counsel and follow up existing infections. • Screen for possible TB disease at every visit (Pg 21). • For those who do not have TB disease, start TPT. Screen for possible TB throughout TPT. • For those with TB disease, defer ART initiation for 2 weeks. • Initiate Cotrimoxazole prophylaxis (Pg 35). • Assess infant's breastfeeding and provide appropriate counseling to the mother/ caregiver (Pg 57 & 65 respectively). • Advise the mother on home care (Pg 66). • Offer routine follow up for growth, nutrition and development. • Immunize as per schedule (Pg 25). • Follow up and offer comprehensive management of HIV as per the national ART guidelines.
• Mother HIV positive and negative DNA PCR in young infant. **OR** • Mother HIV positive, young infant not yet tested. **OR** • Positive antibody test in young infant whose mother is not available	**HIV EXPOSED**	• Treat, counsel and follow up existing infections. • Initiate or continue Cotrimoxazole prophylaxis (Pg 35). • Categorize the mother and child based on MTCT risk (High/Low) (Pg 35): • High risk infant/child should receive triple infant prophylaxis with ABC+3TC+DTG for 14 weeks, after which they should continue with NVP prophylaxis until 6 weeks after complete cessation of breastfeeding • Low risk infant/child should receive NVP and AZT for 6 weeks, after which they should continue with NVP prophylaxis until 6 weeks after complete cessation of breastfeeding. • Repeat DNA PCR test at 6weeks, and if negative at 6 months and at 12 months. If negative, repeat antibody test at 18 months (EID algorithm, Pg 64). • For those who do not have TB disease, start TPT. Screen for possible TB throughout TPT (Pg 21). • Immunize for measles at 6 months, 9 months and boost at 18 months (Pg 25).
Mother's antibody test is NEGATIVE	**HIV NEGATIVE**	• Manage presenting conditions according to IMNCI and other recommended national guidelines. • Advise the mother about breastfeeding (Pg 65) and about her own health (Pg 46).

THEN CHECK FOR FEEDING PROBLEM, LOW WEIGHT FOR AGE OR LOW BIRTH WEIGHT

ASSESS

ASK

- Is the infant breast-fed?
  - If Yes,
    - How many times in 24 hours?
    - During the day?
    - During the night?
  - If No, why?
- Does the infant usually receive any other foods or drinks?

LOOK, LISTEN, FEEL

- Determine weight for age (Pg 73 - 75)*.
- Measure MUAC for young infants 6 weeks and older.
- Look for ulcers or white patches in the mouth (thrush).
- Determine mother's HIV status: Positive, Negative or Unknown.
- Determine HIV Exposed Infant's HIV status: Positive, Negative or Unknown.

ASSESS BREAST FEEDING:

- Has the infant breast-fed in the previous hour?
  - If the infant has not fed in the previous hour, ask the mother to put her infant to the breast. Observe as the infant breastfeeds for 4 minutes.
  - If the infant was fed during the last hour, ask the mother if she can wait and tell you when the infant is willing to feed again.

TO CHECK CORRECT POSITIONING, LOOK FOR:

- Infant's head and body straight
- Infant facing the mother's breast with nose opposite the nipple
- Infants body close to the mother's body
- Mother supporting infant's whole body

Is the child well positioned?

- Well positioned
- Not well positioned

TO CHECK ATTACHMENT, LOOK FOR:

- Chin touching breast
- Mouth wide open
- Lower lip turned outward
- More areola visible above than below the mouth

(All of these signs should be present if the attachment is good.)

- Is the infant able to attach?
  - Good attachment
  - Not well attached
  - No attachment at all
- Is the infant suckling effectively (that is, slow deep sucks, sometimes pausing)?
  - Suckling effectively
  - Not suckling effectively
  - Not sucking at all

- Clear a blocked nose if it interferes with breastfeeding.

CLASSIFY FEEDING PROBLEM AND LOW WEIGHT FOR AGE

CLASSIFY

IDENTIFY TREATMENT

SIGNS	CLASSIFY	TREATMENT (URGENT PRE-REFERRAL TREATMENTS ARE IN BOLD PRINT.)
• Not well attached to breast or not attached at all **OR** • Not suckling effectively or not suckling at all **OR** • Breastfeeds less than 8 times in 24 hours **OR** • Receives other foods or drinks **OR** • Weight for age Z Score < -2SD, or MUAC <11.5cm for infants 6 wks and older **OR** • Thrush (white patches in mouth) **OR** • Mouth ulcers **OR** • Features suggestive of possible TB disease	**FEEDING PROBLEM AND /OR LOW WEIGHT FOR AGE**	• Advise the mother to breastfeed as often and for as long as the infant wants, day and night. • If not well attached or not suckling effectively, teach and show correct positioning and attachment (Pg 65). • If breast feeding less than 8 times in 24 hours advise to increase frequency of feeding. • If receiving other foods or drinks, counsel mother/ caregiver. Refer to current MIYCN policy and Breast Milk Substitute Regulation and Control Act 2012. • If a HIV positive mother has chosen not to breastfeed, give feeding advice (Pg 43). • If thrush or mouth ulcers, teach the mother to treat thrush or mouth ulcers at home (Pg 60). • If mother is HIV positive, follow Guidance on Infant Feeding in HIV context (Pg 43). • If mother's HIV status is unknown, test. • Advise the mother to keep infant warm (Pg 65). • Advise mother to give home care for the young infant (Pg 66). • Screen for possible TB disease (Pg 21) during routine visits. If TB is not diagnosed, treatment as per regimen for TB treatment (Pg 37 - 40). • Follow up on any feeding problem, thrush or mouth ulcers in 2 days. • Follow-up low weight for age in 14 days.
Weight for age Z Score ≥ -2SD and no other sign of inadequate feeding.	**NO FEEDING PROBLEM**	• Praise the mother for feeding the infant well. • Advise the mother to keep infant warm (Pg 59). • Screen for possible TB disease during routine visits. • Advise mother to give home care for the young infant (Pg 66).

Continue next page

* If the young infant was born preterm, use the preterm growth charts on pg 73

CLASSIFY FOR BIRTH WEIGHT FOR INFANT LESS THAN ONE WEEK

CHECK FOR LOW BIRTH WEIGHT IN YOUNG INFANT LESS THAN ONE WEEK

SIGNS	CLASSIFY	TREATMENT (URGENT PRE-REFERRAL TREATMENTS ARE IN BOLD PRINT.)
Weight less than 1500gm	**VERY LOW BIRTH WEIGHT**	• **Test for and treat / prevent low blood sugar** (Pg 63). • **Advise mother to keep the young infant warm on the way to hospital** (Pg 59). • **Check for HIV exposure** (Pg 56)**.** • **Refer urgently to hospital.**
Weight between 1500gm and 2499gm	**LOW BIRTH WEIGHT**	• Manage as for FEEDING PROBLEM OR LOW WEIGHT FOR AGE. • Advise mother to keep the young infant warm (Pg 65). • Check for HIV exposure (Pg 56) and screen for possible TB disease (Pg 21). • Refer to a higher level facility for routine follow-up.
Weight 2500gm or more	**NORMAL BIRTH WEIGHT**	• Advise mother to give home care to the young infant (Pg 66) • Advise the mother to keep infant warm (Pg 59) • Check for HIV exposure (Pg 56) and screen for possible TB disease (Pg 21).

CHECK FOR DEVELOPMENTAL MILESTONES (Pg 22)

CHECK FOR INTERACTION AND / OR COMMUNICATION AND RESPONSIVENESS (Pg 23 - 24)

ASSESS FOR OTHER ILLNESSES (Pg 25)

ASSESS MOTHER'S / CAREGIVER'S NEEDS (Pg 25)

THEN CHECK IMMUNIZATION STATUS

(The schedule below also applies to Preterm babies)

AGE	VACCINES				
BIRTH	BCG*	b-OPV 0			• Give all missed doses on this visit • Immunize sick infants, unless they are being referred • Advise the mother/ caregiver when to return for the next dose
6 WEEKS	PENTAVALENT 1	b-OPV 1	PCV10 - 1	ROTA 1	

* Do not give BCG to a child with symptomatic HIV/AIDS. HIV infected, receiving ART and clinically well and immunologically stable, should be vaccinated. In child exposed to TB disease at birth, do not give BCG, instead give child TPT for 6 months then administer BCG 2 weeks after completion of TPT. No repeat dose of BCG after primary dose is given, even if the scar does not develop.

ROUTINE CARE FOR ALL NEWBORNS AFTER DELIVERY

The routine care described below applies to all newborns, either born in hospital or born outside and brought to the hospital.

- Keep the baby in skin-to-skin contact on the mother's chest or at her side, in a warm, draught-free room.
- Start breastfeeding within the first hour as soon as the baby shows signs of readiness to feed.
- Let the infant breastfeed on demand if able to suck.
- Give IM vitamin K to all newborns.
  - For babies 1500gm and more, give 1mg IM Vitamin K once. For preterm neonates <1500gm, give 0.5 mg IM Vitamin K once.
- Keep umbilical cord clean and dry.
- Apply 7.1% Chlorhexidine Digluconate gel upon cutting the cord. Do this daily (once every 24 hours) for 7 days or until the cord falls off (whichever comes first). For preterm infants, apply once at birth.
- Apply Tetracycline Eye Ointment to both eyes once at birth.
- Give oral polio and BCG vaccines.

KEEP THE YOUNG INFANT WARM

Immediate Kangaroo Mother Care (iKMC) started within 2 hours of birth, is prolonged continuous skin-to-skin contact between a mother or her surrogate and her preterm baby or low birth weight infant. The difference between KMC and skin-to-skin care (which is recommended for all newborns at birth) is in the duration of contact.

Skin-to-skin care is usually done immediately after birth for a short time for every newborn to ensure that all babies stay warm in the first hour of life and helps in early initiation of breast feeding while KMC is prolonged skin to skin contact of at least 8-24 hrs a day with support for exclusive breastfeeding or breast milk feeding.

- All stable babies born below 2500g should be started on iKMC.
- Stable babies between 2000g and 2500g should be evaluated by a health worker, the mother counselled on KMC, the baby initiated and discharged on iKMC.
- All stable babies below 2000g should be admitted into a iKMC unit and started on iKMC:
  - Stable babies weighing between 1800g and 2000g can be started on iKMC soon after birth.
  - Babies weighing between 1200g and 1799g should be stabilized then started on iKMC as soon as possible and referred for assessment.
- Babies weighing less than 1200g should be transferred immediately to a centre that can offer intensive neonatal care. It may take weeks before their condition allows the initiation of iKMC. Depending on the facilities available for the transfer, iKMC can be utilised to keep the baby warm during the transfer to the higher-level facility.

Warm the young infant using skin-to-skin contact (Kangaroo Mother Care)

- Maintain hand and body hygiene.
- Provide privacy to the mother. If mother is not available, skin-to-skin contact can be provided by the father or any other adult.
- Request the mother to sit or recline comfortably.
- Undress the baby gently, except for hat, nappy/diaper and socks.
- Ask the mother to wear light, loose clothing that is comfortable in the ambient temperature, and can accommodate the young infant.
- Place the young infant on the mother's chest prone in an upright and extended posture, between her breasts in skin to skin contact; turn the baby's head to one side to keep airways clear.
- Ensure that the young infant's hips and elbows are flexed into a frog-like position and the head and chest are on the mother's chest, with the head in a slightly extended position.
- Secure the baby on to the mother's chest with a clean lesso, soft cloth or any of the other recommended carrying pouches for KMC babies that are available.
- Wrap the mother-baby duo with an added blanket or shawl. Cover the young infant with mother's clothes.
- After positioning the young infant, allow mother to rest with the young infant.
- Breastfeed the young infant frequently.
- If possible, warm the room (>25ºC) with a heating device.

- **Reassess after 1 hour:**
  - Assess and classify for signs of PSBI / VERY SEVERE DISEASE (Pg 52).
  - Measure axillary temperature by placing the thermometer in the axilla for 5 minutes (or feel for low body temperature):
    - If **any signs of "PSBI / VERY SEVERE DISEASE"**:
      - Refer URGENTLY to hospital after giving pre-referral treatment for " PSBI / VERY SEVERE DISEASE".
    - If **no signs of "PSBI / VERY SEVERE DISEASE"**:
      - Advise how to keep the young infant warm at home.
      - Advise mother to give home care.
    - Advise mother when to return immediately.





TEACH THE MOTHER HOW TO KEEP THE YOUNG INFANT WARM ON THE WAY TO THE HOSPITAL

- Provide skin-to-skin contact **OR**
- Keep the young infant clothed or covered as much as possible all the time. Dress the young infant in 3 - 4 layers of clothes (including hat, gloves, socks) and wrap him/her in a soft dry cloth and cover with a shawl or blanket. Hold the baby close to the mother's/caregiver's body.

TEACH THE CAREGIVER TO TREAT LOCAL INFECTIONS AT HOME

- Explain how the treatment is given.
- Watch her as she does the first treatment in the clinic.
- She should return to the clinic if the infection worsens.
- Check the mother's understanding before she leaves the clinic.

TO TREAT SKIN PUSTULES OR UMBILICAL INFECTION

The mother should give the treatment twice daily for 5 days:

- Wash hands.
- Gently wash off pus and crusts with soap and water.
- Dry the area.
- Paint the skin lesions with Gentian Violet / Mupirocin.
- Wash hands.
- Give oral Flucloxacillin (Pg 60).

TO TREAT THRUSH (WHITE PATCHES IN MOUTH) OR MOUTH ULCERS

The mother should give the treatment 4 times daily for 7 days:

- Wash hands.
- Wash the young infant's mouth with clean soft cloth wrapped around the finger and wet with salt water.
- Instill Nystatin / Gentian Violet.
- Wash hands.
- If breastfed, advise the mother to wash her breast after feeds and apply the same medicine on the areola.

TREAT EYE INFECTION WITH TETRACYCLINE EYE OINTMENT

The mother should treat both eyes 3 times daily for 5 -7 days:

- Wash hands.
- Use clean cloth and water to gently wipe away pus.
- Open the eyes of the young infant.
- Squirt a small amount of tetracycline eye ointment on the inside of the lower lid.
- Wash hands again.
- Then apply tetracycline eye ointment in both eyes 3 times daily.
- Treat until redness is gone.
- Do not use other eye ointments or drops, or put anything else in the eye.

TEACH THE MOTHER/CAREGIVER TO GIVE ORAL DRUGS AT HOME

Follow the instructions below for every oral drug to be given at home. Also follow the instructions listed with each drug's dosage table.

- Determine the appropriate drugs and dosage for the child's age or weight.
- Tell the mother/caregiver the reason for giving the drug to the child.
- Demonstrate how to measure a dose.
- Watch the mother/caregiver practice measuring a dose by themselves.
- Ask the mother/caregiver to give the dose to their child.
- Explain carefully how to give the drug, then label and package the drug.
- If more than one drug will be given, collect, count and package each drug separately.
- Explain that all the oral drug tablets or syrups must be used to complete the course of treatment, even if the child gets better.
- Check the mother's/caregiver's understanding before they leave the clinic.

GIVE ORAL FLUCLOXACILLIN FOR LOCAL BACTERIAL INFECTION

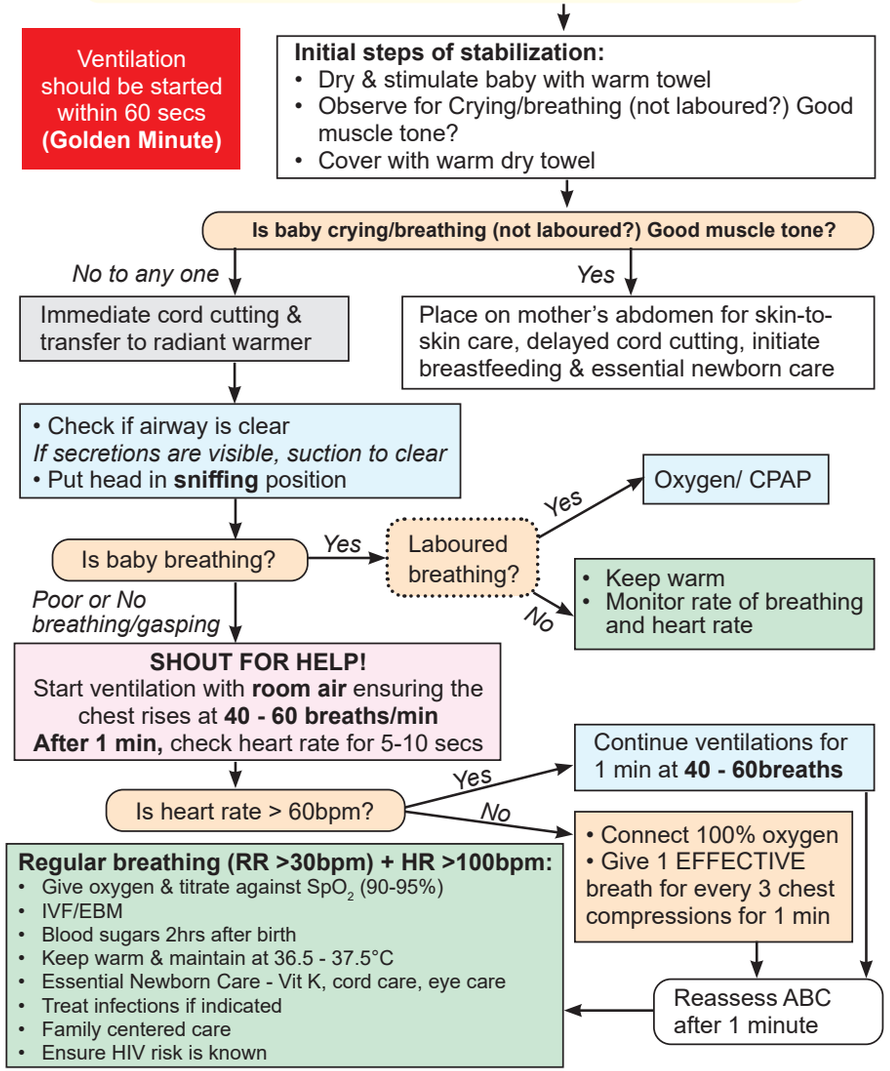
ORAL ANTIBIOTICS AGED <7DAYS. (For young infants ≥ 7days, see the oral antibiotic table Pg 26)	
Oral Antibiotics Aged <7 days 25mg/kg 12 hourly	
Weight (kg)	**Flucloxacillin syrup (125mg/5mls) per dose**
2.0	2.0ml
2.5	3.0ml
3.0	3.0ml
4.0	4.0ml

GIVE THESE TREATMENTS IN A HEALTH FACILITY ONLY

NEONATAL RESUSCITATION

For trained health workers - Anticipate and prepare

Note for all newborns:
- Practice immediate cord clamping and cutting for newborns requiring resuscitation
- For newborns with good heart rate and spontaneous breathing, practice delayed cord clamping and cutting (1-3 mins)



TREATMENT OF POSSIBLE SERIOUS BACTERIAL INFECTION OR VERY SEVERE DISEASE

- Explain to the mother why the drug is being given.
- Determine the dose appropriate for the infant's weight and age.
- Use a sterile needle and sterile syringe to accurately measure the dose.
- Do not mix Benzyl Penicillin with Gentamicin.
- Give the drug as intramuscular injection.
- If the infant cannot be referred, follow instructions provided in the section "**WHERE REFERRAL IS REFUSED OR NOT FEASIBLE**" (Pg 62 - 63).

INTRAVENOUS / INTRAMUSCULAR ANTIBIOTICS FOR NEONATES AGED < 7 DAYS (For young infants ≥ 7days, see the IV/IM antibiotic table on Pg 29)						
Weight (kg)	**Benzyl Penicillin (50,000iu/kg)**	**Ampicillin / Flucloxacillin (50mg/kg)**	**Gentamicin (3mg/kg <2kg, 5mg/kg ≥ 2kg)**	**Ceftriaxone (50mg/kg)**	**Metronidazole (7.5mg/kg)**	**REFERRAL IS THE BEST OPTION FOR A YOUNG INFANT WITH SEVERE DISEASE**
	IV / IM	**IV / IM**	**IV / IM**	**IV / IM**	**IV**	
	12 hrly	**12 hrly**	**24 hrly**	**24 hrly**	**12 hrly**	
1.00	50,000	50	3	50	7.5	
1.25	75,000	60	4	62.5	10	
1.50	75,000	75	5	75	12.5	
1.75	100,000	85	6	75	12.5	
2.00	100,000	100	10	100	15	
2.50	150,000	125	12.5	125	20	
3.00	150,000	150	15	150	22.5	
4.00	200,000	200	20	200	30	

TREAT AT RISK OF CONGENITAL SYPHILIS

- For a child whose mother has syphilis, give baby IM Benzathine Penicillin STAT dose at 50,000 IU/kg.

TREAT DIARRHOEA

- For a sick young infant with diarrhoea, refer to Pg 27 on Zinc Sulphate and Pg 32 - 33 on Rehydration Therapy including ORS.
- On feeding for Diarrhoea refer to Pg 32.

WHERE REFERRAL IS REFUSED OR IS NOT FEASIBLE - FURTHER CLASSIFY THE SICK YOUNG INFANT

FOR A SICK YOUNG INFANT WITH POSSIBLE SERIOUS BACTERIAL INFECTION OR VERY SEVERE DISEASE

CLASSIFY THE SICK YOUNG INFANT

CLASSIFY		IDENTIFY TREATMENT
SIGNS	**CLASSIFY**	**TREATMENT** (URGENT PRE-REFERRAL TREATMENTS ARE IN BOLD PRINT.)
Young infant has any one of the following: • Convulsions or convulsing now • Not able to feed or breastfeed • No movement at all • Weight <2kg	**CRITICAL ILLNESS**	• **Reinforce URGENT referral. Explain to the caregiver that the infant is very sick and needs urgent referral for further evaluation and hospital care.** • **If referral is still refused or not feasible, give once daily IM Gentamicin and twice daily IM Benzyl Penicillin/ Ampicillin for 7 days or until referral is possible** (Pg 63). • **Test for and treat / prevent low blood sugar** (Pg 63). • Teach the mother how to keep the young infant warm at home (Pg 59). • Advise the mother to return immediately if the child becomes worse (Pg 66). • Advise the mother to return for the second injection on day 2. • On day 2 reinforce the need for urgent referral: • If referral is still refused or not feasible, continue giving give once daily IM Gentamicin and twice daily IM Benzyl Penicillin until referral is feasible or for a total of 7 days. • Advise the mother to return for a mandatory review on day 4. • Reassess the young infant at each visit.
Young infant has any one of the following: • Not feeding well on observation • Temperature 37.5°C or more • Temperature less than 35.5°C • Severe lower chest wall indrawing • Movement only when stimulated • Fast breathing (60 breaths per minute or more) in infants less than 7 days old	**CLINICAL SEVERE INFECTION**	• **Give once daily IM Gentamicin for 2 days and oral Amoxicillin dispersible tablet twice a day for 7 days or until referral is possible** (Pg 63). • **Test for and treat / prevent low blood sugar** (Pg 63). • Teach the mother how to keep the young infant warm at home (Pg 59). • Advise the mother to return immediately if the child becomes worse (Pg 66). • Advise the mother to return for the second injection on day 2. • On day 2 reinforce the need for urgent referral: • If referral is still not possible give the second IM Gentamicin and advise to complete oral Amoxicillin for a total of seven days. • Advise the mother to return for a mandatory review on day 4. • Reassess the young infant at each visit.

WHERE REFERRAL IS REFUSED OR NOT FEASIBLE: FOR A SICK YOUNG INFANT WITH POSSIBLE SERIOUS BACTERIAL INFECTION OR VERY SEVERE DISEASE

Give Intramuscular Gentamicin and Benzyl Penicillin / Ampicillin:

- For **CRITICAL ILLNESS**: Give Gentamicin 5 – 7.5 mg/kg/day once daily and Ampicillin 50 mg/kg twice daily till referral is possible or for 7 days. For Benzyl Penicillin doses see Pg 61.
- For **CLINICAL SEVERE INFECTION**: Give Gentamicin 5 – 7.5 mg/kg/day once daily for 7 days.

WEIGHT	GENTAMICIN (Strength 40 mg/ml)	GENTAMICIN (Strength 20 mg/ml)	AMPICILLIN To a vial of 250 mg, add 1.3 ml sterile water (Strength 250 mg/1.5 ml)
	Volume per dose	Volume per dose	Volume per dose
1.5 - 2.4 kg	0.2 ml	0.4 ml	0.8 ml
2.5 - 3.9 kg	0.4 ml	0.8 ml	1.2 ml
4.0 - 5.9 kg	0.6 ml	1.2 ml	1.5 ml

Give Oral Amoxicillin:

- For **CLINICAL SEVERE INFECTION**
- For **PNEUMONIA** (fast breathing alone in infants less than 7 days)

WEIGHT	AMOXICILLIN 75 to 100 mg/kg/day divided into 2 doses. Give twice daily for 7 days		
	Dispersible tablet (250 mg) per dose	Dispersible tablet (125 mg) per dose	Syrup (125 mg in 5 ml) per dose
1.5 - 2.4 kg	1/2 tablet	1 tablet	5 ml
2.5 - 3.9 kg	1/2 tablet	1 tablet	5 ml
4.0 - 5.9 kg	1 tablet	2 tablets	10 ml

TREAT OR PREVENT LOW BLOOD SUGAR IN THE YOUNG INFANT

Normal blood sugar target: 3 – 72 hours of life: ≥ 2.6 mmol/L ; > 72 hours of life: ≥ 3.3 mmol/L

TO PREVENT LOW BLOOD SUGAR IN YOUNG INFANT (HYPOGLYCAEMIA)

- If the child is able to swallow:
  - If breastfed: ask the mother to breastfeed the child.
- If the baby is too sick to feed, give 10ml/kg before referral and 3ml/kg per hour of expressed breastmilk via cup on the way to hospital. Give milk via nasogastric tube if baby cannot swallow.
  - If expressed breastmilk cannot be given, give 10 ml/kg body weight of sugar water **(To make sugar water: Dissolve 4 level teaspoons of sugar (20 g) in a 200-mL cup of clean water).**
- If feeding is contraindicated:
  - Fix an intravenous (IV) line and give 10% Dextrose by slow IV infusion at 3ml/kg/hour **(If 10% Dextrose is not available, add 1 part 50% Dextrose water to 4 parts water for injection to make 10% solution).**

TREAT FOR LOW BLOOD SUGAR (HYPOGLYCAEMIA)

- Suspect low blood sugar in any infant that is convulsing, unconscious or lethargic, or has a low body temperature.
- Test to confirm low blood sugar.
- Keep the baby warm at all times.

Low blood sugar 1.8 to less than < 2.6 mmol/L in a young infant

- Breastfeed or feed expressed breastmilk.
  - If breastfeeding is not possible, give 10ml/kg of replacement milk feed.
- Repeat the blood glucose in 15 minutes while awaiting transport to hospital.
- If the blood sugar remains low, treat for severe hypoglycaemia (see below).
- If the blood glucose is normal, give expressed breast milk and check the blood glucose 2-3 hourly.

Low blood sugar < 1.8 mmol/L in a young infant (severe hypoglycaemia)

- Give a mini-bolus of 10% Dextrose at 2ml/kg.
  - Then continue with the 10% Dextrose infusion at 3ml/kg/hour.
- Repeat the blood glucose in 15 minutes.
- If still low, repeat the bolus of 2ml/kg of 10% Dextrose and continue IV infusion.
- Refer URGENTLY and continue feeds during transfer.

TREAT FOR NEONATAL CONVULSIONS

Note: If young infant is >1 month, use the algorithm on page 31.

Neonatal Convulsions

In the absence of clinical seizures, neonates with hypoxic-ischemic encephalopathy need not to be given prophylactic treatment with phenobarbitone

Any clinically apparent seizure > 3min OR brief serial seizures

- **Check ABC:**
  A Lateral position decubitus, suction if needed.
  B Start on oxygen via NRM.
  C Check for temperature gradient, severe pallor.
  D Treat both fit and hypoglycemia.
  Consider glucose and calcium.
- Phenobarbitone 1M 20mg/kg (loading dose)

Re-assess after 1 hour

Seizure stopped/decreased significantly?

No → Repeat phenobarbitone 10mg/kg (miniload) after 1 hour if needed. Maximum total loading of phenobarbitone 40mg/kg.

Re-assess after 1 hour

Seizure stopped/decreased significantly?

Yes → IM/PO Phenobarbitone 5mg/kg 24hly (maintenance dose **AFTER** at least 12hrs of the loading dose).

No

Is Levetiracetam available?

Yes → Levetiracetam 30mg/kg IV **infusion** over 15mins (loading dose)

Re-assess after 2 hours

Seizure stopped/decreased significantly?

Yes → Levetiracetam 30mg/kg/day divided into 2 doses (maintenance dose)

Start 12 hrs **after** the loading dose.

No → Consult a neurologist. May consider phenytoin /levetiracetam if not previously used.

No → Phenytoin 15mg/kg IV **infusion** over 15mins at 1mg/kg/min (loading dose)

Re-assess after 1 hour

Seizure stopped/decreased significantly?

Yes → PO Phenytoin 5mg/kg/day divided in two doses (maintenance dose).

No → Consult a neurologist. May consider phenytoin /levetiracetam if not previously used.

WHEN TO STOP ANTICONVULSANTS:

1. In neonates with neurological examination and/or normal electroencephalography, consider stopping antiepileptic drugs if neonate has been seizures-free for more than 72hrs; the drug(s) should be reinstituted in case of reoccurence of seizures.
2. In neonates in whom seizure control is achieved with a single antiepileptic drug, the drug can be discontinued abruptly without tapering of the doses.
3. In neonates requiring more than one antiepileptic drug for seizure control, the drugs may be stopped one by one, with phenobarbital being the last drug to be withdrawn.

EARLY INFANT DIAGNOSIS OF HIV IN INFANTS AND CHILDREN

EARLY INFANT DIAGNOSIS OF HIV IN INFANTS AND CHILDREN

Establish HIV Exposure of Infants and Children < 18 months
- Mother known HIV positive
- HTS for mothers with unknown status
- Rapid antibody test on infant/child if mother's HIV status cannot be established

HTS Settings: At L&D, MNCH/FP, IPD, A&E, CCC & Community

Establish HIV infection for HIV Exposed infants at birth or at first contact (first two weeks of birth):
- Collect a sample for HIV DNA PCR test
- Categorise the infant (High/low risk)
- Start infant ARV prophylaxis immediately after birth

If Mother/Infant HIV antibody test negative
Child is HIV unexposed
- Routine under-5 care for the well baby
- Repeat HIV testing for mother as per HTS recommendations

HIV DNA PCR test POSITIVE
Child presumed HIV infected
- Initiate ART immediately
- Book for follow up after 2 weeks
- Offer comprehensive care for HIV infected child
- Collect new sample for confirmatory HIV DNA PCR, baseline viral load and CD4%

HIV DNA PCR Negative
- Continue infant ARV prophylaxis and follow up for HIV exposed infants

Conduct DNA PCR at 6 weeks of age or soonest contact thereafter (or earlier if child develops symptoms suggestive of HIV as per WHO staging criteria)
- Start Cotrimoxazole Prophylaxis Therapy

Confirmatory HIV DNA PCR test POSITIVE

Child confirmed HIV infected
- Continue ART, and comprehensive care and routine under-5 care

Confirmatory HIV DNA PCR test NEGATIVE
Child presumed HIV infected
- Continue ART
- **Collect and send a DBS to NHRL and manage as per the results from NHRL**

HIV DNA PCR result NEGATIVE
Child HIV-exposed
- Continue HEI follow-up
- **Conduct DNA PCR at 6 months of age** or soonest contact (or earlier if child develops symptoms suggestive of HIV as per WHO staging criteria)

HIV DNA PCR result NEGATIVE
Child HIV-exposed
- Continue HEI follow-up
- Continue routine under-5 care
- **Conduct HIV Antibody test at 18 months of age**
- If breastfeeding, do HIV antibody test every 6 months while breastfeeding and 6 weeks after complete cessation of breastfeeding

HIV DNA PCR result NEGATIVE
Child HIV-exposed
- Continue HEI follow-up
- **Conduct DNA PCR at 12 months of age** or soonest contact till 18 months of age (or earlier if child develops symptoms suggestive of HIV as per WHO staging criteria)

If HIV antibody test is POSITIVE
Child confirmed HIV infected
- Start ART and comprehensive care
- Continue CPT
- Continue routine under-5 care

If HIV antibody test is NEGATIVE
Child HIV negative
- Stop CPT*
- Review at age 2 years and document vital status
- Continue routine under-5 care

NOTE:
*Only after final antibody test negative
1. Where Point of Care DNA PCR is available – EID should be done using whole blood at the facility
2. For baseline viral load testing – if available, use point of care machine for viral load; If there is no point of care machine to do viral load, take a plasma sample and send it to the VL testing laboratory

TEACH CORRECT POSITIONING AND ATTACHMENT FOR BREASTFEEDING

Show the mother how to hold her infant:

- With the infant's head and body straight
- Facing her breast, with infant's nose opposite her nipple
- With infant's body close to her body
- Supporting infant's whole body, not just neck and shoulders.

Show her how to help the infant to attach. She should:

- Touch her infant's lips with her nipple
- Wait until her infant's mouth is opening wide
- Move her infant quickly onto her breast, aiming the infant's lower lip well below the nipple.

Look for signs of good attachment and effective suckling.

- If the attachment or suckling is not good, try again.
- If still not suckling effectively, ask the mother to express breast milk and feed with a cup in the clinic.
    - If **able to feed with a cup and spoon**, advise mother to continue breastfeeding the infant and at the end of each feed, express breast milk and feed with a cup.
    - If **not able to feed with a cup**, refer to hospital.

TEACH THE MOTHER TO TREAT BREAST OR NIPPLE PROBLEMS

- If the **nipple is flat or inverted**, evert the nipple several times before each feed and put the baby on the breast.
- If the **nipple is sore**, apply breast milk for soothing effect and ensure correct positioning and attachment of the baby.
    - If mother continues discomfort, feed expressed breast milk with cup.
- If breasts **are engorged**, let the baby continue to suck if possible. If the baby cannot suckle effectively, help the mother to express some milk and then put the young infant to the breast. Putting warm compress on the breast may help.
- If **breasts have an abscess**, advice the mother to feed from the other breast and refer. If the young infant wants more milk give appropriate formula.

TEACH THE MOTHER HOW TO KEEP THE YOUNG INFANT WITH LOW WEIGHT FOR AGE OR LOW BIRTH WEIGHT WARM AT HOME

- Do not bathe the young infant with low weight for age or low body temperature; instead sponge him/her in a warm room with lukewarm water.
- Provide skin-to-skin contact (Kangaroo mother care) as much as possible, day and night.
    - When not in skin-to-skin contact or if this is not possible:
        - Keep the room warm.
        - Clothe the young infant in 3 - 4 layers of warm clothes, cover the head with a hat (include gloves and socks) and wrap him / her in a soft dry cloth and cover with a warm blanket or shawl.
        - Let the baby and mother lie together on a soft, thick bedding.
        - Change clothes including diaper/ napkins whenever they are wet.
        - Breastfeed the infant (or give expressed breast milk by cup) frequently.
- Feel the feet of the baby periodically—baby's feet should be always warm to touch.

TEACH THE MOTHER ON UMBILICAL CORD CARE

- Wash hands with soap and running water.
- Open the container of 7.1% Chlorhexidine Digluconate gel.
- Apply the 7.1% Chlorhexidine Digluconate gel to the base of the umbilical cord, cord stump, and tip of the cord. Ensure the entire cord is covered with 7.1% Chlorhexidine Digluconate.
- Spread the gel using your index finger.
- Wash hands after application.
- Do not clean off the 7.1% Chlorhexidine Digluconate from the umbilicus after application. Do not wrap or bind the umbilical area after 7.1% Chlorhexidine Digluconate application. Do not apply anything else.
- Continue application once daily up to the seventh day or until the umbilical cord falls, whichever happens first.

Counsel the HIV positive mother on feeding recommendations, about feeding problems and guidance on infant feeding including in HIV context (Pg 42 - 43).

ADVISE MOTHER TO GIVE HOME CARE FOR THE YOUNG INFANT

Teach the mother on:

- General hygiene
- Exclusive breastfeeding for 6 months
  - Breastfeed frequently, as often and for as long as the infant wants, day or night, during sickness and health.
  - If the young infant is not being fed as described on page 65, counsel the mother accordingly.
- Cord care
- When to return
- Keeping the young infant warm at all times.
  - In cool weather, cover the infant's head and feet and dress the infant with extra clothing.

- **Remember to counsel the mother on:**
  - Immunization of Every Sick Young Infant, as needed.
  - Management of Diarrhoea (Pg 61).
  - Care for development problems (Pg 44 - 45).
  - Her own health (Pg 46).

WHEN TO RETURN FOR A FOLLOW-UP VISIT

If the infant has:	Return for follow-up in:
• Jaundice	1 day
• Local bacteria infection • Eye infection • Some dehydration • No dehydration • Feeding problem • Thrush or mouth ulcers	2 days
• Pneumonia	3 days
• Low weight for age or low birth weight • Risk of tuberculosis (after completion of TPT) • Poor interaction and/or communication and responsiveness	14 days
• Developmental milestone alert	30 days
• Presumed HIV infection • HIV exposed	Monthly

WHEN TO RETURN IMMEDIATELY

- Breastfeeding or drinking poorly
- Becomes sicker
- Develops a fever
- Fast breathing
- Difficulty breathing
- Yellow palms and soles if infant has jaundice
- Blood in stool
- Feels unusually cold
- Reduced activity



GIVE FOLLOW-UP CARE FOR THE SICK YOUNG INFANT

ASSESS EVERY YOUNG INFANT FOR VERY SEVERE DISEASE DURING FOLLOW UP VISIT.

WHEN REFERRAL WAS REFUSED OR NOT FEASIBLE: CRITICAL ILLNESSS

- At each contact for injection of antibiotics:
  - Explain again to the caregiver that the infant is very sick and should urgently be referred for hospital care.
  - Reassess the young infant (Pg 52).
  - Treat any new problem.
  - If referral is still not feasible, continue giving once-daily intramuscular Gentamicin and twice-daily intramuscular Benzyl Penicillin until referral is feasible or for a total of 7 days.

WHERE REFERRAL WAS REFUSED OR NOT FEASIBLE: CLINICAL SEVERE INFECTION

- Follow up at the next contact for injection (day 2) and on day 4 of treatment.
- At each contact for injection, reassess the young infant (Pg 52).
  - If **the young infant is improving,** complete the 2 days treatment with IM gentamicin.
    - Ask the mother to continue giving the oral amoxicillin twice daily until completion of the 7-day treatment.
    - Ask the mother to bring back the young infant on day 4 of treatment (3 days after initial visit).
  - **Refer young infant if any of the following:**
    - Infant becomes worse after treatment is started (Shows any sign of CRITICAL ILLNESS)
    - Any new sign of CLINICAL SEVERE INFECTION appears while on treatment
    - If no improvement on day 4 after 3 full days of treatment.
    - Any sign of CLINICAL SEVERE INFECTION is still present after day 7 of treatment.

PNEUMONIA

- **After 2 days**
  - Reassess the young infant for POSSIBLE SERIOUS BACTERIAL INFECTION or PNEUMONIA (Pg 52).

  - **Ask:**
    - Is the infant breathing slower?
    - Is the infant breastfeeding better?

  - **Treatment**
    - **Refer urgently to hospital if:**
      - The infant becomes worse or
      - Any new sign of POSSIBLE SERIOUS BACTERIAL INFECTION or VERY SEVERE DISEASE appears while on treatment.
    - If the **young infant is improving**, ask the mother to continue giving the oral amoxicillin twice daily until all the tablets are finished.

LOCAL BACTERIAL INFECTION

- **After 2 days:**
  - Look at the umbilicus. Is it red or draining pus?
  - Does redness extend to the skin?
  - Look at the skin pustules. Are there many or severe pustules?
  - Look at the eyes. Is there pus? Is there associated swelling or redness of the eyelids?

  - **Treatment:**
    - If **pus or redness remains or is worse**, refer to hospital.
    - If **pus and redness are improved,** tell the mother to continue giving the 7 days of antibiotic and continue treating the local infection at home.
    - If there are **many or severe pustules**, refer to hospital.
    - If **skin pustules are improved**, tell the mother to continue giving the 7 days of antibiotic and continue treating the local infection at home.

EYE INFECTION

- **After 2 days:**
  - Look for pus draining from the eyes.

- **Treatment:**
  - If **pus is still draining from the eyes**, ask the mother to describe how she has treated the eye infection.
    - If treatment has been correct, refer to hospital.
    - If treatment has not been correct, teach the mother correct treatment and advise her to give treatment for 5 days.
    - Follow up in 2 days.
  - If **no pus**, ask the mother to continue treatment for 5 days.

DIARRHOEA: SOME DEHYDRATION OR NO DEHYDRATION

- **After 2 days:**
  - **Ask:** Has the diarrhoea stopped?

  - **Treatment:**
    - If the **diarrhoea has not stopped**, Assess and Treat the young infant for DIARRHOEA.
    - If the **diarrhoea has stopped**, tell the mother to continue exclusive breast feeding. Advise her to continue giving zinc to complete 10 days.

JAUNDICE

- **After 1 day:**
  - Look for jaundice.
    - Are palms and soles yellow?

  - If **palms and soles are yellow or age 14 days or more**, refer to hospital.
  - If **palms and soles are not yellow and age is less than 14 days, and infant feeding well**, advise on home care and breastfeeding.
  - If y**ou are concerned about the jaundice,** ask he mother to return after 1-2 days or if jaundice worsens.

FEEDING PROBLEM

- **After 2 days:**
  - **Reassess feeding** (Pg 57).
    - Ask about any feeding problems found on the initial visit.
      - Counsel the mother about any new or continuing feeding problems.
      - If you **counsel the mother to make significant changes in feeding**, ask her to bring the young infant back after 2 days.
      - If the **young infant is low weight for age,** ask the mother to return 14 days after the initial visit to measure the young infant's weight gain.

EXCEPTION:
If you do not think that feeding will improve, or if the young infant has lost weight, refer the child.

LOW WEIGHT FOR AGE

- **After 14 days:**
  - Weigh the young infant and determine if the infant is still low weight for age (Pg 73 - 75).
  - Reassess feeding (Pg 57).
    - If the infant is **no longer low weight for age**, praise the mother and encourage her to continue.
    - If the infant is **still low weight for age, but is feeding well**, praise the mother. Ask her to have her infant weighed again within a month or when she returns for immunization.
    - If the infant is **still low weight for age and still has a feeding problem**, counsel the mother about the feeding problem.
      - Ask the mother to return again in 14 days (or when she returns for immunization, if this is within 2 weeks).
      - Continue to see the young infant every 2 weeks until the infant is feeding well and gaining weight regularly or is no longer low weight for age.

EXCEPTION:
If you do not think that feeding will improve, or if the young infant has lost weight, refer to hospital.

THRUSH OR MOUTH ULCERS

- **After 2 days:**
  - Look for ulcers or white patches in the mouth (thrush). Reassess feeding (Pg 57).
    - If **thrush or mouth ulcers are worse or the infant has problems with attachment or sucking,** refer to hospital.
    - If t**hrush or mouth ulcers are the same or better, and the baby is feeding well,** continue with Gentian Violet or Nystatin for a total of 5 days.

MANAGEMENT OF THE SICK CHILD AGE 2 MONTHS UP TO 5 YEARS

Date of Visit: ..... (DD/MM/YYYY) Facility Name: ..... Caregiver's Name: ..... Caregiver's Tel: .....

Name of Child: ..... Sex: F☐ /M☐, Age: ..... Months, Weight: ..... Kgs, MUAC: ..... cm, Height/Length: ..... cm

Temperature: .....°C, SpO2: .....%, Pulse Rate: .....beats/minute

Ask: What are the child's problems?..... Initial Visit? ☐ Follow up Visit? ☐ Residence: .....

ASSESS THE CHILD (TICK YES IF SIGN IS PRESENT, TICK NO IF SIGN IS ABSENT AND FILL AS APPROPRIATE)	**CLASSIFY**
CHECK FOR GENERAL DANGER SIGNS: UNABLE TO DRINK OR BREASTFEED?: Yes☐ / No☐, VOMITS EVERYTHING?: Yes☐ / No☐, HISTORY OF CONVULSIONS IN CURRENT ILLNESS?: Yes☐ / No☐, CONVULSING NOW?: Yes☐ / No☐, LETHARGIC OR UNCONSCIOUS?: Yes☐ / No☐,	General danger signs present? Yes☐ / No☐,
DOES THE CHILD HAVE A COUGH OR DIFFICULTY BREATHING? Yes☐ / No☐, For how long? ..... days, Count the breaths in one minute: ..... breaths/minute, Is it Fast breathing?: Yes☐ / No☐, Stridor? Yes☐ / No☐, Central Cyanosis? Yes☐ / No☐, Grunting? Yes☐ /No☐, Lower chest wall indrawing? Yes☐ / No☐, Wheeze? Yes☐ / No☐	
DOES THE CHILD HAVE DIARRHOEA? Yes☐ / No☐, For how long? ..... days, Is there blood in the stools? Yes☐ / No☐. **Look for signs of shock**: AVPU <A? Yes☐ / No☐, Weak /Absent Pulse? Yes☐ / No☐, Cold hands+ Temp gradient :Yes☐ / No☐, Capillary refill 3 secs or more? Yes☐ / No☐ **Is the child:** Lethargic or unconscious (AVPU <A)?: Yes☐ / No☐, Restless or irritable? Yes☐ / No☐, Unable to drink or drinking poorly?: Yes☐ / No☐, Drinking eagerly or thirstily? Yes☐ / No☐, Does the child have sunken eyes? Yes☐ / No☐, **Does the skin pinch go back:** Very slowly (≥2 secs)? ☐/ slowly (1 -2secs)? ☐ / immediately? ☐	
DOES THE CHILD HAVE FEVER? (*By history/feels hot/axillary temperature is ≥37.5°C*) Yes☐ / No☐, If yes, for how long? ..... days, Stiff neck? Yes☐ / No☐ Has the child visited a high malaria burden area in the last 1 month? Yes☐ / No☐, Decide Malaria burden: High Malaria risk? Yes☐ / No☐, Low Malaria risk? Yes☐ / No☐, Malaria Test Results: Positive ☐ Negative ☐ **Check for signs of Measles:** Generalized rash? Yes☐ / No☐, if yes look for: Runny nose? Yes☐ / No☐, A cough?: Yes☐ / No☐, Red eyes? Yes☐ / No☐ **If the child has signs of measles now or within the last 3 months? Check for the following complications:** Mouth ulcers? Yes☐ / No☐, Check if deep or extensive: Yes☐ / No☐, Pus draining from the eye? Yes☐ / No☐, Clouding of cornea? Yes☐ / No☐	
CHECK FOR AN EAR PROBLEM? Is there pain? Yes☐ / No☐, is the child rubbing/pulling the ear? Yes☐ / No☐, Is there discharge? Yes☐ / No☐ If yes, for how long ..... days, Is pus draining from ear? Yes☐ / No☐, Tender swelling behind the ear? Yes☐ / No☐	
CHECK INFANTS UNDER 6 MONTHS FOR RISK OF POOR GROWTH AND DEVELOPMENT Does the child have any IMNCI danger sign? Yes☐ / No☐, Presence of bilateral pitting edema? Yes☐ / No☐, Fast breathing? Yes☐ / No☐, Severe chest indrawing? Yes☐ / No☐, Movement only when stimulated or no movement at all? Yes☐ / No☐, Recent weight loss? Yes☐ / No☐ Was the infant born preterm? Yes☐ / No☐, Temperature? _____°C, MUAC _____ cm, WLZ score _____, WAZ score_____, Low Birth weight (less than 2500g)? Yes☐ / No☐, Has neonate regained birthweight by two weeks of age? Yes☐ / No☐, Failure to gain weight? Yes☐ / No☐ Any difficulties in feeding? Yes☐ / No☐, Receives other foods or drinks other than breastmilk? Yes☐ / No☐, Mother has feeding concern or breast problem? Yes☐ / No☐, Mother has illness requiring further assessment? Yes☐ / No☐, Mother indicates she has had difficulties taking care of her infant or herself recently? Yes☐ / No☐, Mother's MUAC <23cm? Yes☐ / No☐	
CHECK FOR ACUTE MALNUTRITION (6-59 MONTHS): **Was the baby born preterm?** Yes☐ / No☐, **Does the child have:** Oedema on both feet? Yes☐ / No☐, Severe wasting? Yes☐ / No☐ Any acute medical problem as per IMNCI severe classification? Yes ☐ / No☐, **Check weight:** Weight for age: Very low ☐/ Not very low☐, Is there growth faltering (For 2 consecutive months)? Yes☐ / No☐, Determine MUAC: ..... cm, Determine Z Score for WFH/L: ..... If z score is ≤ -3 or MUAC ≤11.5 cm, perform appetite test. Perform appetite test: Pass the appetite test? Yes☐ / No☐,	
CHECK FOR ANAEMIA: Does the child have: Severe palmar pallor? Yes☐ / No☐, Some palmar pallor? Yes☐ / No☐, No Palmar pallor? Yes☐ / No☐, (If any pallor, check HB___g/l)	
CHECK FOR HIV EXPOSURE AND INFECTION: HIV exposed? Yes☐ / No☐, Antibody test (≥18mths): Pos ☐ / Neg ☐, DNA PCR Test (<18mths): Pos ☐ /Neg ☐, TB Contact? Yes☐ / No☐, **If child is <18months with a positive HIV antibody test, check for:** Severe Pneumonia? Yes☐ / No☐, Severe sepsis? Yes☐ / No☐, Oral Candidiasis / thrush? Yes☐ / No☐, Other AIDS defining conditions ? Yes☐ / No☐, Recent maternal death?: Yes☐ / No☐,	
CHECK FOR TUBERCULOSIS Cough more than 2 weeks? Yes☐ / No☐, Persistent fever for more than 2weeks? Yes☐ / No☐, Night sweats more than 2weeks? Yes☐ / No☐, Weight loss /poor weight gain /failure to thrive? Yes☐ / No☐, Fatigue/lethargy/reduced playfulness more ≥2 weeks? Yes☐ / No☐, Positive TB contact? Yes☐ / No☐	

CHECK FOR DEVELOPMENTAL MILESTONES

Domain	Age-specific milestones (tick if achieved)			Milestones present or absent
Gross Motor	Holds head steady (2-4 mo) ☐ Rolls over (4-6 mo) ☐ Sits without support (6 mo) ☐	Stands with support (9 mo) ☐ Walks alone (12–15 mo) ☐	Runs / climbs (2 yrs) ☐ Jumps with both feet (3 yrs) ☐	
Fine Motor	Grasps object (3–4 mo) ☐ Transfers hand-to-hand (6 mo) ☐	Pincer grasp (9 mo) ☐ Scribbles (18 mo) ☐	Builds tower of 6 blocks (2 yrs) ☐ Copies circle (3 yrs) ☐	
Language/ Communication	Coos (2 mo) ☐ Babbles (6 mo) ☐	Says 1–2 words (12 mo) ☐ Combines 2 words (2 yrs) ☐	Speaks in sentences (3 yrs) ☐	
Social/Adaptive	Social smile (6–8 wks) ☐ Responds to name (6–9 mo) ☐	Waves "bye-bye" (9–12 mo) ☐ Plays interactive games (2 yrs) ☐	Toilet trained (3 yrs) ☐	

CHECK FOR INTERACTION, COMMUNICATION AND RESPONSIVENESS

Caregiver Action (Skills)	Yes	No
Does the caregiver move child's arms or legs or gently strokes the child?		
Does the caregiver get child's attention with shaker toy or other objects?		
Does the caregiver talk to child?		
Does the caregiver respond to child's sounds/gestures to get child to smile?		
Does the caregiver play with child?		

Classify for milestones and interaction

CHECK THE CHILD'S IMMUNIZATION STATUS: *Tick immunizations that have been given. Circle immunizations needed today*

At Birth	6 weeks	10 weeks	14 weeks	6 months	7 months	9 months	18 months	24 months
BCG ☐	bOPV 1 ☐	bOPV 2 ☐	bOPV 3 ☐	Measles, rubella* ☐	**Malaria vaccine ☐	Measles, rubella ☐	Measles, rubella ☐	**Malaria vaccine ☐
bOPV 0 ☐ (birth –2wks)	Penta 1 ☐	Penta 2 ☐	Penta 3 ☐	**Malaria vaccine ☐		**Malaria vaccine ☐		
	PCV10 1 ☐	PCV10 2 ☐	PCV 10 3 ☐			TCV ☐		
	ROTA 1 ☐	ROTA 2 ☐	ROTA 3 ☐			Yellow Fever*** ☐		
			IPV1 ☐			IPV2 ☐		

Vitamin A + Deworming

Tick doses that have been given. Circle doses needed today

6 mo: Vitamin A ☐	**36 mo:** Vitamin A ☐ + Deworming ☐
12 mo: Vitamin A ☐ + Deworming ☐	**42 mo:** Vitamin A ☐ + Deworming ☐
18 mo: Vitamin A ☐ + Deworming ☐	**48 mo:** Vitamin A ☐ + Deworming ☐
24 mo: Vitamin A ☐ + Deworming ☐	**54 mo:** Vitamin A ☐ + Deworming ☐
30 mo: Vitamin A ☐ + Deworming ☐	**60 mo:** Vitamin A ☐ + Deworming ☐

Indicate next return date (DD/MM/YYYY):

Immunization: ..... Vitamin A: .....

Growth monitoring: ..... Deworming: .....

**Measles Rubella vaccine at 6 months for HIV exposed/infected and immunocompromised children. Repeat at 9 and 18mos as per the immunization schedule. ** Malaria Vaccine is only offered in Busia, Bungoma, Kakamega, Vihiga, Kisumu, Siaya, Migori and Homabay.*
****Yellow fever is only offered in the following counties: Turkana, West Pokot, Baringo and Elgeyo Marakwet.*

ASSESS FOR OTHER CHILDHOOD ILLNESSESS:	Any other problems?	**ASSESS MOTHER'S/CAREGIVERS NEEDS:**	**Advise when to return:** *DD/MM/YYYY*

TREATMENT

Indicate treatment given for each classification, advice given and return date. Admit or refer any child who has a danger sign and no other severe classification

Feeding advice:

Indicate next return date: Development milestones ..... Stimulation: .....
DD/MM/YYYY *DD/MM/YYYY*

Give any immunizations or Vitamin A needed

Advise Caregiver when to return immediately. Return for follow-up in ..... days.

MANAGEMENT OF THE SICK YOUNG INFANT AGE UP TO 2 MONTHS

Date of Visit: ..... *DD/MM/YYYY* ..... Facility Name: ..... Caregiver's Name: ..... Caregiver's Tel: .....

Name of Young infant: ..... Sex: F ☐ /M☐, Age: ....., Birth Weight: ..... Kgs, Current Weight: ..... Kgs, Gestation at Birth .....wks,

Length: ..... cm, Temperature: .....°C, SpO_2: .....%, Pulse Rate: .....

Ask: What are the infant's problems?..... Initial Visit? ☐, Follow up Visit? ☐, Residence: .....

ASSESS THE INFANT (TICK YES IF SIGN IS PRESENT, TICK NO IF SIGN IS ABSENT AND FILL AS APPROPRIATE)	**CLASSIFY**
CHECK FOR POSSIBLE SERIOUS BACTERIAL INFECTION OR VERY SEVERE DISEASE, PNEUMONIA, LOCAL BACTERIAL INFECTION Has the infant had convulsions? Yes☐ / No☐, Is the young infant able to feed or breastfeed? Yes☐ / No☐, Breath counts per minute? ..... B/Min Fast breathing (60 breaths or more per minute)? Yes☐ / No☐, Movement only when stimulated or no movement at all? Yes☐ / No☐, Severe lower chest wall indrawing? Yes☐ / No☐, Fever (axillary temp ≥38°C)? Yes☐ / No☐, Low body temperature(<35.5°C)? Yes☐ / No☐, Umbilicus red? Yes☐ / No☐, Umbilicus draining pus? Yes☐ / No☐, Skin pustules? Yes☐ / No☐	
CHECK FOR JAUNDICE: Does the infant have yellow discolouration of the skin? Yes☐ / No☐, If yes, for how many days? ..... days, Are the infant's eyes yellow? Yes☐ / No☐ Are the infant's palms yellow? Yes☐ / No☐, Are the infant's soles yellow? Yes☐ / No☐	
CHECK FOR EYE CONDITION: Is there eye discharge? Yes☐ / No☐, If yes, for how long?..... days, Eyes draining pus? Yes☐ / No☐, Eyes swollen? Yes☐ / No☐ Excessive tearing? Yes☐ / No☐, White spot on the pupil? Yes☐ / No☐, Crossed eyes? Yes☐ / No☐, Red eyes? Yes☐ / No☐, Clouding of the cornea and no signs of Measles? Yes☐ / No☐, Fear of light? Yes☐ / No☐	
ASSESS FOR DIARRHOEA: Does the infant have diarrhoea?Yes☐ / No☐, Is the infant: Restless and irritable? Yes☐ / No☐, Is the infant moving only when stimulated or has no movement at all? Yes☐ / No☐, Does the infant have sunken eyes? Yes☐ / No☐, Does the skin pinch go back: Very slowly (>2 secs)? ☐ / slowly (1 -2secs)? ☐ / Immediately? ☐	
CHECK FOR SPECIAL TREATMENT NEEDS: **IF INFANT LESS THAN 72 HRS: Has the mother had any of the following during labour or within 24 hours before delivery:** Fever ≥38°C? Yes☐ / No☐, Infection treated with antibiotics? Yes☐ / No☐ Untreated maternal Infections? Yes☐ / No☐, Membranes ruptured >18 hours before delivery? Yes☐ / No☐, Foul smelling amniotic fluid? Yes☐ / No☐ **FOR ALL YOUNG INFANTS:** Mother tested VDRL positive? Yes☐ / No☐, Mother tested positive for HBsAg?: Yes☐ / No☐, Has the mother/caregiver/close household contact received TB treatment in the last 2 months? Yes☐ / No☐, Is the infant receiving other foods or drinks? Yes☐ / No☐,	
CHECK FOR HIV EXPOSURE & INFECTION: Is the infant HIV Exposed? Yes☐ / No☐ Unknown ☐, DNA PCR Test (<18mths): Pos☐ Neg ☐ Unknown ☐ ***(Check HIV status for all unknown)*** Mother tested HIV positive?: Yes☐ / No☐, if yes, has the mother been on ARVs? Yes☐ / No☐ **If child is <18months with a positive HIV antibody test, check for:** Severe Pneumonia? Yes☐ / No☐, Severe sepsis? Yes☐ / No☐, Oral Candidiasis / thrush? Yes☐ / No☐, Other AIDS defining conditions ? Yes☐ / No☐, Recent maternal death?: Yes☐ / No☐,	
CHECK FOR TB EXPOSURE & INFECTION: Ask for history of mother/caregiver or close household contact started TB treatment <2months before delivery of the baby: Yes☐ / No☐ **Does the infant have 2 or more:** Cough? Yes☐ / No☐, Persistent fever? Yes☐ / No☐, Night sweats? Yes☐ / No☐, Weight loss /poor weight gain /failure to thrive? Yes☐ / No☐, Fatigue/lethargy/reduced playfulness? Yes☐ / No☐	
CHECK FOR FEEDING PROBLEMS, LOW WEIGHT FOR AGE OR LOW BIRTH WEIGHT: Determine weight for age Z score ....., MUAC (for infants ≥6 weeks) ..... cm, Receives other foods or drinks? Yes☐ / No☐, Mouth ulcers? Yes☐ / No☐, Thrush (white patches in mouth)? Yes☐ / No☐, Breastfeeds less than 8 times in 24 hours? Yes☐ / No☐ ***If the infant has no indications for urgent referral to hospital from previous assessment: Proceed with breastfeeding assessment.***	
ASSESS BREASTFEEDING: *Ask mother to put infant to the breast. Observe the breastfeed for 4 minutes* **Infant Positioning:** Is infant's head and body straight? Yes☐ / No☐, Is infant facing the mother with nose opposite to the nipple? Yes☐ / No☐ Is infant's body close to the mother's body? Yes☐ / No☐, Is mother supporting infant's whole body and not just neck and shoulders? Yes☐ / No☐ *Circle Positioning Outcome:* Correct positioning / Incorrect positioning **Breast Attachment:** Is the infant's chin touching breast? Yes☐ / No☐, Is infant's mouth wide open? Yes☐ / No☐, Is infant's lower lip turned outward? Yes☐ / No☐, Is there more areola above than below the infant's mouth? Yes☐ / No☐, *Circle Attachment Outcome:* Good attachment / Not well attached / No attachment at all **Is the infant suckling effectively** (slow deep sucks, sometimes pausing)? *Circle Outcome:* Suckling effectively / Not suckling effectively / Not suckling at all	
CHECK FOR LOW BIRTH WEIGHT: INFANT LESS THAN 1 WEEK Is the infant less than 1 week old? Yes☐ / No☐ If yes, determine infant's birth weight: Less than 1500gms? ☐ Between 1500gms and 2499 gms? ☐ 2500gms or more? ☐	

CHECK FOR DEVELOPMENTAL MILESTONES

Age	Assess for the following milestones (tick if achieved)		Milestones present OR delayed
UNDER 1 MONTH	☐ Moro reflex ☐ Blinking reflex ☐ Sucking reflex	☐ Stepping reflex ☐ Grasping reflex	
0-2 MONTHS	☐ Vocalizing ☐ Alternate kicking ☐ Social smile (baby smiles back)	☐ Baby follows a colourful object dangled before their eyes ☐ Startles to loud sound	

CHECK FOR INTERACTION, COMMUNICATION AND RESPONSIVENESS

Caregiver Action (Skills)	Yes	No
Does the caregiver move baby's arms or legs or gently strokes the baby?		
Does the caregiver get baby's attention with shaker toy or other objects?		
Does the caregiver talk to baby (Copies baby's sounds)?		
Does the caregiver respond to baby's sounds and gestures to get baby smile?		

Classify for milestones and interaction

CHECK THE YOUNG INFANT'S IMMUNIZATION STATUS:
Tick immunizations that have been given. Circle immunizations needed today

At Birth	6 weeks
BCG	bOPV 1
bOPV 0 (birth –2wks)	Penta 1
	PCV10 1
	ROTA 1

Indicate next return date (DD/MM/YYYY):

Immunization: .....

Growth monitoring: .....

ASSESS FOR OTHER CHILDHOOD ILLNESSES:	Any other problems?	**ASSESS MOTHER'S/CAREGIVERS NEEDS:**	**Advise when to return:** *DD/MM/YYYY*

TREATMENT

Indicate treatment given for each classification, advice given and return date. Admit or refer any child who has a danger sign and no other severe classification

Feeding advice:

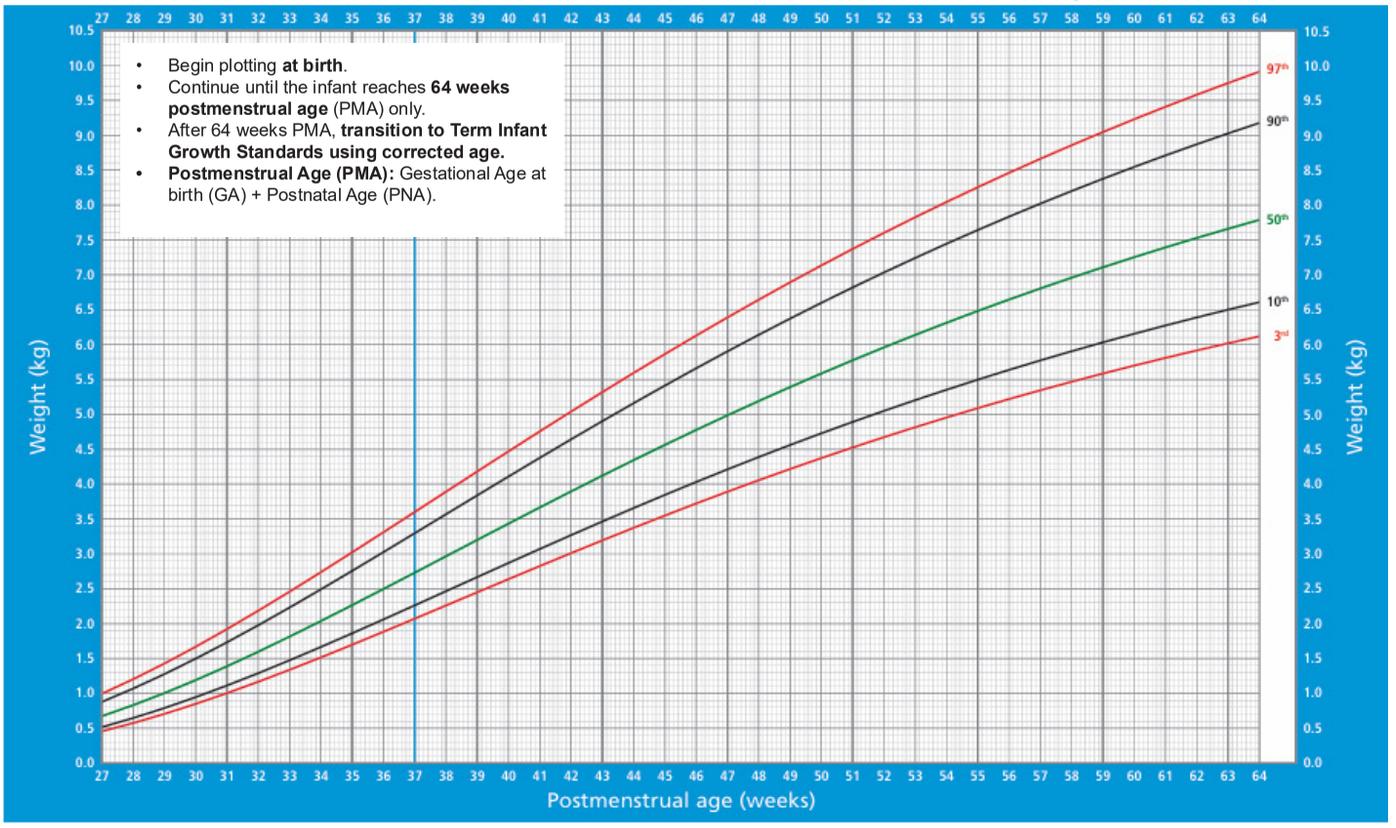
Indicate next return date: Development milestones ..... Stimulation: .....
DD/MM/YYYY *DD/MM/YYYY*

Give any immunizations needed

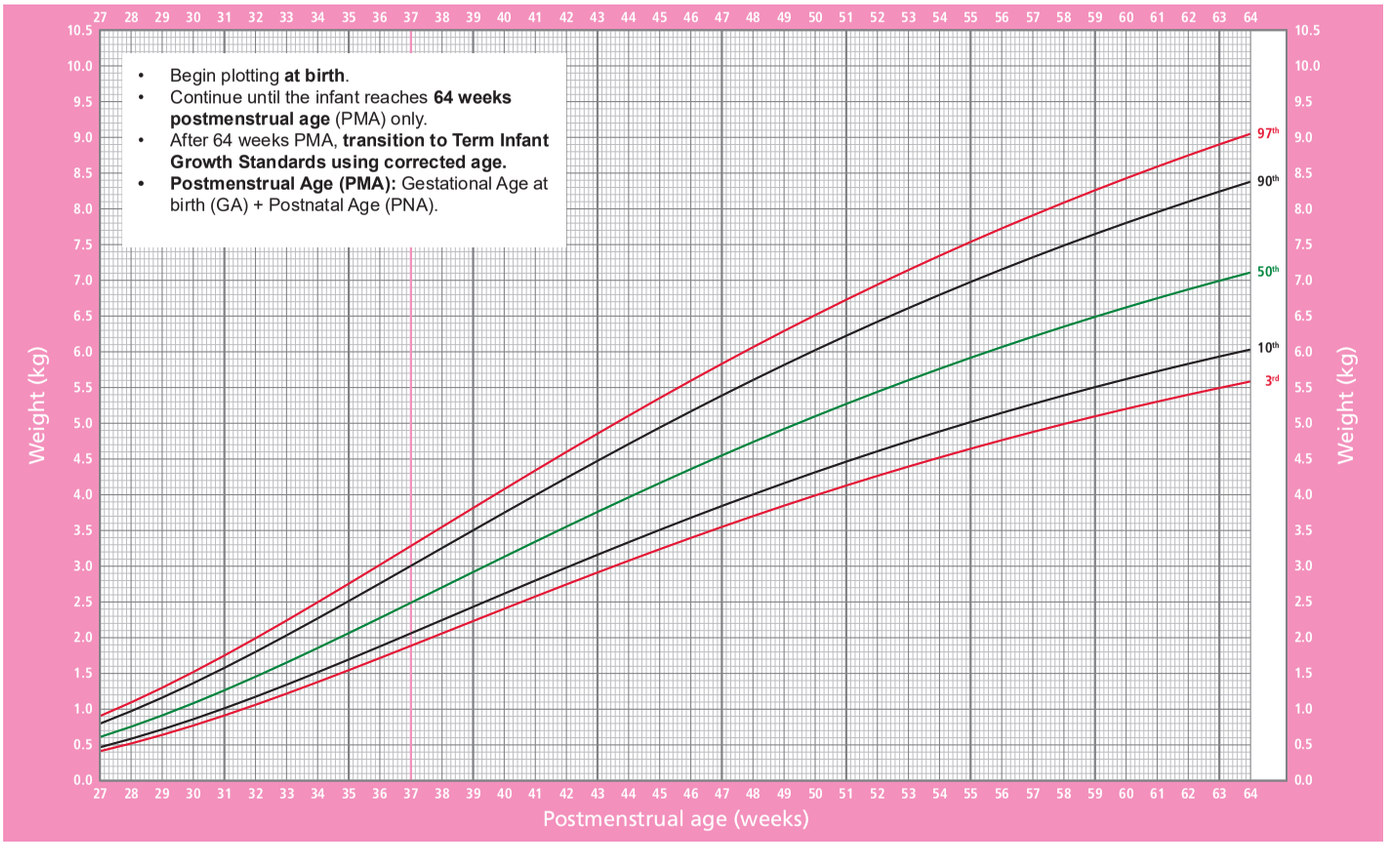
Advise Caregiver when to return immediately. Return for follow-up in ..... days.

MUAC is appropriate for screening but weight for height Z scores is the gold standard for growth monitoring. The following section provides charts for plotting preterm growth, weight for age boys/girls, length/height for age boys/girls, weight for length/height boys/girls.

Preterm Growth Charts (International Postnatal Growth Standards weight - BOYS)



Preterm Growth Charts (International Postnatal Growth Standards weight - GIRLS)

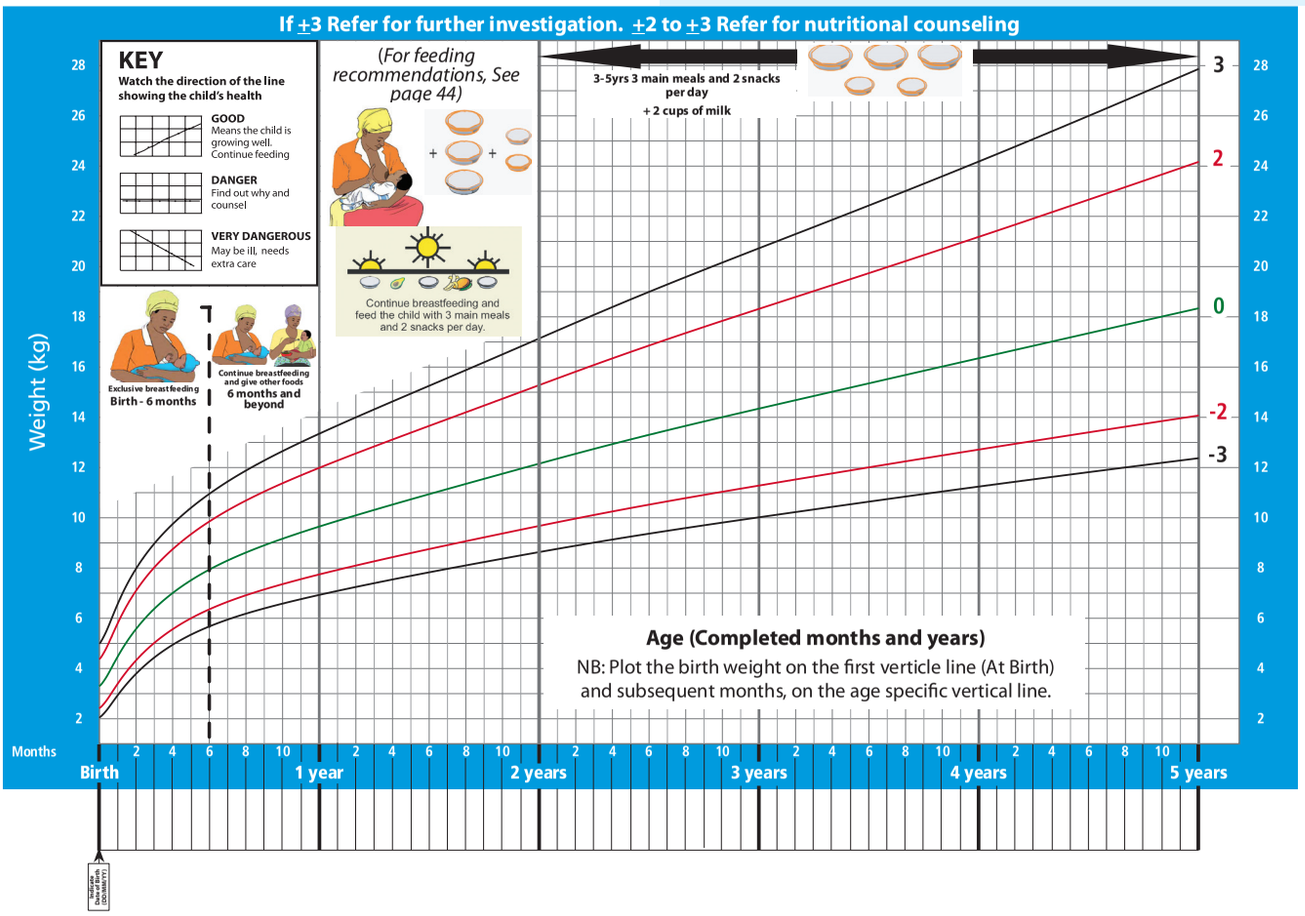


Weight-for-age BOYS

BIRTH TO 5 YEARS (Z-SCORES)

This Weight-for-age chart shows body weight relative to age in comparison to the median (0 line).

- **A child whose weight-forage is below the line –2 is underweight.**
- **Below –3 is severely underweight.**

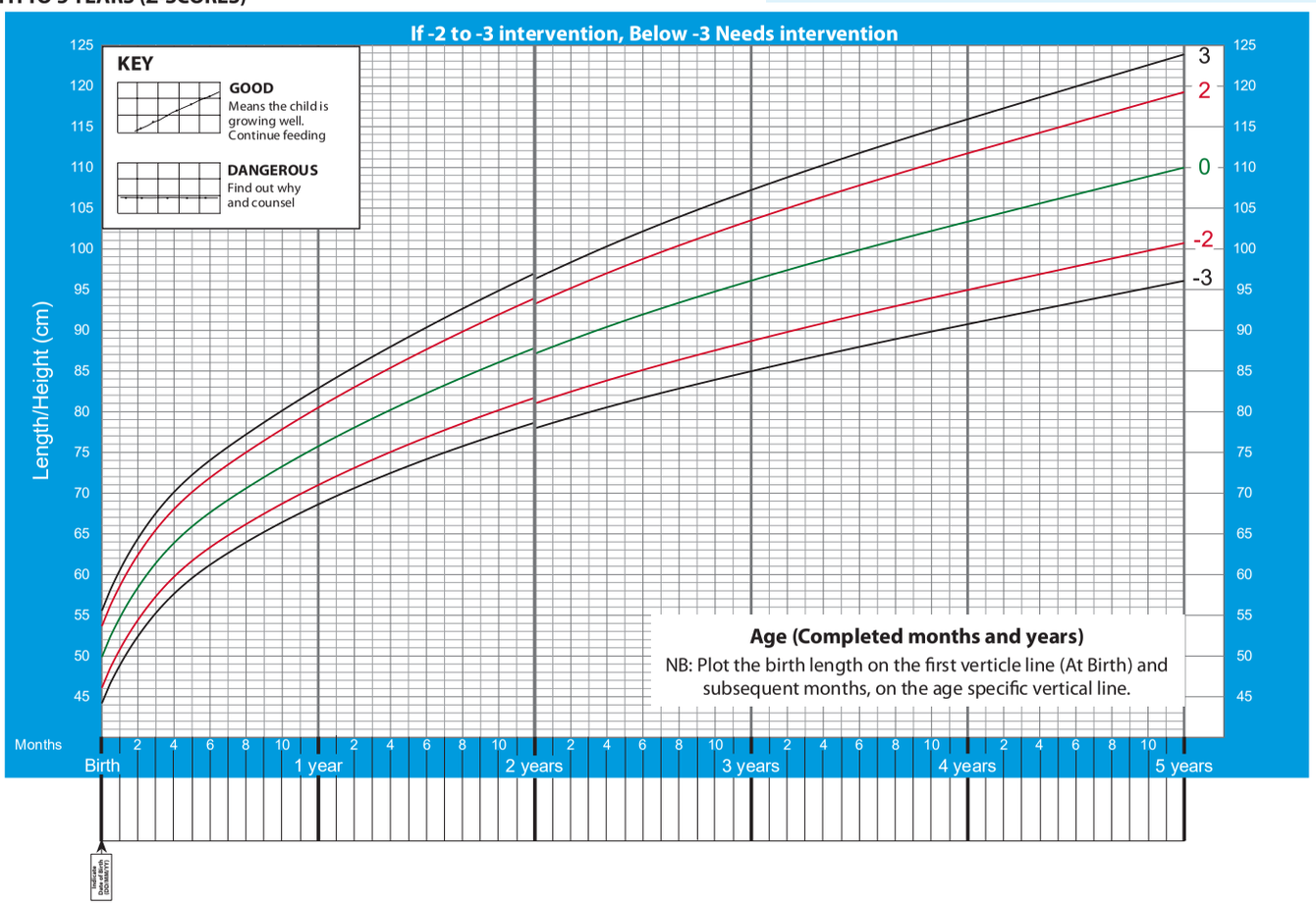


Length/Height-for-age BOYS

BIRTH TO 5 YEARS (Z-SCORES)

This Length-for-age chart shows attained length relative to age in comparison to the median (0 line).

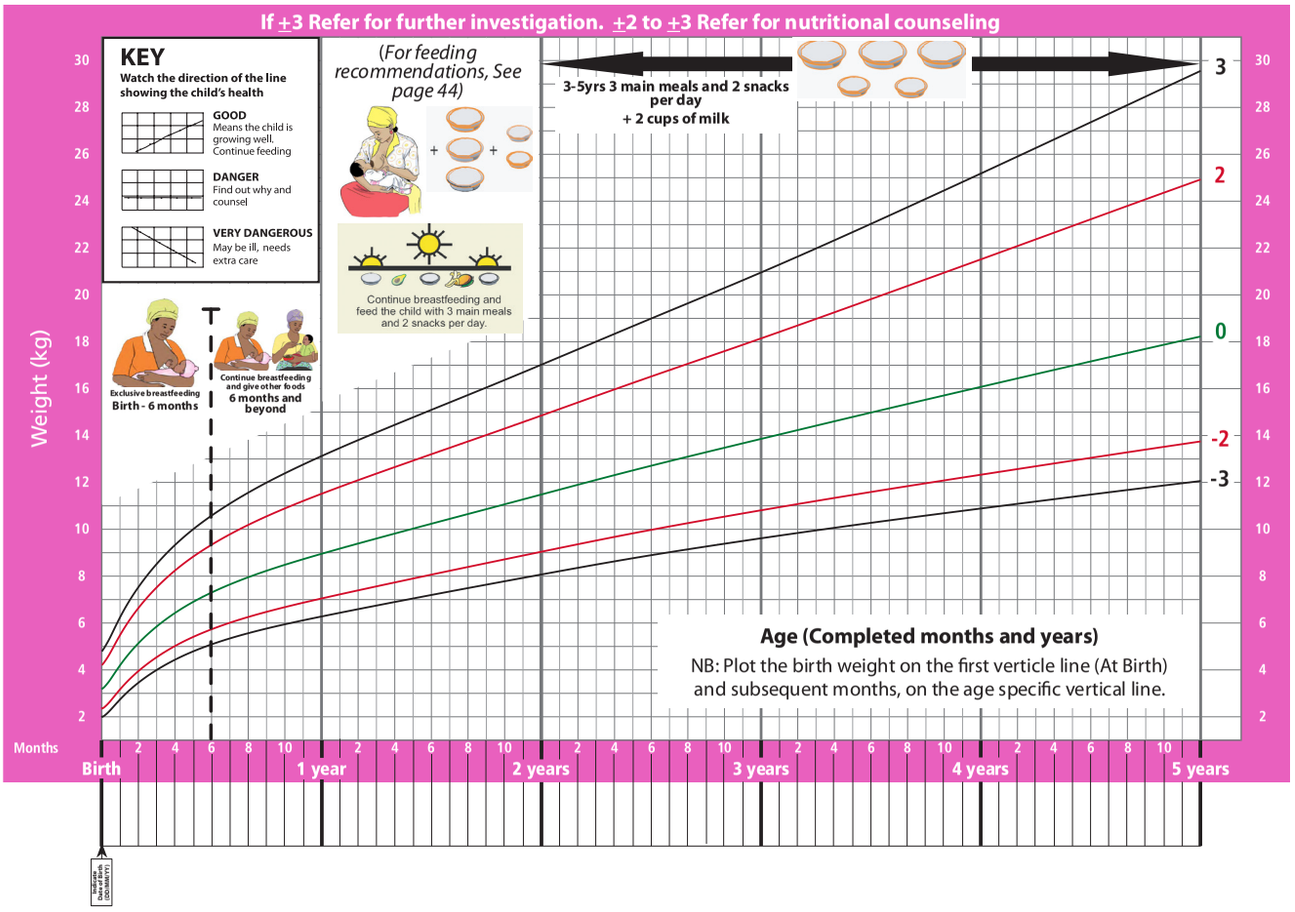
- **A child whose length-forage is below the line –2 is stunted.**
- **Below –3 is severely stunted.**



Weight-for-age GIRLS

BIRTH TO 5 YEARS (Z-SCORES)

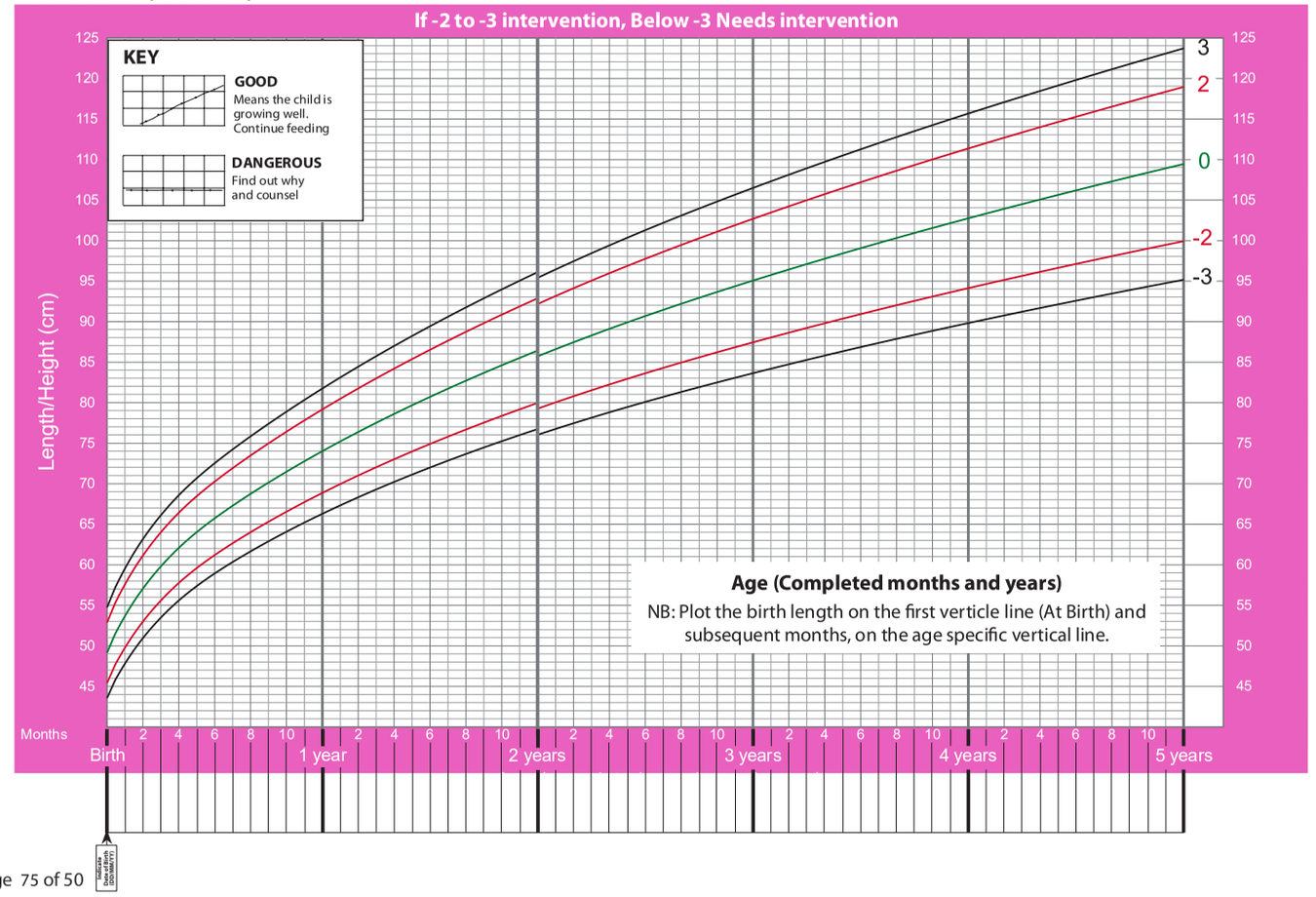
This Weight-for-age chart shows body weight relative to age in comparison to the median (0 line).
- A child whose weight-forage is below the line –2 is underweight.
- Below –3 is severely underweight.



Length/Height-for-age GIRLS

BIRTH TO 5 YEARS (Z-SCORES)

This Length-for-age chart shows attained length relative to age in comparison to the median (0 line).
- A child whose length-forage is below the line –2 is stunted.
- Below –3 is severely stunted.

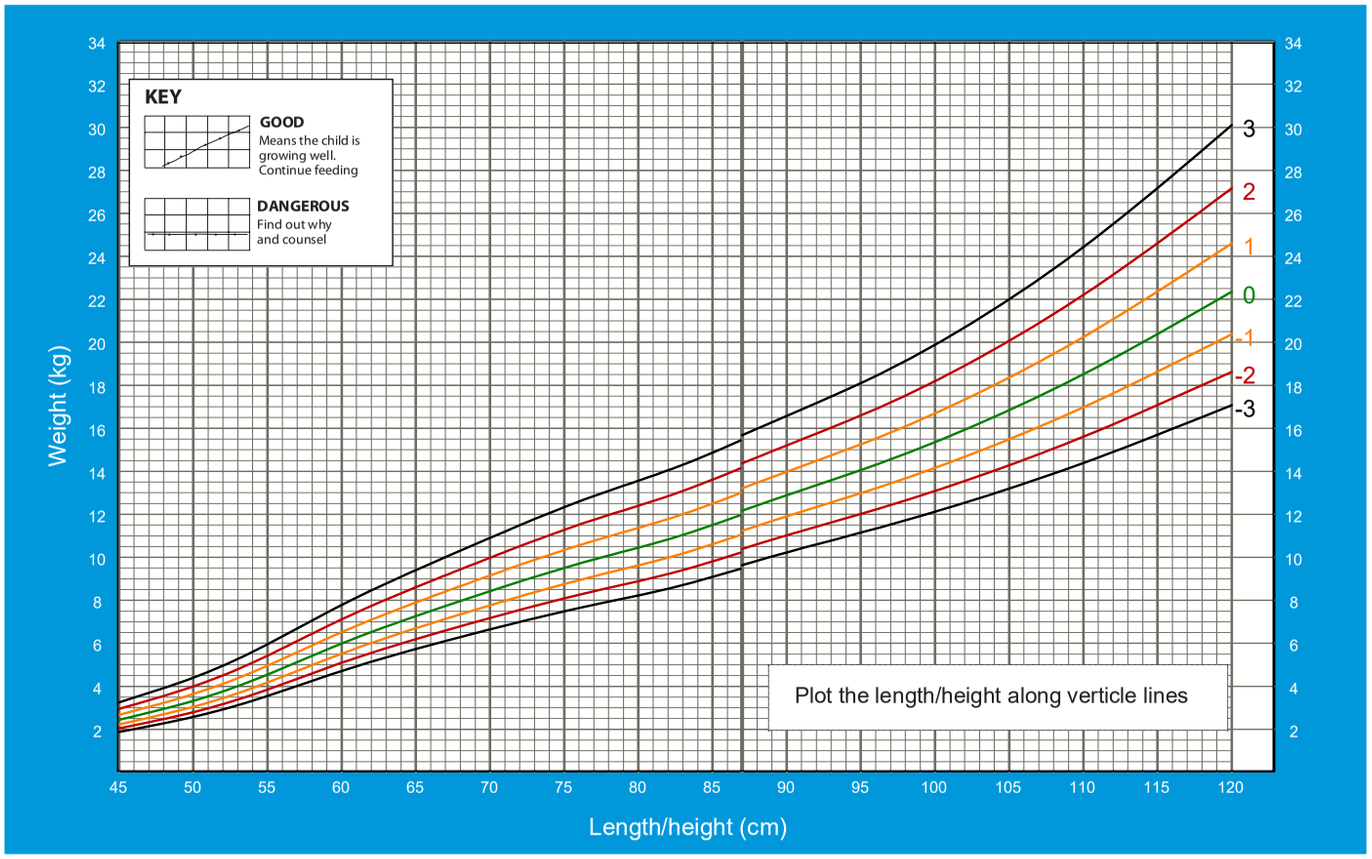


Weight-for-length/height BOYS

Birth to 5 years (z-scores)

This Weight-for-length chart shows body weight relative to length in comparison to the median (0 line).
- A child whose weight-forlength is above the line 3 is obese.
- Above 2 is overweight.
- Above 1 shows possible risk of overweight.
- Below the line –2 is wasted.
- Below –3 is severely wasted.

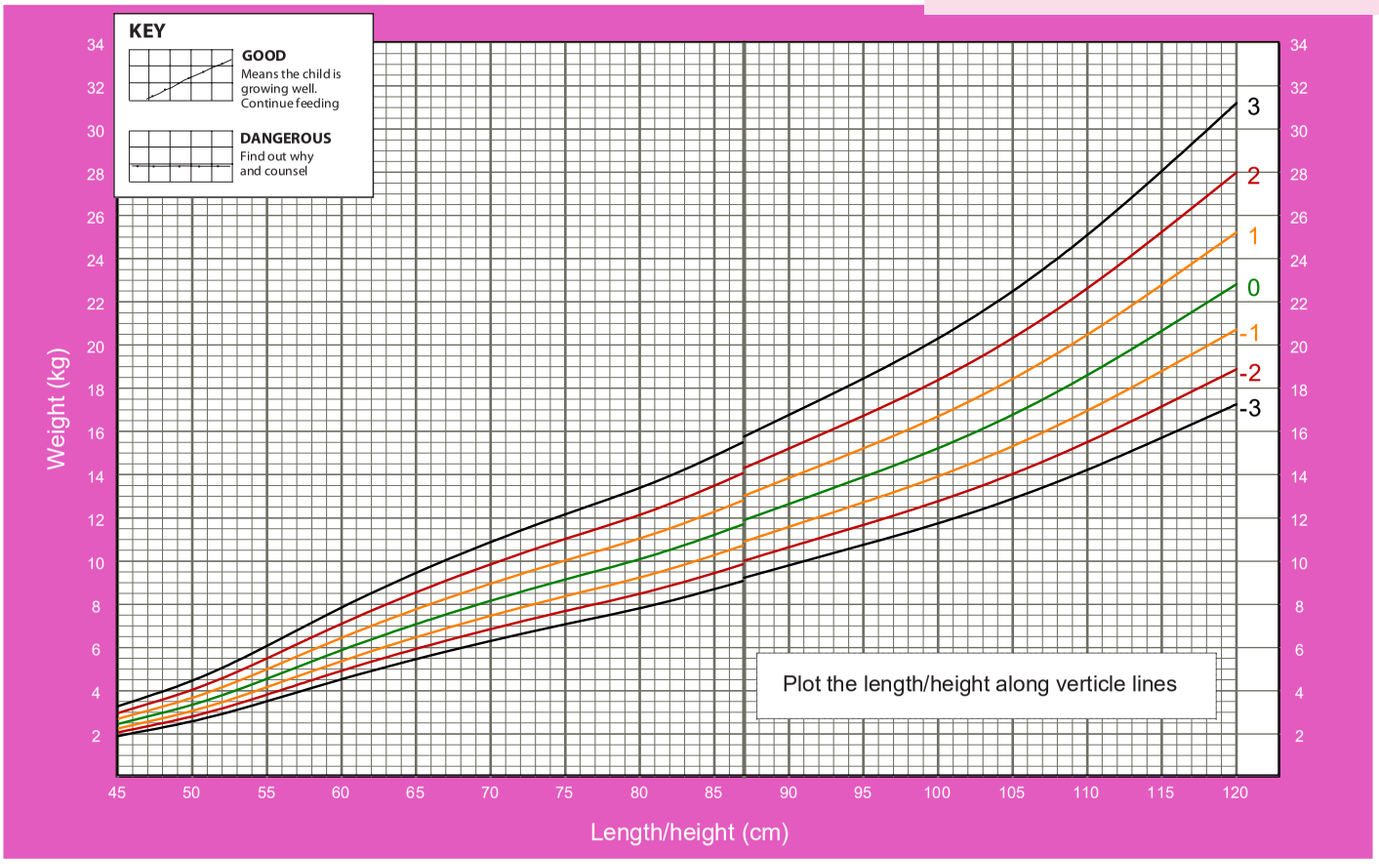
Refer for urgent specialized care.



Weight-for-length/height GIRLS

Birth to 5 years (z-scores)

- This Weight-for-length chart shows body weight
- relative to length in comparison to the median (0 line).
- A child whose weight-forlength is above the line 3 is obese.
- Above 2 is overweight.
- Above 1 shows possible risk of overweight.
- Below the line –2 is wasted.
- Below –3 is severely wasted.
- Refer for urgent specialized care.



Weight-for-length GIRLS
Birth to 2 years (z-scores)

World Health Organization

cm	-3 SD	-2 SD	-1 SD	Median	+1 SD	+2 SD	+3 SD
45.0	1.9	2.1	2.3	2.5	2.7	3.0	3.3
45.5	2.0	2.1	2.3	2.5	2.8	3.1	3.4
46.0	2.0	2.2	2.4	2.6	2.9	3.2	3.5
46.5	2.1	2.3	2.5	2.7	3.0	3.3	3.6
47.0	2.2	2.4	2.6	2.8	3.1	3.4	3.7
47.5	2.2	2.4	2.6	2.9	3.2	3.5	3.8
48.0	2.3	2.5	2.7	3.0	3.3	3.6	4.0
48.5	2.4	2.6	2.8	3.1	3.4	3.7	4.1
49.0	2.4	2.6	2.9	3.2	3.5	3.8	4.2
49.5	2.5	2.7	3.0	3.3	3.6	3.9	4.3
50.0	2.6	2.8	3.1	3.4	3.7	4.0	4.5
50.5	2.7	2.9	3.2	3.5	3.8	4.2	4.6
51.0	2.8	3.0	3.3	3.6	3.9	4.3	4.8
51.5	2.8	3.1	3.4	3.7	4.0	4.4	4.9
52.0	2.9	3.2	3.5	3.8	4.2	4.6	5.1
52.5	3.0	3.3	3.6	3.9	4.3	4.7	5.2
53.0	3.1	3.4	3.7	4.0	4.4	4.9	5.4
53.5	3.2	3.5	3.8	4.2	4.6	5.0	5.5
54.0	3.3	3.6	3.9	4.3	4.7	5.2	5.7
54.5	3.4	3.7	4.0	4.4	4.8	5.3	5.9
55.0	3.5	3.8	4.2	4.5	5.0	5.5	6.1
55.5	3.6	3.9	4.3	4.7	5.1	5.7	6.3
56.0	3.7	4.0	4.4	4.8	5.3	5.8	6.4
56.5	3.8	4.1	4.5	5.0	5.4	6.0	6.6
57.0	3.9	4.3	4.6	5.1	5.6	6.1	6.8
57.5	4.0	4.4	4.8	5.2	5.7	6.3	7.0
58.0	4.1	4.5	4.9	5.4	5.9	6.5	7.1
58.5	4.2	4.6	5.0	5.5	6.0	6.6	7.3
59.0	4.3	4.7	5.1	5.6	6.2	6.8	7.5
59.5	4.4	4.8	5.3	5.7	6.3	6.9	7.7

Weight-for-length GIRLS
Birth to 2 years (z-scores)

World Health Organization

cm	-3 SD	-2 SD	-1 SD	Median	+1 SD	+2 SD	+3 SD
60.0	4.5	4.9	5.4	5.9	6.4	7.1	7.8
60.5	4.6	5.0	5.5	6.0	6.6	7.3	8.0
61.0	4.7	5.1	5.6	6.1	6.7	7.4	8.2
61.5	4.8	5.2	5.7	6.3	6.9	7.6	8.4
62.0	4.9	5.3	5.8	6.4	7.0	7.7	8.5
62.5	5.0	5.4	5.9	6.5	7.1	7.8	8.7
63.0	5.1	5.5	6.0	6.6	7.3	8.0	8.8
63.5	5.2	5.6	6.2	6.7	7.4	8.1	9.0
64.0	5.3	5.7	6.3	6.9	7.5	8.3	9.1
64.5	5.4	5.8	6.4	7.0	7.6	8.4	9.3
65.0	5.5	5.9	6.5	7.1	7.8	8.6	9.5
65.5	5.5	6.0	6.6	7.2	7.9	8.7	9.6
66.0	5.6	6.1	6.7	7.3	8.0	8.8	9.8
66.5	5.7	6.2	6.8	7.4	8.1	9.0	9.9
67.0	5.8	6.3	6.9	7.5	8.3	9.1	10.0
67.5	5.9	6.4	7.0	7.6	8.4	9.2	10.2
68.0	6.0	6.5	7.1	7.7	8.5	9.4	10.3
68.5	6.1	6.6	7.2	7.9	8.6	9.5	10.5
69.0	6.1	6.7	7.3	8.0	8.7	9.6	10.6
69.5	6.2	6.8	7.4	8.1	8.8	9.7	10.7
70.0	6.3	6.9	7.5	8.2	9.0	9.9	10.9
70.5	6.4	6.9	7.6	8.3	9.1	10.0	11.0
71.0	6.5	7.0	7.7	8.4	9.2	10.1	11.1
71.5	6.5	7.1	7.7	8.5	9.3	10.2	11.3
72.0	6.6	7.2	7.8	8.6	9.4	10.3	11.4
72.5	6.7	7.3	7.9	8.7	9.5	10.5	11.5
73.0	6.8	7.4	8.0	8.8	9.6	10.6	11.7
73.5	6.9	7.4	8.1	8.9	9.7	10.7	11.8
74.0	6.9	7.5	8.2	9.0	9.8	10.8	11.9
74.5	7.0	7.6	8.3	9.1	9.9	10.9	12.0
75.0	7.1	7.7	8.4	9.1	10.0	11.0	12.2

Weight-for-length GIRLS
Birth to 2 years (z-scores)

World Health Organization

cm	-3 SD	-2 SD	-1 SD	Median	+1 SD	+2 SD	+3 SD
75.5	7.1	7.8	8.5	9.2	10.1	11.1	12.3
76.0	7.2	7.8	8.5	9.3	10.2	11.2	12.4
76.5	7.3	7.9	8.6	9.4	10.3	11.4	12.5
77.0	7.4	8.0	8.7	9.5	10.4	11.5	12.6
77.5	7.4	8.1	8.8	9.6	10.5	11.6	12.8
78.0	7.5	8.2	8.9	9.7	10.6	11.7	12.9
78.5	7.6	8.2	9.0	9.8	10.7	11.8	13.0
79.0	7.7	8.3	9.1	9.9	10.8	11.9	13.1
79.5	7.7	8.4	9.1	10.0	10.9	12.0	13.3
80.0	7.8	8.5	9.2	10.1	11.0	12.1	13.4
80.5	7.9	8.6	9.3	10.2	11.2	12.3	13.5
81.0	8.0	8.7	9.4	10.3	11.3	12.4	13.7
81.5	8.1	8.8	9.5	10.4	11.4	12.5	13.8
82.0	8.1	8.8	9.6	10.5	11.5	12.6	13.9
82.5	8.2	8.9	9.7	10.6	11.6	12.8	14.1
83.0	8.3	9.0	9.8	10.7	11.8	12.9	14.2
83.5	8.4	9.1	9.9	10.9	11.9	13.1	14.4
84.0	8.5	9.2	10.1	11.0	12.0	13.2	14.5
84.5	8.6	9.3	10.2	11.1	12.1	13.3	14.7
85.0	8.7	9.4	10.3	11.2	12.3	13.5	14.9
85.5	8.8	9.5	10.4	11.3	12.4	13.6	15.0
86.0	8.9	9.7	10.5	11.5	12.6	13.8	15.2
86.5	9.0	9.8	10.6	11.6	12.7	13.9	15.4
87.0	9.1	9.9	10.7	11.7	12.8	14.1	15.5
87.5	9.2	10.0	10.9	11.8	13.0	14.2	15.7
88.0	9.3	10.1	11.0	12.0	13.1	14.4	15.9
88.5	9.4	10.2	11.1	12.1	13.2	14.5	16.0
89.0	9.5	10.3	11.2	12.2	13.4	14.7	16.2
89.5	9.6	10.4	11.3	12.3	13.5	14.8	16.4
90.0	9.7	10.5	11.4	12.5	13.7	15.0	16.5
90.5	9.8	10.6	11.5	12.6	13.8	15.1	16.7

Weight-for-length GIRLS
Birth to 2 years (z-scores)

World Health Organization

cm	-3 SD	-2 SD	-1 SD	Median	+1 SD	+2 SD	+3 SD
91.0	9.9	10.7	11.7	12.7	13.9	15.3	16.9
91.5	10.0	10.8	11.8	12.8	14.1	15.5	17.0
92.0	10.1	10.9	11.9	13.0	14.2	15.6	17.2
92.5	10.1	11.0	12.0	13.1	14.3	15.8	17.4
93.0	10.2	11.1	12.1	13.2	14.5	15.9	17.5
93.5	10.3	11.2	12.2	13.3	14.6	16.1	17.7
94.0	10.4	11.3	12.3	13.5	14.7	16.2	17.9
94.5	10.5	11.4	12.4	13.6	14.9	16.4	18.0
95.0	10.6	11.5	12.6	13.7	15.0	16.5	18.2
95.5	10.7	11.6	12.7	13.8	15.2	16.7	18.4
96.0	10.8	11.7	12.8	14.0	15.3	16.8	18.6
96.5	10.9	11.8	12.9	14.1	15.4	17.0	18.7
97.0	11.0	12.0	13.0	14.2	15.6	17.1	18.9
97.5	11.1	12.1	13.1	14.4	15.7	17.3	19.1
98.0	11.2	12.2	13.3	14.5	15.9	17.5	19.3
98.5	11.3	12.3	13.4	14.6	16.0	17.6	19.5
99.0	11.4	12.4	13.5	14.8	16.2	17.8	19.6
99.5	11.5	12.5	13.6	14.9	16.3	18.0	19.8
100.0	11.6	12.6	13.7	15.0	16.5	18.1	20.0
100.5	11.7	12.7	13.9	15.2	16.6	18.3	20.2
101.0	11.8	12.8	14.0	15.3	16.8	18.5	20.4
101.5	11.9	13.0	14.1	15.5	17.0	18.7	20.6
102.0	12.0	13.1	14.3	15.6	17.1	18.9	20.8
102.5	12.1	13.2	14.4	15.8	17.3	19.0	21.0
103.0	12.3	13.3	14.5	15.9	17.5	19.2	21.3
103.5	12.4	13.5	14.7	16.1	17.6	19.4	21.5
104.0	12.5	13.6	14.8	16.2	17.8	19.6	21.7
104.5	12.6	13.7	15.0	16.4	18.0	19.8	21.9
105.0	12.7	13.8	15.1	16.5	18.2	20.0	22.2
105.5	12.8	14.0	15.3	16.7	18.4	20.2	22.4
106.0	13.0	14.1	15.4	16.9	18.5	20.5	22.6

Weight-for-length GIRLS
Birth to 2 years (z-scores)

World Health Organization

cm	-3 SD	-2 SD	-1 SD	Median	+1 SD	+2 SD	+3 SD
106.5	13.1	14.3	15.6	17.1	18.7	20.7	22.9
107.0	13.2	14.4	15.7	17.2	18.9	20.9	23.1
107.5	13.3	14.5	15.9	17.4	19.1	21.1	23.4
108.0	13.5	14.7	16.0	17.6	19.3	21.3	23.6
108.5	13.6	14.8	16.2	17.8	19.5	21.6	23.9
109.0	13.7	15.0	16.4	18.0	19.7	21.8	24.2
109.5	13.9	15.1	16.5	18.1	20.0	22.0	24.4
110.0	14.0	15.3	16.7	18.3	20.2	22.3	24.7

WHO Child Growth Standards

Weight-for-height GIRLS
2 to 5 years (z-scores)

World Health Organization

cm	-3 SD	-2 SD	-1 SD	Median	+1 SD	+2 SD	+3 SD
65.0	5.6	6.1	6.6	7.2	7.9	8.7	9.7
65.5	5.7	6.2	6.7	7.4	8.1	8.9	9.8
66.0	5.8	6.3	6.8	7.5	8.2	9.0	10.0
66.5	5.8	6.4	6.9	7.6	8.3	9.1	10.1
67.0	5.9	6.4	7.0	7.7	8.4	9.3	10.2
67.5	6.0	6.5	7.1	7.8	8.5	9.4	10.4
68.0	6.1	6.6	7.2	7.9	8.7	9.5	10.5
68.5	6.2	6.7	7.3	8.0	8.8	9.7	10.7
69.0	6.3	6.8	7.4	8.1	8.9	9.8	10.8
69.5	6.3	6.9	7.5	8.2	9.0	9.9	10.9
70.0	6.4	7.0	7.6	8.3	9.1	10.0	11.1
70.5	6.5	7.1	7.7	8.4	9.2	10.1	11.2
71.0	6.6	7.1	7.8	8.5	9.3	10.3	11.3
71.5	6.7	7.2	7.9	8.6	9.4	10.4	11.5
72.0	6.7	7.3	8.0	8.7	9.5	10.5	11.6
72.5	6.8	7.4	8.1	8.8	9.7	10.6	11.7
73.0	6.9	7.5	8.1	8.9	9.8	10.7	11.8
73.5	7.0	7.6	8.2	9.0	9.9	10.8	12.0
74.0	7.0	7.6	8.3	9.1	10.0	11.0	12.1
74.5	7.1	7.7	8.4	9.2	10.1	11.1	12.2
75.0	7.2	7.8	8.5	9.3	10.2	11.2	12.3
75.5	7.2	7.9	8.6	9.4	10.3	11.3	12.5
76.0	7.3	8.0	8.7	9.5	10.4	11.4	12.6
76.5	7.4	8.0	8.7	9.6	10.5	11.5	12.7
77.0	7.5	8.1	8.8	9.6	10.6	11.6	12.8
77.5	7.5	8.2	8.9	9.7	10.7	11.7	12.9
78.0	7.6	8.3	9.0	9.8	10.8	11.8	13.1
78.5	7.7	8.4	9.1	9.9	10.9	12.0	13.2
79.0	7.8	8.4	9.2	10.0	11.0	12.1	13.3
79.5	7.8	8.5	9.3	10.1	11.1	12.2	13.4

Weight-for-height GIRLS
2 to 5 years (z-scores)

World Health Organization

cm	-3 SD	-2 SD	-1 SD	Median	+1 SD	+2 SD	+3 SD
80.0	7.9	8.6	9.4	10.2	11.2	12.3	13.6
80.5	8.0	8.7	9.5	10.3	11.3	12.4	13.7
81.0	8.1	8.8	9.6	10.4	11.4	12.6	13.9
81.5	8.2	8.9	9.7	10.6	11.6	12.7	14.0
82.0	8.3	9.0	9.8	10.7	11.7	12.8	14.1
82.5	8.4	9.1	9.9	10.8	11.8	13.0	14.3
83.0	8.5	9.2	10.0	10.9	11.9	13.1	14.5
83.5	8.5	9.3	10.1	11.0	12.1	13.3	14.6
84.0	8.6	9.4	10.2	11.1	12.2	13.4	14.8
84.5	8.7	9.5	10.3	11.3	12.3	13.5	14.9
85.0	8.8	9.6	10.4	11.4	12.5	13.7	15.1
85.5	8.9	9.7	10.6	11.5	12.6	13.8	15.3
86.0	9.0	9.8	10.7	11.6	12.7	14.0	15.4
86.5	9.1	9.9	10.8	11.8	12.9	14.2	15.6
87.0	9.2	10.0	10.9	11.9	13.0	14.3	15.8
87.5	9.3	10.1	11.0	12.0	13.2	14.5	15.9
88.0	9.4	10.2	11.1	12.1	13.3	14.6	16.1
88.5	9.5	10.3	11.2	12.3	13.4	14.8	16.3
89.0	9.6	10.4	11.4	12.4	13.6	14.9	16.4
89.5	9.7	10.5	11.5	12.5	13.7	15.1	16.6
90.0	9.8	10.6	11.6	12.6	13.8	15.2	16.8
90.5	9.9	10.7	11.7	12.8	14.0	15.4	16.9
91.0	10.0	10.9	11.8	12.9	14.1	15.5	17.1
91.5	10.1	11.0	11.9	13.0	14.3	15.7	17.3
92.0	10.2	11.1	12.0	13.1	14.4	15.8	17.4
92.5	10.3	11.2	12.1	13.3	14.5	16.0	17.6
93.0	10.4	11.3	12.3	13.4	14.7	16.1	17.8
93.5	10.5	11.4	12.4	13.5	14.8	16.3	17.9
94.0	10.6	11.5	12.5	13.6	14.9	16.4	18.1
94.5	10.7	11.6	12.6	13.8	15.1	16.6	18.3
95.0	10.8	11.7	12.7	13.9	15.2	16.7	18.5

Weight-for-height GIRLS
2 to 5 years (z-scores)

World Health Organization

cm	-3 SD	-2 SD	-1 SD	Median	+1 SD	+2 SD	+3 SD
95.5	10.8	11.8	12.8	14.0	15.4	16.9	18.6
96.0	10.9	11.9	12.9	14.1	15.5	17.0	18.8
96.5	11.0	12.0	13.1	14.3	15.6	17.2	19.0
97.0	11.1	12.1	13.2	14.4	15.8	17.4	19.2
97.5	11.2	12.2	13.3	14.5	15.9	17.5	19.3
98.0	11.3	12.3	13.4	14.7	16.1	17.7	19.5
98.5	11.4	12.4	13.5	14.8	16.2	17.9	19.7
99.0	11.5	12.5	13.7	14.9	16.4	18.0	19.9
99.5	11.6	12.7	13.8	15.1	16.5	18.2	20.1
100.0	11.7	12.8	13.9	15.2	16.7	18.4	20.3
100.5	11.9	12.9	14.1	15.4	16.9	18.6	20.5
101.0	12.0	13.0	14.2	15.5	17.0	18.7	20.7
101.5	12.1	13.1	14.3	15.7	17.2	18.9	20.9
102.0	12.2	13.3	14.5	15.8	17.4	19.1	21.1
102.5	12.3	13.4	14.6	16.0	17.5	19.3	21.4
103.0	12.4	13.5	14.7	16.1	17.7	19.5	21.6
103.5	12.5	13.6	14.9	16.3	17.9	19.7	21.8
104.0	12.6	13.8	15.0	16.4	18.1	19.9	22.0
104.5	12.8	13.9	15.2	16.6	18.2	20.1	22.3
105.0	12.9	14.0	15.3	16.8	18.4	20.3	22.5
105.5	13.0	14.2	15.5	16.9	18.6	20.5	22.7
106.0	13.1	14.3	15.6	17.1	18.8	20.8	23.0
106.5	13.3	14.5	15.8	17.3	19.0	21.0	23.2
107.0	13.4	14.6	15.9	17.5	19.2	21.2	23.5
107.5	13.5	14.7	16.1	17.7	19.4	21.4	23.7
108.0	13.7	14.9	16.3	17.8	19.6	21.7	24.0
108.5	13.8	15.0	16.4	18.0	19.8	21.9	24.3
109.0	13.9	15.2	16.6	18.2	20.0	22.1	24.5
109.5	14.1	15.4	16.8	18.4	20.3	22.4	24.8
110.0	14.2	15.5	17.0	18.6	20.5	22.6	25.1
110.5	14.4	15.7	17.1	18.8	20.7	22.9	25.4

Weight-for-height GIRLS
2 to 5 years (z-scores)

World Health Organization

cm	-3 SD	-2 SD	-1 SD	Median	+1 SD	+2 SD	+3 SD
111.0	14.5	15.8	17.3	19.0	20.9	23.1	25.7
111.5	14.7	16.0	17.5	19.2	21.2	23.4	26.0
112.0	14.8	16.2	17.7	19.4	21.4	23.6	26.2
112.5	15.0	16.3	17.9	19.6	21.6	23.9	26.5
113.0	15.1	16.5	18.0	19.8	21.8	24.2	26.8
113.5	15.3	16.7	18.2	20.0	22.1	24.4	27.1
114.0	15.4	16.8	18.4	20.2	22.3	24.7	27.4
114.5	15.6	17.0	18.6	20.5	22.6	25.0	27.8
115.0	15.7	17.2	18.8	20.7	22.8	25.2	28.1
115.5	15.9	17.3	19.0	20.9	23.0	25.5	28.4
116.0	16.0	17.5	19.2	21.1	23.3	25.8	28.7
116.5	16.2	17.7	19.4	21.3	23.5	26.1	29.0
117.0	16.3	17.8	19.6	21.5	23.8	26.3	29.3
117.5	16.5	18.0	19.8	21.7	24.0	26.6	29.6
118.0	16.6	18.2	19.9	22.0	24.2	26.9	29.9
118.5	16.8	18.4	20.1	22.2	24.5	27.2	30.3
119.0	16.9	18.5	20.3	22.4	24.7	27.4	30.6
119.5	17.1	18.7	20.5	22.6	25.0	27.7	30.9
120.0	17.3	18.9	20.7	22.8	25.2	28.0	31.2

WHO Child Growth Standards

Weight-for-length BOYS
Birth to 2 years (z-scores)

World Health Organization

cm	-3 SD	-2 SD	-1 SD	Median	+1 SD	+2 SD	+3 SD
45.0	1.9	2.0	2.2	2.4	2.7	3.0	3.3
45.5	1.9	2.1	2.3	2.5	2.8	3.1	3.4
46.0	2.0	2.2	2.4	2.6	2.9	3.1	3.5
46.5	2.1	2.3	2.5	2.7	3.0	3.2	3.6
47.0	2.1	2.3	2.5	2.8	3.0	3.3	3.7
47.5	2.2	2.4	2.6	2.9	3.1	3.4	3.8
48.0	2.3	2.5	2.7	2.9	3.2	3.6	3.9
48.5	2.3	2.6	2.8	3.0	3.3	3.7	4.0
49.0	2.4	2.6	2.9	3.1	3.4	3.8	4.2
49.5	2.5	2.7	3.0	3.2	3.5	3.9	4.3
50.0	2.6	2.8	3.0	3.3	3.6	4.0	4.4
50.5	2.7	2.9	3.1	3.4	3.8	4.1	4.5
51.0	2.7	3.0	3.2	3.5	3.9	4.2	4.7
51.5	2.8	3.1	3.3	3.6	4.0	4.4	4.8
52.0	2.9	3.2	3.5	3.8	4.1	4.5	5.0
52.5	3.0	3.3	3.6	3.9	4.2	4.6	5.1
53.0	3.1	3.4	3.7	4.0	4.4	4.8	5.3
53.5	3.2	3.5	3.8	4.1	4.5	4.9	5.4
54.0	3.3	3.6	3.9	4.3	4.7	5.1	5.6
54.5	3.4	3.7	4.0	4.4	4.8	5.3	5.8
55.0	3.6	3.8	4.2	4.5	5.0	5.4	6.0
55.5	3.7	4.0	4.3	4.7	5.1	5.6	6.1
56.0	3.8	4.1	4.4	4.8	5.3	5.8	6.3
56.5	3.9	4.2	4.6	5.0	5.4	5.9	6.5
57.0	4.0	4.3	4.7	5.1	5.6	6.1	6.7
57.5	4.1	4.5	4.9	5.3	5.7	6.3	6.9
58.0	4.3	4.6	5.0	5.4	5.9	6.4	7.1
58.5	4.4	4.7	5.1	5.6	6.1	6.6	7.2
59.0	4.5	4.8	5.3	5.7	6.2	6.8	7.4
59.5	4.6	5.0	5.4	5.9	6.4	7.0	7.6

Weight-for-length BOYS
Birth to 2 years (z-scores)

World Health Organization

cm	-3 SD	-2 SD	-1 SD	Median	+1 SD	+2 SD	+3 SD
45.0	1.9	2.0	2.2	2.4	2.7	3.0	3.3
45.5	1.9	2.1	2.3	2.5	2.8	3.1	3.4
46.0	2.0	2.2	2.4	2.6	2.9	3.1	3.5
46.5	2.1	2.3	2.5	2.7	3.0	3.2	3.6
47.0	2.1	2.3	2.5	2.8	3.0	3.3	3.7
47.5	2.2	2.4	2.6	2.9	3.1	3.4	3.8
48.0	2.3	2.5	2.7	2.9	3.2	3.6	3.9
48.5	2.3	2.6	2.8	3.0	3.3	3.7	4.0
49.0	2.4	2.6	2.9	3.1	3.4	3.8	4.2
49.5	2.5	2.7	3.0	3.2	3.5	3.9	4.3
50.0	2.6	2.8	3.0	3.3	3.6	4.0	4.4
50.5	2.7	2.9	3.1	3.4	3.8	4.1	4.5
51.0	2.7	3.0	3.2	3.5	3.9	4.2	4.7
51.5	2.8	3.1	3.3	3.6	4.0	4.4	4.8
52.0	2.9	3.2	3.5	3.8	4.1	4.5	5.0
52.5	3.0	3.3	3.6	3.9	4.2	4.6	5.1
53.0	3.1	3.4	3.7	4.0	4.4	4.8	5.3
53.5	3.2	3.5	3.8	4.1	4.5	4.9	5.4
54.0	3.3	3.6	3.9	4.3	4.7	5.1	5.6
54.5	3.4	3.7	4.0	4.4	4.8	5.3	5.8
55.0	3.6	3.8	4.2	4.5	5.0	5.4	6.0
55.5	3.7	4.0	4.3	4.7	5.1	5.6	6.1
56.0	3.8	4.1	4.4	4.8	5.3	5.8	6.3
56.5	3.9	4.2	4.6	5.0	5.4	5.9	6.5
57.0	4.0	4.3	4.7	5.1	5.6	6.1	6.7
57.5	4.1	4.5	4.9	5.3	5.7	6.3	6.9
58.0	4.3	4.6	5.0	5.4	5.9	6.4	7.1
58.5	4.4	4.7	5.1	5.6	6.1	6.6	7.2
59.0	4.5	4.8	5.3	5.7	6.2	6.8	7.4
59.5	4.6	5.0	5.4	5.9	6.4	7.0	7.6

Weight-for-length BOYS
Birth to 2 years (z-scores)

World Health Organization

cm	-3 SD	-2 SD	-1 SD	Median	+1 SD	+2 SD	+3 SD
60.0	4.7	5.1	5.5	6.0	6.5	7.1	7.8
60.5	4.8	5.2	5.6	6.1	6.7	7.3	8.0
61.0	4.9	5.3	5.8	6.3	6.8	7.4	8.1
61.5	5.0	5.4	5.9	6.4	7.0	7.6	8.3
62.0	5.1	5.6	6.0	6.5	7.1	7.7	8.5
62.5	5.2	5.7	6.1	6.7	7.2	7.9	8.6
63.0	5.3	5.8	6.2	6.8	7.4	8.0	8.8
63.5	5.4	5.9	6.4	6.9	7.5	8.2	8.9
64.0	5.5	6.0	6.5	7.0	7.6	8.3	9.1
64.5	5.6	6.1	6.6	7.1	7.8	8.5	9.3
65.0	5.7	6.2	6.7	7.3	7.9	8.6	9.4
65.5	5.8	6.3	6.8	7.4	8.0	8.7	9.6
66.0	5.9	6.4	6.9	7.5	8.2	8.9	9.7
66.5	6.0	6.5	7.0	7.6	8.3	9.0	9.9
67.0	6.1	6.6	7.1	7.7	8.4	9.2	10.0
67.5	6.2	6.7	7.2	7.9	8.5	9.3	10.2
68.0	6.3	6.8	7.3	8.0	8.7	9.4	10.3
68.5	6.4	6.9	7.5	8.1	8.8	9.6	10.5
69.0	6.5	7.0	7.6	8.2	8.9	9.7	10.6
69.5	6.6	7.1	7.7	8.3	9.0	9.8	10.8
70.0	6.6	7.2	7.8	8.4	9.2	10.0	10.9
70.5	6.7	7.3	7.9	8.5	9.3	10.1	11.1
71.0	6.8	7.4	8.0	8.6	9.4	10.2	11.2
71.5	6.9	7.5	8.1	8.8	9.5	10.4	11.3
72.0	7.0	7.6	8.2	8.9	9.6	10.5	11.5
72.5	7.1	7.6	8.3	9.0	9.8	10.6	11.6
73.0	7.2	7.7	8.4	9.1	9.9	10.8	11.8
73.5	7.2	7.8	8.5	9.2	10.0	10.9	11.9
74.0	7.3	7.9	8.6	9.3	10.1	11.0	12.1
74.5	7.4	8.0	8.7	9.4	10.2	11.2	12.2
75.0	7.5	8.1	8.8	9.5	10.3	11.3	12.3

Weight-for-length BOYS
Birth to 2 years (z-scores)

World Health Organization

cm	-3 SD	-2 SD	-1 SD	Median	+1 SD	+2 SD	+3 SD
75.5	7.6	8.2	8.8	9.6	10.4	11.4	12.5
76.0	7.6	8.3	8.9	9.7	10.6	11.5	12.6
76.5	7.7	8.3	9.0	9.8	10.7	11.6	12.7
77.0	7.8	8.4	9.1	9.9	10.8	11.7	12.8
77.5	7.9	8.5	9.2	10.0	10.9	11.9	13.0
78.0	7.9	8.6	9.3	10.1	11.0	12.0	13.1
78.5	8.0	8.7	9.4	10.2	11.1	12.1	13.2
79.0	8.1	8.7	9.5	10.3	11.2	12.2	13.3
79.5	8.2	8.8	9.5	10.4	11.3	12.3	13.4
80.0	8.2	8.9	9.6	10.4	11.4	12.4	13.6
80.5	8.3	9.0	9.7	10.5	11.5	12.5	13.7
81.0	8.4	9.1	9.8	10.6	11.6	12.6	13.8
81.5	8.5	9.1	9.9	10.7	11.7	12.7	13.9
82.0	8.5	9.2	10.0	10.8	11.8	12.8	14.0
82.5	8.6	9.3	10.1	10.9	11.9	13.0	14.2
83.0	8.7	9.4	10.2	11.0	12.0	13.1	14.3
83.5	8.8	9.5	10.3	11.2	12.1	13.2	14.4
84.0	8.9	9.6	10.4	11.3	12.2	13.3	14.6
84.5	9.0	9.7	10.5	11.4	12.4	13.5	14.7
85.0	9.1	9.8	10.6	11.5	12.5	13.6	14.9
85.5	9.2	9.9	10.7	11.6	12.6	13.7	15.0
86.0	9.3	10.0	10.8	11.7	12.8	13.9	15.2
86.5	9.4	10.1	11.0	11.9	12.9	14.0	15.3
87.0	9.5	10.2	11.1	12.0	13.0	14.2	15.5
87.5	9.6	10.4	11.2	12.1	13.2	14.3	15.6
88.0	9.7	10.5	11.3	12.2	13.3	14.5	15.8
88.5	9.8	10.6	11.4	12.4	13.4	14.6	15.9
89.0	9.9	10.7	11.5	12.5	13.5	14.7	16.1
89.5	10.0	10.8	11.6	12.6	13.7	14.9	16.2
90.0	10.1	10.9	11.8	12.7	13.8	15.0	16.4
90.5	10.2	11.0	11.9	12.8	13.9	15.1	16.5

Weight-for-length BOYS
Birth to 2 years (z-scores)

World Health Organization

cm	-3 SD	-2 SD	-1 SD	Median	+1 SD	+2 SD	+3 SD
91.0	10.3	11.1	12.0	13.0	14.1	15.3	16.7
91.5	10.4	11.2	12.1	13.1	14.2	15.4	16.8
92.0	10.5	11.3	12.2	13.2	14.3	15.6	17.0
92.5	10.6	11.4	12.3	13.3	14.4	15.7	17.1
93.0	10.7	11.5	12.4	13.4	14.6	15.8	17.3
93.5	10.7	11.6	12.5	13.5	14.7	16.0	17.4
94.0	10.8	11.7	12.6	13.7	14.8	16.1	17.6
94.5	10.9	11.8	12.7	13.8	14.9	16.3	17.7
95.0	11.0	11.9	12.8	13.9	15.1	16.4	17.9
95.5	11.1	12.0	12.9	14.0	15.2	16.5	18.0
96.0	11.2	12.1	13.1	14.1	15.3	16.7	18.2
96.5	11.3	12.2	13.2	14.3	15.5	16.8	18.4
97.0	11.4	12.3	13.3	14.4	15.6	17.0	18.5
97.5	11.5	12.4	13.4	14.5	15.7	17.1	18.7
98.0	11.6	12.5	13.5	14.6	15.9	17.3	18.9
98.5	11.7	12.6	13.6	14.8	16.0	17.5	19.1
99.0	11.8	12.7	13.7	14.9	16.2	17.6	19.2
99.5	11.9	12.8	13.9	15.0	16.3	17.8	19.4
100.0	12.0	12.9	14.0	15.2	16.5	18.0	19.6
100.5	12.1	13.0	14.1	15.3	16.6	18.1	19.8
101.0	12.2	13.2	14.2	15.4	16.8	18.3	20.0
101.5	12.3	13.3	14.4	15.6	16.9	18.5	20.2
102.0	12.4	13.4	14.5	15.7	17.1	18.7	20.4
102.5	12.5	13.5	14.6	15.9	17.3	18.8	20.6
103.0	12.6	13.6	14.8	16.0	17.4	19.0	20.8
103.5	12.7	13.7	14.9	16.2	17.6	19.2	21.0
104.0	12.8	13.9	15.0	16.3	17.8	19.4	21.2
104.5	12.9	14.0	15.2	16.5	17.9	19.6	21.5
105.0	13.0	14.1	15.3	16.6	18.1	19.8	21.7
105.5	13.2	14.2	15.4	16.8	18.3	20.0	21.9
106.0	13.3	14.4	15.6	16.9	18.5	20.2	22.1

Weight-for-length BOYS
Birth to 2 years (z-scores)

World Health Organization

cm	-3 SD	-2 SD	-1 SD	Median	+1 SD	+2 SD	+3 SD
106.5	13.4	14.5	15.7	17.1	18.6	20.4	22.4
107.0	13.5	14.6	15.9	17.3	18.8	20.6	22.6
107.5	13.6	14.7	16.0	17.4	19.0	20.8	22.8
108.0	13.7	14.9	16.2	17.6	19.2	21.0	23.1
108.5	13.8	15.0	16.3	17.8	19.4	21.2	23.3
109.0	14.0	15.1	16.5	17.9	19.6	21.4	23.6
109.5	14.1	15.3	16.6	18.1	19.8	21.7	23.8
110.0	14.2	15.4	16.8	18.3	20.0	21.9	24.1

WHO Child Growth Standards

Weight-for-height BOYS
2 to 5 years (z-scores)

World Health Organization

cm	-3 SD	-2 SD	-1 SD	Median	1 SD	2 SD	3 SD
65.0	5.9	6.3	6.9	7.4	8.1	8.8	9.6
65.5	6.0	6.4	7.0	7.6	8.2	8.9	9.8
66.0	6.1	6.5	7.1	7.7	8.3	9.1	9.9
66.5	6.1	6.6	7.2	7.8	8.5	9.2	10.1
67.0	6.2	6.7	7.3	7.9	8.6	9.4	10.2
67.5	6.3	6.8	7.4	8.0	8.7	9.5	10.4
68.0	6.4	6.9	7.5	8.1	8.8	9.6	10.5
68.5	6.5	7.0	7.6	8.2	9.0	9.8	10.7
69.0	6.6	7.1	7.7	8.4	9.1	9.9	10.8
69.5	6.7	7.2	7.8	8.5	9.2	10.0	11.0
70.0	6.8	7.3	7.9	8.6	9.3	10.2	11.1
70.5	6.9	7.4	8.0	8.7	9.5	10.3	11.3
71.0	6.9	7.5	8.1	8.8	9.6	10.4	11.4
71.5	7.0	7.6	8.2	8.9	9.7	10.6	11.6
72.0	7.1	7.7	8.3	9.0	9.8	10.7	11.7
72.5	7.2	7.8	8.4	9.1	9.9	10.8	11.8
73.0	7.3	7.9	8.5	9.2	10.0	11.0	12.0
73.5	7.4	7.9	8.6	9.3	10.2	11.1	12.1
74.0	7.4	8.0	8.7	9.4	10.3	11.2	12.2
74.5	7.5	8.1	8.8	9.5	10.4	11.3	12.4
75.0	7.6	8.2	8.9	9.6	10.5	11.4	12.5
75.5	7.7	8.3	9.0	9.7	10.6	11.6	12.6
76.0	7.7	8.4	9.1	9.8	10.7	11.7	12.8
76.5	7.8	8.5	9.2	9.9	10.8	11.8	12.9
77.0	7.9	8.5	9.2	10.0	10.9	11.9	13.0
77.5	8.0	8.6	9.3	10.1	11.0	12.0	13.1
78.0	8.0	8.7	9.4	10.2	11.1	12.1	13.3
78.5	8.1	8.8	9.5	10.3	11.2	12.2	13.4
79.0	8.2	8.8	9.6	10.4	11.3	12.3	13.5
79.5	8.3	8.9	9.7	10.5	11.4	12.4	13.6

Weight-for-height BOYS
2 to 5 years (z-scores)

World Health Organization

cm	-3 SD	-2 SD	-1 SD	Median	1 SD	2 SD	3 SD
65.0	5.9	6.3	6.9	7.4	8.1	8.8	9.6
65.5	6.0	6.4	7.0	7.6	8.2	8.9	9.8
66.0	6.1	6.5	7.1	7.7	8.3	9.1	9.9
66.5	6.1	6.6	7.2	7.8	8.5	9.2	10.1
67.0	6.2	6.7	7.3	7.9	8.6	9.4	10.2
67.5	6.3	6.8	7.4	8.0	8.7	9.5	10.4
68.0	6.4	6.9	7.5	8.1	8.8	9.6	10.5
68.5	6.5	7.0	7.6	8.2	9.0	9.8	10.7
69.0	6.6	7.1	7.7	8.4	9.1	9.9	10.8
69.5	6.7	7.2	7.8	8.5	9.2	10.0	11.0
70.0	6.8	7.3	7.9	8.6	9.3	10.2	11.1
70.5	6.9	7.4	8.0	8.7	9.5	10.3	11.3
71.0	6.9	7.5	8.1	8.8	9.6	10.4	11.4
71.5	7.0	7.6	8.2	8.9	9.7	10.6	11.6
72.0	7.1	7.7	8.3	9.0	9.8	10.7	11.7
72.5	7.2	7.8	8.4	9.1	9.9	10.8	11.8
73.0	7.3	7.9	8.5	9.2	10.0	11.0	12.0
73.5	7.4	7.9	8.6	9.3	10.2	11.1	12.1
74.0	7.4	8.0	8.7	9.4	10.3	11.2	12.2
74.5	7.5	8.1	8.8	9.5	10.4	11.3	12.4
75.0	7.6	8.2	8.9	9.6	10.5	11.4	12.5
75.5	7.7	8.3	9.0	9.7	10.6	11.6	12.6
76.0	7.7	8.4	9.1	9.8	10.7	11.7	12.8
76.5	7.8	8.5	9.2	9.9	10.8	11.8	12.9
77.0	7.9	8.5	9.2	10.0	10.9	11.9	13.0
77.5	8.0	8.6	9.3	10.1	11.0	12.0	13.1
78.0	8.0	8.7	9.4	10.2	11.1	12.1	13.3
78.5	8.1	8.8	9.5	10.3	11.2	12.2	13.4
79.0	8.2	8.8	9.6	10.4	11.3	12.3	13.5
79.5	8.3	8.9	9.7	10.5	11.4	12.4	13.6

Weight-for-height BOYS
2 to 5 years (z-scores)

World Health Organization

cm	-3 SD	-2 SD	-1 SD	Median	1 SD	2 SD	3 SD
80.0	8.3	9.0	9.7	10.6	11.5	12.6	13.7
80.5	8.4	9.1	9.8	10.7	11.6	12.7	13.8
81.0	8.5	9.2	9.9	10.8	11.7	12.8	14.0
81.5	8.6	9.3	10.0	10.9	11.8	12.9	14.1
82.0	8.7	9.3	10.1	11.0	11.9	13.0	14.2
82.5	8.7	9.4	10.2	11.1	12.1	13.1	14.4
83.0	8.8	9.5	10.3	11.2	12.2	13.3	14.5
83.5	8.9	9.6	10.4	11.3	12.3	13.4	14.6
84.0	9.0	9.7	10.5	11.4	12.4	13.5	14.8
84.5	9.1	9.9	10.7	11.5	12.5	13.7	14.9
85.0	9.2	10.0	10.8	11.7	12.7	13.8	15.1
85.5	9.3	10.1	10.9	11.8	12.8	13.9	15.2
86.0	9.4	10.2	11.0	11.9	12.9	14.1	15.4
86.5	9.5	10.3	11.1	12.0	13.1	14.2	15.5
87.0	9.6	10.4	11.2	12.2	13.2	14.4	15.7
87.5	9.7	10.5	11.3	12.3	13.3	14.5	15.8
88.0	9.8	10.6	11.5	12.4	13.5	14.7	16.0
88.5	9.9	10.7	11.6	12.5	13.6	14.8	16.1
89.0	10.0	10.8	11.7	12.6	13.7	14.9	16.3
89.5	10.1	10.9	11.8	12.8	13.9	15.1	16.4
90.0	10.2	11.0	11.9	12.9	14.0	15.2	16.6
90.5	10.3	11.1	12.0	13.0	14.1	15.3	16.7
91.0	10.4	11.2	12.1	13.1	14.2	15.5	16.9
91.5	10.5	11.3	12.2	13.2	14.4	15.6	17.0
92.0	10.6	11.4	12.3	13.4	14.5	15.8	17.2
92.5	10.7	11.5	12.4	13.5	14.6	15.9	17.3
93.0	10.8	11.6	12.6	13.6	14.7	16.0	17.5
93.5	10.9	11.7	12.7	13.7	14.9	16.2	17.6
94.0	11.0	11.8	12.8	13.8	15.0	16.3	17.8
94.5	11.1	11.9	12.9	13.9	15.1	16.5	17.9
95.0	11.1	12.0	13.0	14.1	15.3	16.6	18.1

Weight-for-height BOYS
2 to 5 years (z-scores)

World Health Organization

cm	-3 SD	-2 SD	-1 SD	Median	1 SD	2 SD	3 SD
95.5	11.2	12.1	13.1	14.2	15.4	16.7	18.3
96.0	11.3	12.2	13.2	14.3	15.5	16.9	18.4
96.5	11.4	12.3	13.3	14.4	15.7	17.0	18.6
97.0	11.5	12.4	13.4	14.6	15.8	17.2	18.8
97.5	11.6	12.5	13.6	14.7	15.9	17.4	18.9
98.0	11.7	12.6	13.7	14.8	16.1	17.5	19.1
98.5	11.8	12.8	13.8	14.9	16.2	17.7	19.3
99.0	11.9	12.9	13.9	15.1	16.4	17.9	19.5
99.5	12.0	13.0	14.0	15.2	16.5	18.0	19.7
100.0	12.1	13.1	14.2	15.4	16.7	18.2	19.9
100.5	12.2	13.2	14.3	15.5	16.9	18.4	20.1
101.0	12.3	13.3	14.4	15.6	17.0	18.5	20.3
101.5	12.4	13.4	14.5	15.8	17.2	18.7	20.5
102.0	12.5	13.6	14.7	15.9	17.3	18.9	20.7
102.5	12.6	13.7	14.8	16.1	17.5	19.1	20.9
103.0	12.8	13.8	14.9	16.2	17.7	19.3	21.1
103.5	12.9	13.9	15.1	16.4	17.8	19.5	21.3
104.0	13.0	14.0	15.2	16.5	18.0	19.7	21.6
104.5	13.1	14.2	15.4	16.7	18.2	19.9	21.8
105.0	13.2	14.3	15.5	16.8	18.4	20.1	22.0
105.5	13.3	14.4	15.6	17.0	18.5	20.3	22.2
106.0	13.4	14.5	15.8	17.2	18.7	20.5	22.5
106.5	13.5	14.7	15.9	17.3	18.9	20.7	22.7
107.0	13.7	14.8	16.1	17.5	19.1	20.9	22.9
107.5	13.8	14.9	16.2	17.7	19.3	21.1	23.2
108.0	13.9	15.1	16.4	17.8	19.5	21.3	23.4
108.5	14.0	15.2	16.5	18.0	19.7	21.5	23.7
109.0	14.1	15.3	16.7	18.2	19.8	21.8	23.9
109.5	14.3	15.5	16.8	18.3	20.0	22.0	24.2
110.0	14.4	15.6	17.0	18.5	20.2	22.2	24.4
110.5	14.5	15.8	17.1	18.7	20.4	22.4	24.7

Weight-for-height BOYS
2 to 5 years (z-scores)

World Health Organization

cm	-3 SD	-2 SD	-1 SD	Median	1 SD	2 SD	3 SD
111.0	14.6	15.9	17.3	18.9	20.7	22.7	25.0
111.5	14.8	16.0	17.5	19.1	20.9	22.9	25.2
112.0	14.9	16.2	17.6	19.2	21.1	23.1	25.5
112.5	15.0	16.3	17.8	19.4	21.3	23.4	25.8
113.0	15.2	16.5	18.0	19.6	21.5	23.6	26.0
113.5	15.3	16.6	18.1	19.8	21.7	23.9	26.3
114.0	15.4	16.8	18.3	20.0	21.9	24.1	26.6
114.5	15.6	16.9	18.5	20.2	22.1	24.4	26.9
115.0	15.7	17.1	18.6	20.4	22.4	24.6	27.2
115.5	15.8	17.2	18.8	20.6	22.6	24.9	27.5
116.0	16.0	17.4	19.0	20.8	22.8	25.1	27.8
116.5	16.1	17.5	19.2	21.0	23.0	25.4	28.0
117.0	16.2	17.7	19.3	21.2	23.3	25.6	28.3
117.5	16.4	17.9	19.5	21.4	23.5	25.9	28.6
118.0	16.5	18.0	19.7	21.6	23.7	26.1	28.9
118.5	16.7	18.2	19.9	21.8	23.9	26.4	29.2
119.0	16.8	18.3	20.0	22.0	24.1	26.6	29.5
119.5	16.9	18.5	20.2	22.2	24.4	26.9	29.8
120.0	17.1	18.6	20.4	22.4	24.6	27.2	30.1

WHO Child Growth Standards

MINISTRY OF HEALTH

REPUBLIC OF KENYA

Integrated MCH Flow Chart

ENTRY

Registration

- MCH Handbook

Antenatal

- Birth plan
- Antenatal profile
- Tetanus Toxoid
- Mother knows HIV status
- Mother on ART
- Retested after 3 months
- IPTp in malaria endemic areas
- Blood Pressure (BP)
- Oedema
- TB Screening
- Anaemia
- Slept under LLIN in malaria endemic areas
- Danger signs during pregnancy assessed
- De-worming

Post Natal

- Postnatal Physical Examination
- Vitamin A
- Mother Knows HIV Status
- Cervical Cancer Screening
- Mother on any FP

Child

- Birth Registration
- Immunization up-to-date
- Vitamin A up-to-date
- De-worming in the last 6 months
- Care for child development
- Length/Height
- Weight
- Temperature
- Slept under LLIN in malaria endemic areas
- If Child Exposed
  - NVP
  - Cotrimoxazole
  - DBS
  - TB screen
- If Child HIV Positive
  - on ART
- If 6 months EBF
- If > 6 months appropriate complementary feeding

If NO to any

Counsel and take necessary action

To appropriate MCH services

Positive encouragement with appropriate counseling

If Yes to all

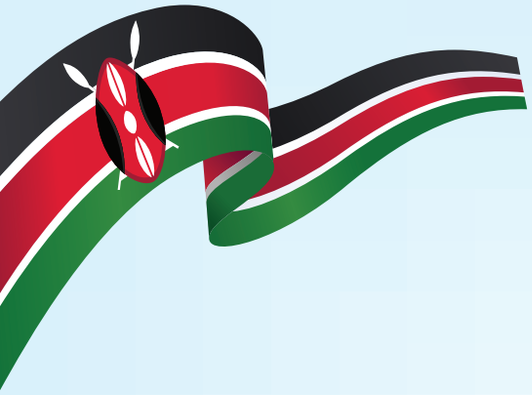
Service offered at MCH

1. MCH Handbook
2. ANC - PMTCT
3. PNC
4. FP
5. HIV Testing/ DBS & Psycho-social support
   - -adherence
   - -disclosure
   - -condom use
6. Immunization
7. Growth Monitoring
8. Vitamin A supplementation
9. Iron and folate supplementation
10. De-worming
11. Sick child consultation (IMNCI)
12. LLIN provision
13. Nutrition
    Counselling on EBF, Complementary feeding
14. Laboratory Services
    RDT/BS, Blood sugar, Blood group, Rh Factor, CD4 Viral Load, HIV Testing, DBS, VDRL/RPR, Urinalysis, Hb, Pregnancy Diagnostic Test.
15. Pharmacy Services
    IMNCI drugs, Cotrimoxazole, NVP, INH, ART, IFAS

EXIT

Abbreviations

TT	Tetanus Toxoid	**MNCH**	Maternal, Newborn and Child Health
IPTp	Intermittent Preventive Treatment of Malaria in pregnancy	**ANC**	Antenatal Care
LLIN	Long Lasting Insecticidal Nets	**PNC**	Postnatal Care
NVP	Nevirapine	**PMTCT**	Prevention of Mother-to-Child Transmission of HIV
DBS	Dried Blood Spot	**IMNCI**	Intergrated Management of Newborn & Childhood Illness
EBF	Exclusive Breastfeeding		
FP	Family Planning		



REPUBLIC OF KENYA

Ministry of Health